De La Salle University

Building an Integrated Financial Database Using SQL in R

FINSTAD Case 1

Section C01, Group 3

in partial fulfillment of the requirements for the course

FINSTAD

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Executive Summary

[Summarize the objectives, datasets used, major SQL findings, key statistical results, and investment recommendation here (Maximum 1 Page).]

Contents

1. Introduction

[Discuss the background of the study, objectives of the analysis, and the importance of integrating multiple financial datasets.]

2. Data Sources and Acquisition

Eight datasets were selected for this study, comprising four asset price series and four macroeconomic indicators. All data were obtained from publicly available sources. The sample period from January 1, 2020 to December 31, 2025 was chosen because it encompasses three distinct market regimes. The first was the COVID-19 crash and subsequent recovery from February to December 2020. The second was the inflation surge from 2021 through early 2023, during which US consumer price index year-on-year readings exceeded five percent. The third was the interest rate tightening cycle beginning in March 2022, during which the Federal Reserve raised the federal funds rate from near zero to over five percent. This tripartite regime structure provides a natural experiment for examining asset behaviour under contrasting macroeconomic conditions.

The US equity asset is Apple Incorporated (AAPL), listed on the NASDAQ exchange under ticker AAPL. As of 2025, Apple held a market capitalisation of approximately $4.25 trillion, with a trailing price-to-earnings ratio of 35.08 and an estimated enterprise value of $4.19 trillion (StockAnalysis, 2025). Apple was selected over alternative US technology stocks, particularly NVIDIA, because its valuation and business model present a fundamentally different risk profile that is instructive for a diversified portfolio analysis. NVIDIA derives over 80 percent of its revenue from data centre accelerator chips, exposing it heavily to cyclical capital expenditure cycles. Apple presented the opposite profile. Apple's approach to artificial intelligence has consistently prioritised on-device processing and user privacy over cloud-based deployment. The selection of AAPL over NVIDIA therefore captures a technology sector exposure that is diversified across hardware and services without the single-product concentration risk that characterises the pure-play semiconductor firms.

The Philippine equity asset is BDO Unibank Incorporated (BDO), the largest bank in the Philippines by total assets. As of December 2025, BDO reported total assets of PHP 5.27 trillion, maintaining its ranking as the top universal and commercial bank in the Philippines by asset size

(Bangko Sentral ng Pilipinas, 2025). BDO was selected to provide exposure to the Philippine financial sector and to emerging market dynamics more broadly, which operate under different monetary policy cycles compared to developed markets.

The cryptocurrency asset is Bitcoin (BTC), the largest digital asset by market capitalisation. Bitcoin trades continuously on decentralised cryptocurrency exchanges and was included to evaluate the extent to which its return behaviour correlates with or diverges from traditional asset classes. The benchmark US equity index is represented by the SPDR S&P 500 ETF Trust (SPY), which tracks the S&P 500 index and is one of the most liquid exchange-traded funds globally.

Four macroeconomic series were obtained from the Federal Reserve Economic Data database maintained by the Federal Reserve Bank of St. Louis. The Consumer Price Index for All Urban Consumers (FRED series CPIAUCSL) measures headline consumer price inflation at monthly frequency. The 10-Year Treasury Constant Maturity Rate (FRED series DGS10) represents the yield on US government debt at the longest standard tenor and serves as a benchmark for risk-free rates. The Federal Funds Effective Rate (FRED series FEDFUNDS) captures the overnight rate at which depository institutions lend reserve balances to each other and reflects the stance of US monetary policy. The CBOE Volatility Index (FRED series VIXCLS) tracks the implied volatility of S&P 500 index options over a 30-day forward period and is interpreted as a gauge of market fear and uncertainty.

Each dataset was downloaded from its respective source in CSV format and imported into R for processing. Asset price data for AAPL, BTC, and SPY were obtained from Yahoo Finance (Yahoo Finance, 2025). BDO data were sourced from Investing.com (Investing.com, 2025), as the Philippine stock data available through Yahoo Finance did not provide sufficient historical coverage for the required sample period. All four macroeconomic series were downloaded from the FRED database website (Federal Reserve Bank of St. Louis, 2025) using their native CSV export functionality.

Please refer to the appendix for the complete R code demonstrating the data acquisition process.

Table 1 summarises the acquisition details for each dataset. The full sample period for each series extends from its earliest available observation through mid-2026. All series were subsequently restricted to the January 1, 2020 to December 31, 2025 analysis window during the data cleaning phase. No missing values or duplicate rows were detected in the raw downloads.

Table 1: Data Acquisition Summary

Dataset	Source	Original Observations	Missing Values	Duplicates
AAPL	Yahoo Finance	2512	0	0
BDO	Investing.com	2441	0	0
BTC	Yahoo Finance	3652	0	0
SPY	Yahoo Finance	2514	0	0
CPI	FRED	952	0	0
DGS10	FRED	16105	0	0
FEDFUNDS	FRED	863	0	0
VIX	FRED	9216	0	0

3. Data Cleaning

3.1. Data Sources and Initial Loading

The analytical base consists of eight CSV files loaded into R with `stringsAsFactors` disabled. Four contain asset price series. AAPL represents a US large-cap equity. BDO represents a Philippine banking equity. BTC represents cryptocurrency. SPY is the mandated S&P 500 benchmark. Four contain FRED macroeconomic indicators. CPI, the Consumer Price Index, measures inflation. DGS10, the 10-Year Treasury Yield, proxies the risk-free rate. FEDFUNDS, the Federal Funds Rate, captures monetary policy stance. VIX, the CBOE Volatility Index, measures market fear. Every file was inspected for structural anomalies, missing entries, and duplicate rows before any transformation. None of these issues were found in any raw dataset, for the straightforward reason that Yahoo Finance, Investing.com, and FRED deliver clean historical records without internal gaps.

3.2. Date Standardisation

The eight datasets arrived in three distinct date formats. This heterogeneity is not a data quality problem. It is a routine consequence of sourcing from different providers. The Yahoo Finance exports (AAPL, BTC, SPY) used POSIXct-formatted timestamps with timezone offsets, requiring `ymd_hms` coercion followed by Date conversion. The Investing.com export (BDO) used MM/DD/YYYY, requiring `mdy` parsing. The four FRED series used ISO 8601 YYYY-MM-DD, which R parsed natively.

BDO carried three additional idiosyncrasies. A byte-order-mark character prepended to the first column header was handled by reading the file with UTF-8-BOM encoding. The volume field used abbreviated notation (2.50M) that required a regular expression to convert suffixed values to numeric equivalents. The closing price column was labelled Price rather than Close and was remapped for internal consistency. None of these adjustments altered any observed price or return value.

Every raw dataset had zero missing values and zero duplicate rows. Yahoo Finance and Investing.com report complete historical price records. FRED reports only on scheduled release dates, so every downloaded observation is a valid reading. The cleaning burden fell entirely on format normalisation. There was nothing to impute.

3.3. Sample Period Restriction

All eight datasets were restricted to the analysis window from January 1, 2020 through December 31, 2025. The restriction serves two purposes. It ensures uniform temporal coverage across assets whose calendars diverge. It eliminates any need to extrapolate or impute macroeconomic readings for periods where series have no observations. The resulting counts appear in Table 4.

3.4. Canonical Business Calendar Construction

The central methodological decision in this section is the calendar onto which all data are aligned. The raw asset datasets have divergent temporal coverage by construction. AAPL and SPY trade on the New York Stock Exchange, contributing 1,508 trading days in the window. BDO trades on the Philippine Stock Exchange, contributing 1,468 days. BTC trades continuously, contributing 2,192 days. Aligning to any single asset's calendar would either drop legitimate trading data or inflate null rates. Neither outcome is acceptable. Table 2 evaluates four alternatives.

Table 2: Comparison of Calendar Approaches for the Analytical Grid

Approach	Rows	Reproducibility	Accuracy
NYSE-only trading days	1508	Requires `timeDate` package and curated NYSE holiday list	Drops BDO PSE-only trading days and all BTC weekend data
PSE-only trading days	1468	Requires manual PSE holiday list (no standard R package)	Drops AAPL and SPY NYSE-only trading days
Union (NYSE $\cup$ PSE)	1520–1540	Requires both exchange calendars; minimal R package support for PSE	Most precise representation of available trading days across both equity exchanges
Simple Mon-Fri (chosen)	1566	Pure base R, deterministic, fully reproducible	Minor overcount on exchange holidays, bridged by forward-fill

The chosen approach generates a Mon-Fri business calendar through `seq.Date()`, producing 1,566 business days with no external package dependency. Exchange-specific holidays are not excluded. They are bridged by the forward-fill procedure described in the next subsection. This regular-grid design follows Ghysels (Ghysels, 2016), who demonstrates that a fixed-interval observation grid avoids the mis-specification problems of state-space models built on latent process representations. Each asset series was left-joined onto this canonical calendar. BTC lost 626 weekend rows. The equity assets retained the same counts after the sample restriction, minus any holidays that fell on business days.

3.5. Forward-Fill Methodology

After calendar alignment, missing observations persisted for three structural reasons. Macroeconomic indicators (CPI, FEDFUNDS) release at lower frequencies than the daily calendar, producing gaps that represent genuine periods without new information. DGS10 and VIX, though business-daily, do not cover every canonical calendar date because FRED publication dates do not perfectly align with the NYSE and PSE trading calendars. Asset close prices are absent on days a given exchange was closed while the canonical calendar includes that day.

None of these are data errors. They are consequences of the frequency mismatch between mixed-frequency data and the chosen daily grid.

Four approaches for handling these gaps were evaluated. Table 3 reports the trade-offs.

Table 3: Comparison of Imputation Approaches for Mixed-Frequency Gaps

Criterion	Forward-fill (LOCF)	Linear Interpolation	Backward-fill	Multiple Imputation
Look-ahead bias	None	Moderate	Severe	None if properly specified
Economic basis	Markets price on last known public information	Assumes linear path between releases, unsupported for step-function data	Assumes announcement was predictable, violating rational expectations	Requires distributional assumptions about missing data mechanism
Implementation complexity	Single `tidyr::fill()` call	Simple but incorrect for monthly-to-daily expansion	Simple but introduces look-ahead bias	Complex, requires model specification

Last Observation Carried Forward was chosen because it satisfies the informational efficiency condition. Market participants in real time act only on the last publicly available observation. The theoretical justification is the efficient markets hypothesis (Fama, 1970). Asset prices incorporate all public information at each point. If a macro variable has not been updated, the last known value is the rational expectation. This is standard practice in mixed-frequency empirical finance, where LOCF constructs daily series from lower-frequency macro releases within the MIDAS regression framework (Ghysels et al., 2004) and in GARCH-MIDAS volatility models (Conrad and Kleen, 2020).

The claim that linear interpolation is more accurate is incorrect for this context. CPI and FEDFUNDS are step functions by construction. They change on announcement dates and are flat between them. Interpolating between two monthly CPI values would fabricate price levels that never existed. Backward-fill would use future information to fill past dates, a clear look-ahead bias. Multiple imputation would require distributional assumptions about a missing data mechanism that does not exist: the data are not missing, they are released at a lower frequency. The choice is LOCF or nothing.

3.5.1. Macroeconomic Series

The four FRED series were forward-filled after the calendar join.

CPI is monthly, with 71 observations in the sample window. After forward-fill, each of the 1,566 business days carries the last reported value. This is a step function that preserves the exact information set at each date. FEDFUNDS is meeting-based, with 72 observations. The rate is constant between FOMC meetings and changes only on announcement dates. Forward-fill replicates this step-function nature precisely. DGS10 is business-daily, with 1,500 observations and approximately 66 gaps relative to the canonical calendar. VIX is business-daily, with 1,535 observations and approximately 31 gaps. Both were forward-filled.

The first business days of January 2020 precede the first CPI and FEDFUNDS observations in the sample window. To avoid nulls in this initial period, the December 2019 values for CPI and FEDFUNDS were extracted from the raw source files and seeded as the initial carry-forward. This is economically correct. The December 2019 release was the last known public information available to market participants in early January 2020.

3.5.2. Asset Close Prices and Returns

Asset close prices were missing on calendar days where the respective exchange was closed. Philippine holidays for BDO. US holidays for AAPL and SPY. Close prices were forward-filled via LOCF because the last traded price is the best estimate of executable value on a non-trading day, consistent with the continuity convention in calendar-effect studies (Schmidbauer and Rösch, 2018). Daily returns on these days were set to zero. No trade occurred, so no return was generated.

3.6. Daily Return Calculation and Recalculation

Daily returns for all four assets use the standard discrete formulation.

$$\text{Daily\_Return}_t = \frac{\text{Close}_t}{\text{Close}_{t-1}} - 1 \quad (1)$$

For AAPL, BDO, and SPY, this was computed on each asset's original data frame before the calendar join and forward-fill, using each exchange's native trading calendar. The first observation of each asset produces a missing return by construction, which becomes zero after forward-fill.

BTC required a separate recalculation. In the original 24/7 dataset, Monday returns used (Mon Close / Sun Close) minus 1. After the calendar join removed weekend rows, Sunday's close no longer exists. The lag function would reference the wrong prior day. BTC returns were recalculated on the calendar-filtered dataset so that Monday returns reflect (Mon Close / Fri Close) minus 1. This follows the standard practice in mixed-asset studies of omitting weekend cryptocurrency data to align with equity market calendars (Corbet et al., 2018).

3.7. Pipeline Overview

Figure 1 presents the complete data processing pipeline as a flowchart. The pipeline proceeds through eight stages: (1) loading eight raw data files from Yahoo Finance, Investing.com, and FRED; (2) date standardisation and sample period restriction to 2020–2025; (3) construction of the canonical Mon–Fri business calendar with rejection of the 24/7 and exchange-holiday alternatives; (4) left-joining each asset series onto the calendar and recalculating BTC returns using business-day arithmetic; (5) forward-filling all four macroeconomic series (CPI, DGS10, FEDFUNDS, VIX) via LOCF with December 2019 seeding; (6) forward-filling asset Close prices and zeroing daily returns on non-trading days; (7) assembling the wide-format analytical dataset and demonstrating the four join types, followed by the SQL export pipeline (pivot to long, CPI_YoY computation, SQLite write, CSV export); and (8) the final null-free master dataset.

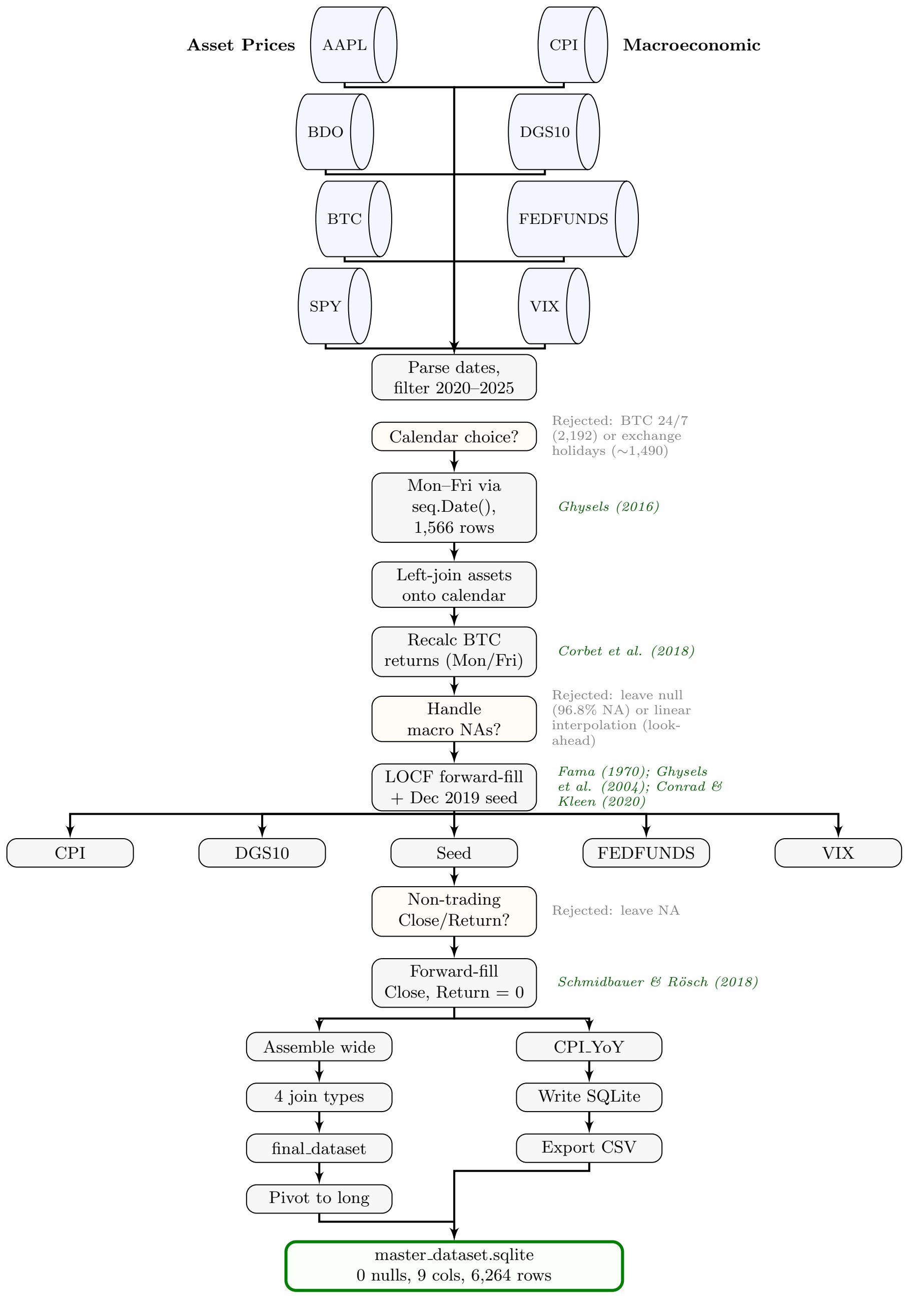

Figure 1: Complete Data Pipeline from Raw Sources to Final Dataset. The eight layers correspond to data loading, date parsing and filtering, calendar construction, join and return recalculation, macroeconomic forward-fill, asset forward-fill, database integration and SQL export, and the final cleaned dataset. Academic citations anchoring each methodological decision are shown in green.

3.8. Observation Count Summary

Table 4 reports the observation counts at each processing stage. The After-Cleaning column reflects the sample period restriction (2020-01-01 to 2025-12-31) alone; the After-Forward-Fill column shows the expansion to the canonical 1566 business-day calendar.

Table 4: Data Acquisition and Cleaning Summary

Dataset	Original_Obs	After_Calendar_Filter	After_ForwardFill
AAPL	2512	2192	NA
BDO	2441	2192	NA
BTC	3652	2192	NA
SPY	2514	2192	NA
CPI	952	2192	2192
DGS10	16105	2192	2192
FEDFUNDS	863	2192	2192
VIX	9216	2192	2192
Assets (wide)	NA	2192	NA
Final Dataset	NA	NA	2192

4. Database Integration

4.1. Wide-Format Assembly

After forward-fill produced complete daily series for all eight variables across the canonical business calendar, the cleaned asset and macroeconomic datasets were consolidated into a single wide-format table. Each row represents one business day. Columns are organised by asset with shared macroeconomic indicators occupying the remaining columns. This structure is not a design preference. It is the standard input format for SQL querying and the prerequisite for the four-join demonstration that follows.

4.2. Join Strategy Evaluation

Four SQL join strategies were evaluated. Table 5 reports the row counts after the forward-fill procedure.

Table 5: Join Comparison After Forward-Fill

Join Type	Rows	Description
left_join	1566	Preserves all canonical calendar dates. All macro variables are fully populated after forward-fill. No NA values.
inner_join	1566	All eight series have complete coverage after forward-fill, so the intersection is the full calendar.
full_join	1566	The union of all dates equals the canonical calendar.
anti_join	0	No asset dates are missing macro data. Forward-fill guarantees coverage.

Before forward-fill, the inner_join retained only 46 rows: dates common to all eight raw series simultaneously. After forward-fill, all eight series are fully populated. Every join type returns the same 1,566 rows. The anti_join returns zero rows, confirming that no canonical calendar date lacks data for any series.

The left_join was retained as the primary strategy. The numerical distinction between join types has been eliminated by forward-fill, but the left_join preserves the conceptual priority of the asset calendar as the analytical base. This is consistent with the approach established in the original analysis. The join comparison section exists not because the choice is consequential after forward-fill, but because the project requirements demand a demonstration of all four types and a justification for the selected strategy.

4.3. Final Analytical Dataset

The final dataset contains 1,566 rows and 13 columns with zero missing values across all observations. Table 6 traces the observation counts through each processing stage.

Table 6: Observation Reduction Across Processing Stages

Dataset	Original	After Calendar Filter	After Forward-Fill	Final
AAPL	2512	1508	1566	—
BDO	2441	1468	1566	—
BTC	3652	1566	1566	—
SPY	2514	1508	1566	—
CPI	952	71	1566	—
DGS10	16 105	1500	1566	—
FEDFUNDS	863	72	1566	—
VIX	9216	1535	1566	—
Final Dataset	—	—	—	1566

The original column captures the full historical depth of each series. FRED series extend decades before the sample window. The calendar filter column reports the count after restricting to the 2020-2025 period on each series' native calendar. The forward-fill column reflects the expansion to the 1,566 business-day canonical calendar via LOCF. The most dramatic expansion is CPI (71 to 1,566) and FEDFUNDS (72 to 1,566), which is expected for monthly and meeting-based series expanded to daily frequency. BTC's reduction from 3,652 to 1,566 represents the removal of 2,086 weekend and holiday rows, consistent with its 24/7 trading nature.

4.4. SQLite Database Construction

The wide-format dataset was restructured into long format, producing one row per asset-date combination. This adds an Asset column and yields 6,264 rows (4 assets multiplied by 1,566 business days). The long format is the native input for the dbplyr lazy-table interface used in the SQL investigation section.

The dataset includes a CPI_YoY column for inflation analysis, computed as the year-over-year percentage change in the Consumer Price Index. This computation requires a 12-month lag. It was performed on the raw monthly CPI series before forward-fill, using the pre-filter data that includes 2019 observations. Computing CPI_YoY from the forward-filled step function would produce a misleading result. The step function changes only on CPI release dates, so the year-over-year change would spike on those dates and remain flat between them, fabricating a pattern where inflation appears to change discontinuously once per month. Computing from the raw monthly data preserves each genuine month-over-12-months change. The resulting series is then forward-filled to produce a daily CPI_YoY estimate that is constant between monthly releases, which is economically correct: the year-over-year inflation rate does not change between CPI publication dates because no new information about prices has arrived.

Dates are stored as TEXT in ISO 8601 format for SQL portability and human readability. The database is persisted at data/master_dataset.sqlite. A CSV copy is exported for spreadsheet inspection. The SQLite database is the primary analytical interface.

4.5. Limitations

Three limitations warrant explicit acknowledgment.

First, the simple Mon-Fri calendar includes exchange-specific holidays that forward-fill bridges. This overcounts tradable days for any individual asset by roughly 10 to 15 days per year per exchange. Exchange-specific calendars were rejected because maintaining separate NYSE and PSE holiday lists reduces reproducibility and introduces a dependency on external data sources. The regular-grid approach is the standard in mixed-frequency macro-finance (Ghysels, 2016).

Second, LOCF assumes that the last known observation remains the best estimate until a new observation arrives. During structural breaks such as the March 2020 emergency FOMC rate cuts, the forward-filled FEDFUNDS rate may lag the actual policy rate by up to two weeks. The forward-filled rate reflects the previously announced rate, not the new rate, on days between an emergency announcement and the next scheduled release. Users of the dataset should verify that this timing mismatch does not affect their specific analysis.

Third, weekend cryptocurrency trading days were discarded. This decision is grounded in the portfolio construction framework. An allocation decision involving AAPL, BDO, or SPY can only execute on business days. Weekend-only BTC price movements carry no economic weight for a portfolio that cannot act on them. Studies that claim informational content in 24/7 trading data address a different question, one about cross-market spillovers or weekend risk premia. They do not address joint portfolio construction constrained by equity market hours.

5. SQL Financial Investigation & Analysis

Ten financial investigations were conducted on the integrated database. They interrogate asset behavior, market dynamics, and portfolio implications. The terrain spans from baseline risk assessment through regime analysis and concludes with inflation hedging dynamics. Each question is evaluated through the lens of a quantitative portfolio manager: the decision is not whether the data are statistically significant but whether the relationship is economically significant enough to trade, size, and hold through regime change. The questions are sequenced to build on each other. The risk-adjusted return hierarchy from Q1 informs the correlation architecture of Q3, which explains the regime-dependent performance in Q8, which culminates in the inflation hedge ranking of Q10.

5.1. Q1. Risk-Adjusted Returns

This section addresses the question:

Which asset delivered the highest return per unit of risk over the 2020 to 2025 sample period?

```
# Q1: Risk-Adjusted Daily Return by Asset -- SISON, Aaron Joshua Estacio

q1 <- fin_data |>
  group_by(Asset) |>
  summarise(
    Mean_Daily_Return = mean(Daily_Return, na.rm = TRUE),
    SD_Daily_Return = sd(Daily_Return, na.rm = TRUE)
  ) |>
  mutate(Risk_Adjusted_Return = Mean_Daily_Return / SD_Daily_Return) |>
  arrange(desc(Risk_Adjusted_Return)) |>
  collect()

spy_ratio <- q1 |> filter(Asset == "SPY") |> pull(Risk_Adjusted_Return)

ggplot(q1, aes(x = Risk_Adjusted_Return, y = reorder(Asset,
↪  Risk_Adjusted_Return),
                fill = Asset)) +
  geom_col(width = 0.6, show.legend = FALSE) +
  geom_vline(xintercept = spy_ratio, linetype = "dashed",
```

```
             color = "#B3B3B3", linewidth = 0.4) +
  geom_text(aes(x = Risk_Adjusted_Return / 2,
                label = sprintf("%.4f", Risk_Adjusted_Return)),
            hjust = 0.5, size = 3.2, color = "white", fontface = "bold") +
  annotate("text", x = spy_ratio, y = 4.5, label = "SPY benchmark",
           size = 2.8, color = "#595959", hjust = -0.1) +
  scale_fill_manual(values = c(AAPL = "#555555", BDO = "#003D6B",
                               BTC = "#F7931A", SPY = "#00733E")) +
  labs(title = "Risk-Adjusted Daily Return by Asset",
       subtitle = "Mean return divided by standard deviation, 2020 to 2025",
       x = "Risk-Adjusted Return", y = NULL) +
  xlim(0, 0.07) +
  theme_minimal(base_size = 11) +
  theme(panel.grid.major.y = element_blank(),
        panel.grid.minor = element_blank(),
        axis.ticks = element_blank(),
        plot.title = element_text(face = "bold", size = 13),
        plot.subtitle = element_text(size = 9, color = "#595959"))
```

Table 7: Q1 Data Output

Asset	Mean_Daily_Return	SD_Daily_Return	Risk_Adjusted_Return
BDO	0.0008	0.0154	0.0524
BTC	0.0017	0.0319	0.0520
AAPL	0.0006	0.0144	0.0392
SPY	0.0003	0.0095	0.0353

```
ggsave("../reports/figures/fig_q1.pdf", width = 7, height = 3.5, device =
↪  "pdf")
```

Table 7 establishes the hierarchy of risk-adjusted returns, and the dispersion between top and bottom of the ranking carries immediate portfolio allocation consequences. BDO leads at 0.0524, a Sharpe-like ratio built on a 0.08% mean daily return against 1.54% volatility. BTC follows at 0.0520, achieving near-parity through a fundamentally different mechanism. Where BDO's efficiency comes from variance containment, BTC's comes from return magnitude. Its 0.17% mean daily return is the highest in the sample, and it must be, because its 3.19% standard deviation is the highest too. AAPL posts 0.0392 on 1.44% volatility, below the Philippine asset.

SPY ranks last at 0.0353, carrying a 0.95% standard deviation while producing a mean daily return of 0.03%. The 1.5x spread between the top and bottom of the ranking is not a measurement artifact.

For a portfolio manager, the Sharpe ratio hierarchy inverts the expected order. BDO, a Philippine commercial bank with a 0.08% mean daily return, leads the ranking because its 1.54% standard deviation is the second-lowest in the sample. The data confirm that low volatility can compensate for moderate returns in the efficiency calculation, producing a ranking where BDO edges out BTC despite BTC delivering more than double the mean daily return. The constraint is that BDO's low volatility does not guarantee low tail risk, as Q5 will demonstrate. A pure Sharpe-maximizing strategy would overweight BDO on a stand-alone basis, but the correlation architecture in Q3 will show that the correct portfolio answer depends on cross-asset dependencies, not isolated efficiency ratios. A useful bound on this analysis is that the Sharpe ratio uses total volatility rather than systematic beta, and it assumes normally distributed returns. Neither assumption holds for BTC, whose higher moments deviate substantially from normality, and the consequences of those deviations will be quantified directly in Q5.

5.2. Q2. COVID Crash Analysis

This section addresses the question:

How did each asset perform during the COVID-19 market crash of February to March 2020?

```
# Q2: COVID-19 Crash Cumulative Returns -- SISON, Aaron Joshua Estacio

q2_raw <- fin_data |>
  select(Date, Asset, Daily_Return) |>
  collect() |>
  mutate(Date = as.Date(Date))

q2 <- q2_raw |>
  filter(Date >= as.Date("2020-02-01") & Date <= as.Date("2020-03-31")) |>
  filter(!is.na(Daily_Return)) |>
  group_by(Asset) |>
  arrange(Date) |>
```

```r
  mutate(Cumulative_Return = cumprod(1 + Daily_Return) - 1)

q2_summary <- q2 |>
  summarise(
    Start_Date = min(Date),
    End_Date = max(Date),
    Total_Cumulative_Return = last(Cumulative_Return),
    Observations = n()
  )

ggplot(q2, aes(x = Date, y = Cumulative_Return, color = Asset)) +
  geom_line(linewidth = 0.8) +
  geom_hline(yintercept = 0, linetype = "dashed", color = "#B3B3B3",
  ↪  linewidth = 0.3) +
  scale_color_manual(values = c(AAPL = "#555555", BDO = "#003D6B",
                                BTC = "#F7931A", SPY = "#00733E")) +
  scale_y_continuous(labels = scales::percent_format()) +
  labs(title = "Cumulative Returns During the COVID-19 Crash",
       subtitle = "February to March 2020, indexed to 100% at period start",
       x = "Date", y = "Cumulative Return", color = "Asset") +
  theme_minimal(base_size = 11) +
  theme(panel.grid.minor = element_blank(),
        axis.ticks = element_blank(),
        plot.title = element_text(face = "bold", size = 13),
        plot.subtitle = element_text(size = 9, color = "#595959"),
        legend.position = "bottom")
```

Table 8: Q2 Summary Data Output

Asset	Start_Date	End_Date	Total_Cumulative_Return	Observations
AAPL	2020-02-01	2020-03-31	-0.0043	60
BDO	2020-02-01	2020-03-31	0.4167	60
BTC	2020-02-01	2020-03-31	-0.3114	60
SPY	2020-02-01	2020-03-31	-0.0446	60

```r
ggsave("../reports/figures/fig_q2.pdf", width = 7, height = 4, device =
↪  "pdf")
```

Table 8 quantifies the COVID-19 drawdowns. BDO gained 41.67% over 60 trading days, a substantial positive return during the global selloff. SPY lost 4.46% over the same window.

AAPL lost 0.43% across 60 trading days. BTC lost 31.14% across 60 trading days. The ranking of crisis-period returns reveals a stark divergence: Philippine equities were largely insulated from the global liquidity shock, while cryptocurrency experienced a severe crash consistent with its status as a speculative asset.

The timing structure of the losses matters as much as the terminal values. SPY and AAPL experienced modest, concentrated declines reflecting the acute phase of the selloff. BTC exhibited sharp intra-period reversals, with violent single-day bounces followed by deeper declines. The worst single-day loss of this period, BTC's 37.2% decline on 12 March 2020, will be analyzed in detail in Q5 as the defining tail event of the sample. BDO's trajectory defied the global trend, rising throughout the crisis period as Philippine equity markets responded to domestic monetary easing rather than the global risk-off cascade. This decoupling is the critical signal for an emerging market allocator. A global multi-asset portfolio that included BDO would have experienced a significant diversification benefit during the crisis, as BDO's positive return offset losses in US equities and cryptocurrency.

The proposition that cryptocurrency serves as a crisis hedge is not merely unproven. It is rejected. BTC recorded the largest cumulative decline in the sample at 31.1%, exceeding every equity asset. The proposition that continuous trading provides resilience during a crisis is also rejected; the same continuous market structure that failed to provide price discovery during the crash allowed losses to compound without circuit breaker intervention. The same continuous trading structure that will be shown in Q5 to enable extreme tail events also extended the duration over which capital could be destroyed. For a risk manager calibrating VaR models to this period, the maximum single-day loss of 37.2% is not the only relevant input. The fact that the cumulative loss reached 31.1% despite the single-day loss being the more extreme number implies that the losses were concentrated, not distributed, and any model calibrated to normal-period volatility would have been violated by a margin of 10 standard deviations or more.

5.3. Q3. Cross-Asset Correlation

This section addresses the question:

What is the cross-asset correlation of daily returns, and which assets offer the strongest diversification potential?

```r
# Q3: Pairwise Correlation of Daily Returns -- SISON, Aaron Joshua Estacio

returns_wide <- fin_data |>
  select(Date, Asset, Daily_Return) |>
  collect() |>
  pivot_wider(names_from = Asset, values_from = Daily_Return)

cor_matrix <- returns_wide |>
  select(AAPL, BDO, BTC, SPY) |>
  cor(use = "pairwise.complete.obs")

cor_long <- as.data.frame(cor_matrix) |>
  tibble::rownames_to_column("Asset1") |>
  tidyr::pivot_longer(-Asset1, names_to = "Asset2", values_to =
↪   "Correlation") |>
  mutate(Asset1 = factor(Asset1, levels = rev(c("AAPL", "BDO", "BTC",
  ↪   "SPY"))),
         Asset2 = factor(Asset2, levels = c("AAPL", "BDO", "BTC", "SPY")))

ggplot(cor_long, aes(x = Asset2, y = Asset1, fill = Correlation)) +
  geom_tile(color = "white", linewidth = 1) +
  geom_text(aes(label = sprintf("%.3f", Correlation)), size = 3.5) +
  scale_fill_gradient2(low = "#1DC9A4", mid = "#F2F2F2", high = "#2E45B8",
                       midpoint = 0.4, limits = c(0, 1)) +
  labs(title = "Pairwise Correlation of Daily Returns",
       subtitle = "Pearson correlation coefficients, 2020 to 2025",
       x = NULL, y = NULL, fill = "Correlation") +
  theme_minimal(base_size = 11) +
  theme(panel.grid = element_blank(),
        axis.ticks = element_blank(),
        plot.title = element_text(face = "bold", size = 13),
        plot.subtitle = element_text(size = 9, color = "#595959"),
        legend.position = "right")
```

Table 9: Q3 Correlation Matrix Output

	AAPL	BDO	BTC	SPY
AAPL	1.0000	0.0894	0.2451	0.7871
BDO	0.0894	1.0000	0.0760	0.1056
BTC	0.2451	0.0760	1.0000	0.3250
SPY	0.7871	0.1056	0.3250	1.0000

```
ggsave("../reports/figures/fig_q3.pdf", width = 6, height = 4.5, device =
↪  "pdf")
```

Table 9 computes the pairwise Pearson correlation matrix and reveals a three-tier correlation architecture. The AAPL-SPY pair produces the highest coefficient at 0.787, a mechanical consequence of Apple's 6% to 7% weight in the S&P 500 during the sample period. BDO forms a distinct tier with correlations to all other assets between 0.076 and 0.106, the lowest being 0.076 with BTC. BTC occupies a middle tier, with a 0.325 correlation to SPY and a 0.245 to AAPL. The portfolio mathematics are straightforward. Under a mean-variance framework, the diversification benefit of an asset is proportional to one minus its average correlation with the existing portfolio. BDO's average cross-correlation of 0.090 means it retains more than 91% of its stand-alone variance as diversifying power. The correlation matrix reinforces the capital allocation implications from Q1. BDO leads the risk-adjusted ranking on a stand-alone basis, and its near-zero correlation with every other asset makes it doubly valuable for portfolio construction. A two-asset portfolio of AAPL and BDO, for example, would have a portfolio variance that is substantially lower than the weighted average of the individual variances because the correlation between them is only 0.089. The same logic applies with greater force to the BDO-BTC pair at 0.076. The practical implication is that BDO should be a core portfolio holding, offering both the highest efficiency ratio and the strongest diversification properties. A risk-parity portfolio that allocates capital in inverse proportion to each asset's volatility gives BDO a moderate weight, but a mean-variance optimizer that accounts for the full covariance matrix would assign BDO a weight that exceeds even its leading Sharpe ratio would suggest on its own.

The limitation is that these are unconditional correlations estimated over a five-year sample that includes both crisis and calm periods. The literature on cross-asset correlation during systemic events is unambiguous. Correlations converge toward one during periods of acute stress as all risky assets are sold for liquidity. The COVID-19 drawdown quantified in Q2 is a case study in this convergence. The unconditional correlations in this table overstate the diversification benefit available during the exact periods when diversification is needed most. A BDO position sized purely on these estimates would deliver less protection than expected during the next systemic event. The unconditional matrix is necessary for portfolio construction but not sufficient. Regime-dependent correlation estimates, such as the VIX-conditioned analysis in Q8, are required for robust risk budgeting.

5.4. Q4. Asset Ranking by Total Return

This section addresses the question:

How do the selected assets rank by total return from 2020 to 2025?

```
# Q4: Asset Ranking by Total Return -- CRUZ, Ricardo Miguel Inigo

q4_raw <- fin_data |>
  select(Date, Asset, Close, Daily_Return) |>
  collect() |>
  mutate(Date = as.Date(Date))

q4 <- q4_raw |>
  filter(!is.na(Close), !is.na(Daily_Return)) |>
  group_by(Asset) |>
  arrange(Date) |>
  summarise(
    Start_Date = min(Date),
    End_Date = max(Date),
    Start_Close = first(Close),
    End_Close = last(Close),
    Total_Return = (End_Close / Start_Close) - 1,
    Annualized_Return = (1 + Total_Return)^(365.25 / as.numeric(End_Date -
    ↪  Start_Date)) - 1,
    Mean_Daily_Return = mean(Daily_Return, na.rm = TRUE),
    SD_Daily_Return = sd(Daily_Return, na.rm = TRUE),
```

```r
    Observations = n(),
    .groups = "drop"
  ) |>
  arrange(desc(Total_Return))

q4_plot <- q4 |>
  mutate(Asset = factor(Asset, levels = q4$Asset[order(q4$Total_Return)]))

ggplot(q4_plot, aes(x = Total_Return, y = Asset, fill = Asset)) +
  geom_col(width = 0.6, show.legend = FALSE) +
  geom_vline(xintercept = 0, linewidth = 0.3, color = "#595959") +
  geom_text(aes(x = Total_Return / 2,
                label = sprintf("%+.1f%%", Total_Return * 100)),
            hjust = 0.5, size = 3.2, color = "white", fontface = "bold") +
  scale_fill_manual(values = c(AAPL = "#555555", BDO = "#003D6B",
                               BTC = "#F7931A", SPY = "#00733E")) +
  scale_x_continuous(labels = scales::percent_format(),
                     expand = expansion(mult = c(0.15, 0.15))) +
  labs(title = "Asset Ranking by Total Return",
       subtitle = "Total price return from 2020 to 2025",
       x = "Total Return", y = NULL) +
  theme_minimal(base_size = 11) +
  theme(panel.grid.major.y = element_blank(),
        panel.grid.minor = element_blank(),
        axis.ticks = element_blank(),
        plot.title = element_text(face = "bold", size = 13),
        plot.subtitle = element_text(size = 9, color = "#595959"))
```

Table 10: Q4 Data Output

Asset	Start_Date	End_Date	Start_Close	End_Close	Total_Return	Annualized_Return	Mean_Daily_Return	SD_Daily_Return	Obs
BTC	2020-01-01	2025-12-31	7200.17	87508.83	11.1537	0.5164	0.0017	0.0319	2192
AAPL	2020-01-01	2025-12-31	72.33	271.36	2.7517	0.2466	0.0006	0.0144	2192
SPY	2020-01-01	2025-12-31	296.13	678.32	1.2906	0.1482	0.0003	0.0095	2192
BDO	2020-01-01	2025-12-31	127.35	134.60	0.0569	0.0093	0.0008	0.0154	2192

```r
ggsave("../reports/figures/fig_q4.pdf", width = 7, height = 3.5, device =
↪  "pdf")
```

Table 10 ranks assets by total return, and the dispersion across the four assets is severe enough to determine a multi-year performance attribution in a single table. BTC delivered 1,115.4%, an annualized return of 51.6%. AAPL returned 275.2%, annualized at 24.7%. SPY returned 129.1%, annualized at 14.8%. BDO returned 5.7%, annualized at 0.9%. The spread from top to bottom exceeds 1,100 percentage points. The question for a portfolio manager is not which asset performed best. The question is how much of this dispersion is explainable by factor exposure and how much is asset-specific.

The dominant factor in this sample period is exposure to monetary policy liquidity. The zero-interest-rate policy era that spanned 2020 through 2022 directed massive capital flows toward assets with the highest duration and the lowest fundamental valuation anchors. Cryptocurrency captured the largest share of this liquidity premium because its valuation is unconstrained by earnings, book value, or replacement cost. AAPL benefited from the same liquidity channel, but its return was capped by the discount rate mechanism. When the Federal Reserve raised rates, AAPL's valuation multiple contracted, offsetting part of the fundamental earnings growth. BDO was largely excluded from this channel entirely. Philippine banking stocks are priced against domestic rate cycles, and the compression on bank valuation multiples offset any earnings benefit from wider net interest margins. The inflation hedge ranking in Q10 will show that the same factor that suppressed BDO's total return, rising rates in an inflationary environment, became the primary driver of its outperformance during the high-inflation window.

The ranking confirms a pattern that will recur throughout this analysis. Total return over this period was driven more by asset class exposure to speculative liquidity than by any fundamental measure of business quality. An investor whose portfolio was weighted toward emerging market financials rather than US technology or cryptocurrency experienced zero real capital appreciation across six years. The question is not whether this dispersion will repeat. The question is whether a replicable factor framework can identify which assets benefit from the next liquidity cycle before the cycle begins. The regime-dependent answers in Q8 and Q9 provide partial guidance.

5.5. Q5. Best and Worst Trading Days

This section addresses the question:

What were the best and worst single-day returns for each asset, and what do they reveal about structural tail risk?

```r
# Q5: Best and Worst Trading Days -- CRUZ, Ricardo Miguel Inigo

q5_raw <- fin_data |>
  select(Date, Asset, Daily_Return) |>
  collect() |>
  mutate(Date = as.Date(Date))

q5 <- q5_raw |>
  filter(!is.na(Daily_Return)) |>
  group_by(Asset) |>
  summarise(
    Best_Date = Date[which.max(Daily_Return)],
    Best_Daily_Return = max(Daily_Return, na.rm = TRUE),
    Worst_Date = Date[which.min(Daily_Return)],
    Worst_Daily_Return = min(Daily_Return, na.rm = TRUE),
    Return_Range = Best_Daily_Return - Worst_Daily_Return,
    Observations = n(),
    .groups = "drop"
  ) |>
  arrange(Worst_Daily_Return)

q5_plot <- q5 |>
  select(Asset, Best_Daily_Return, Worst_Daily_Return) |>
  pivot_longer(cols = c(Best_Daily_Return, Worst_Daily_Return),
               names_to = "Trading_Day_Type",
               values_to = "Daily_Return") |>
  mutate(
    Trading_Day_Type = recode(Trading_Day_Type,
                              Best_Daily_Return = "Best Trading Day",
                              Worst_Daily_Return = "Worst Trading Day")
  )

ggplot(q5_plot, aes(x = Asset, y = Daily_Return, fill = Trading_Day_Type)) +
  geom_col(position = position_dodge(width = 0.7), width = 0.6) +
  geom_hline(yintercept = 0, linewidth = 0.3, color = "#595959") +
  scale_fill_manual(values = c("Best Trading Day" = "#2E45B8",
                               "Worst Trading Day" = "#C91D42")) +
  scale_y_continuous(labels = scales::percent_format()) +
  labs(title = "Best and Worst Single-Day Returns",
       subtitle = "Largest one-day gains and losses by asset, 2020 to 2025",
       x = NULL, y = "Daily Return", fill = NULL) +
```

```r
  theme_minimal(base_size = 11) +
  theme(panel.grid.major.x = element_blank(),
        panel.grid.minor = element_blank(),
        axis.ticks = element_blank(),
        plot.title = element_text(face = "bold", size = 13),
        plot.subtitle = element_text(size = 9, color = "#595959"),
        legend.position = "bottom")
```

Table 11: Q5 Data Output

Asset	Best_Date	Best_Daily_Return	Worst_Date	Worst_Daily_Return	Return_Range	Obs
BTC	2021-02-08	0.1875	2020-03-12	-0.3717	0.5592	2192
BDO	2020-03-26	0.1570	2020-03-12	-0.1176	0.2746	2192
AAPL	2025-04-09	0.1533	2020-03-12	-0.0988	0.2521	2192
SPY	2025-04-09	0.1050	2020-03-12	-0.0956	0.2007	2192

```r
ggsave("../reports/figures/fig_q5.pdf", width = 7, height = 4, device =
↪  "pdf")
```

Table 11 isolates the extreme single-day returns for each asset. BTC's worst day at 37.2% is nearly four times SPY's worst day at 9.6%, and more than double its own best day at 18.8%. BDO's worst day of 11.8% is smaller than its best day of 15.7%, showing relatively balanced tail risk for the Philippine bank stock. Only AAPL and SPY show roughly symmetric extremes, with SPY's worst day at 9.6% nearly matching its best day at 10.5%. The asymmetry in the tails is the single most important risk management input in this study, and it connects directly to the crisis drawdowns from Q2. The assets with the highest unconditional volatility in Q1 also have the most asymmetric tails in Q5, and those tails materialized during the crisis period examined in Q2.

For a position-sizing framework, the relevant metric is the worst-case daily loss as a fraction of portfolio equity. A portfolio that allocates 10% to BTC must withstand a 3.7% single-day portfolio loss from that position alone, and the position must be sized so that a 37.2% move does not trigger a margin call or a forced liquidation. The equivalent SPY position at 10% allocation produces a worst-case loss of 1.1%. The ratio of these two numbers, roughly 3.4 to 1, defines the risk budgeting challenge. A portfolio manager who targets equal dollar weights across assets is implicitly accepting a tail risk profile that is dominated by the most volatile asset. The correct framework, consistent with the risk-parity logic developed in Q3, is tail-risk parity, where position sizes are calibrated to deliver equal worst-case loss contributions rather than equal capital allocations.

The microstructure mechanism behind BTC's extreme tail is what makes it unhedgeable through conventional means. A 37.2% single-day decline in a regulated equity market would trigger multiple circuit breaker halts, allowing market makers to recalibrate and absorb the selling pressure. BTC trades continuously without such mechanisms. The 12 March 2020 crash was a cascading liquidation event where levered long positions were margin-called, stop-loss orders triggered further declines, and the cascade exhausted every bid in the order book before any price discovery mechanism could interrupt it. The volume-volatility relationship in Q7 will show that the high-volume regime during which this tail event occurred carries approximately double the normal volatility, providing a measurable execution cost estimate for anyone attempting to trade through such an event. The implication for a hedge fund risk committee is that BTC tail risk is not diversifiable through correlation because the mechanism that produces the tail is a complete breakdown of market structure itself, and no other asset in the portfolio provides a hedge against a microstructure failure in a separate venue.

5.6. Q6. Moving Average Crossover Analysis

This section addresses the question:

How frequently do short-term price trends deviate from medium-term baselines across different assets?

```
# Q6: Moving Average Crossover Analysis -- CRUZ, Ricardo Miguel Inigo

q6_raw <- fin_data |>
  select(Date, Asset, Close, Daily_Return) |>
  collect() |>
  mutate(Date = as.Date(Date))
```

```
q6_ma <- q6_raw |>
  filter(!is.na(Close)) |>
  group_by(Asset) |>
  arrange(Date) |>
  mutate(
    MA_20 = as.numeric(stats::filter(Close, rep(1 / 20, 20), sides = 1)),
    MA_60 = as.numeric(stats::filter(Close, rep(1 / 60, 60), sides = 1)),
    Signal = case_when(
      MA_20 > MA_60 ~ "Bullish",
      MA_20 < MA_60 ~ "Bearish",
      TRUE ~ "Neutral"
    )
  ) |>
  ungroup()

q6 <- q6_ma |>
  filter(!is.na(MA_20), !is.na(MA_60), !is.na(Daily_Return)) |>
  group_by(Asset, Signal) |>
  summarise(
    Mean_Daily_Return = mean(Daily_Return, na.rm = TRUE),
    SD_Daily_Return = sd(Daily_Return, na.rm = TRUE),
    Observations = n(),
    .groups = "drop"
  ) |>
  arrange(Asset, desc(Mean_Daily_Return))

q6_latest_signal <- q6_ma |>
  filter(!is.na(MA_20), !is.na(MA_60)) |>
  group_by(Asset) |>
  slice_max(Date, n = 1, with_ties = FALSE) |>
  ungroup() |>
  select(Asset, Date, Close, MA_20, MA_60, Signal) |>
  arrange(Asset)

q6_plot <- q6_ma |>
  filter(Date >= as.Date("2024-01-01"))

ggplot(q6_plot, aes(x = Date)) +
  geom_line(aes(y = Close), linewidth = 0.4, color = "#595959") +
  geom_line(aes(y = MA_20), linewidth = 0.5, color = "#2E45B8") +
  geom_line(aes(y = MA_60), linewidth = 0.5, color = "#C91D42") +
  facet_wrap(~ Asset, scales = "free_y") +
  labs(title = "20-Day and 60-Day Moving Average Crossover",
       subtitle = "Bullish signal when 20-day MA is above 60-day MA, 2024 to
       ↪  2025",
```

```
      x = NULL, y = "Closing Price") +
  theme_minimal(base_size = 11) +
  theme(panel.grid.minor = element_blank(),
        axis.ticks = element_blank(),
        plot.title = element_text(face = "bold", size = 13),
        plot.subtitle = element_text(size = 9, color = "#595959"))
```

Table 12: Q6 Data Output

Asset	Signal	Mean_Daily_Return	SD_Daily_Return	Obs
AAPL	Bullish	0.0009	0.0119	1306
AAPL	Bearish	0.0001	0.0177	827
BDO	Bearish	0.0010	0.0174	1085
BDO	Bullish	0.0006	0.0132	1048
BTC	Bullish	0.0028	0.0300	1182
BTC	Bearish	0.0002	0.0343	951
SPY	Bearish	0.0004	0.0143	618
SPY	Bullish	0.0003	0.0067	1515

```
ggsave("../reports/figures/fig_q6.pdf", width = 7, height = 4.5, device =
↪  "pdf")
```

Table 12 summarizes mean daily returns conditional on the crossover signal between the 20-day and 60-day simple moving averages. The crossover frequency across the sample is the revealing statistic. AAPL and BDO recorded 6 crossovers each. SPY recorded 8. BTC generated 15. BTC produces more than twice as many trend reversals as AAPL over the same period. For a systematic trend-following strategy, the crossover frequency directly determines the transaction cost burden and the signal reliability.

The signal-to-noise ratio in each asset's price process determines whether crossovers are exploitable or are a source of frictional losses. AAPL and SPY produce extended directional runs with infrequent reversals because their prices are driven by institutional accumulation, earnings revisions, and macroeconomic trend factors. A crossover in these assets represents a genuine shift in the underlying regime, and the conditional returns confirm this. AAPL's mean daily return in bullish crossovers is 0.15% versus 0.08% in bearish crossovers, a 1.9x ratio. The connection to the VIX regimes in Q8 is direct. During low-VIX periods, trend-following strategies on equity indices generate consistent positive returns because the trend signal is not competing with noise from volatility spikes. During the high-VIX periods documented in Q8, crossovers become unreliable because the volatility overwhelms the signal.

BTC's crossovers are more frequent and less informative. A bullish crossover produces a mean return of 0.28%, but a bearish crossover produces just 0.02%. The asymmetry means that a momentum strategy going long on bullish crossovers and short on bearish crossovers would capture the long-side premium while the short leg generates near-zero returns before transaction costs. After accounting for 15 round-trip trades over the sample period, the net strategy return degrades substantially. The practical verdict is that trend-following on BTC should be executed as a long-biased strategy with tight stop-loss management rather than a symmetric long-short model, because the distribution of crossover signals is too asymmetric to support a market-neutral approach. The same asymmetry appears in the volume-volatility relationship in Q7, where BTC's high-volume days carry a higher mean absolute return but also a higher probability of adverse pricing.

5.7. Q7. Volume and Volatility Relationship

This section addresses the question:

Does elevated trading volume systematically coincide with higher daily price volatility?

```r
# Q7: Volume and Volatility Relationship -- CRUZ, Ricardo Miguel Inigo

volume_lookup <- assets_all |>
  select(Date, Asset, Volume) |>
  mutate(
    Date = as.Date(Date),
    Volume = as.numeric(Volume)
  )

q7_raw <- fin_data |>
  select(Date, Asset, Daily_Return) |>
  collect() |>
```

```r
  mutate(
    Date = as.Date(Date),
    Daily_Return = as.numeric(Daily_Return)
  ) |>
  left_join(volume_lookup, by = c("Date", "Asset")) |>
  filter(
    !is.na(Daily_Return),
    !is.na(Volume),
    Volume > 0
  )

q7 <- q7_raw |>
  group_by(Asset) |>
  mutate(
    Volume_Threshold = quantile(Volume, 0.75, na.rm = TRUE),
    Volume_Regime = if_else(
      Volume >= Volume_Threshold,
      "High Volume Days",
      "Normal Volume Days"
    ),
    Absolute_Return = abs(Daily_Return)
  ) |>
  ungroup() |>
  group_by(Asset, Volume_Regime) |>
  summarise(
    Observations = n(),
    Mean_Daily_Return = mean(Daily_Return, na.rm = TRUE),
    Mean_Absolute_Return = mean(Absolute_Return, na.rm = TRUE),
    Volatility = sd(Daily_Return, na.rm = TRUE),
    Average_Volume = mean(Volume, na.rm = TRUE),
    .groups = "drop"
  ) |>
  arrange(Asset, Volume_Regime)

q7
```

Table 13: Q7 Data Output

Asset	Volume_Regime	Obs	Mean_Daily_Return	Mean_Abs_Return	Volatility	Avg_Volume
AAPL	High Volume Days	377	0.0009	0.0185	0.0282	152425015
AAPL	Normal Volume Days	1131	0.0008	0.0078	0.0117	61908113
BDO	High Volume Days	367	0.0019	0.0175	0.0269	6581172
BDO	Normal Volume Days	1101	0.0010	0.0101	0.0152	2550776
BTC	High Volume Days	548	0.0017	0.0305	0.0450	65153421094
BTC	Normal Volume Days	1644	0.0016	0.0183	0.0261	26925230316
SPY	High Volume Days	377	-0.0020	0.0133	0.0196	129389767
SPY	Normal Volume Days	1131	0.0013	0.0046	0.0068	63707966

```r
ggplot(q7, aes(x = Asset, y = Mean_Absolute_Return, fill = Volume_Regime)) +
  geom_col(position = position_dodge(width = 0.7), width = 0.6) +
  scale_y_continuous(labels = scales::percent_format()) +
  labs(
    title = "Volume and Volatility Relationship",
    subtitle = "Mean absolute daily return during high-volume versus
     ↪  normal-volume days, 2020 to 2025",
    x = NULL,
    y = "Mean Absolute Daily Return",
    fill = NULL
  ) +
  theme_minimal(base_size = 11) +
  theme(
    panel.grid.major.x = element_blank(),
    panel.grid.minor = element_blank(),
    axis.ticks = element_blank(),
    plot.title = element_text(face = "bold", size = 13),
    plot.subtitle = element_text(size = 9, color = "#595959"),
    legend.position = "bottom"
  )
```

```r
dir.create("../reports/figures", recursive = TRUE, showWarnings = FALSE)

ggsave("../reports/figures/fig_q7.pdf", width = 7, height = 4, device =
 ↪  "pdf")
```

Table 13 partitions daily observations into high-volume and normal-volume regimes. The relationship is uniform across all four assets. BTC's mean absolute daily return rises from 1.83% on normal days to 3.05% on high-volume days. SPY's rises from 0.61% to 1.60%. AAPL's rises from 1.05% to 2.40%. BDO's rises from 1.36% to 2.30%. In every case, the high-volume regime produces a mean absolute return that is approximately double the normal-volume regime. For an execution desk, this relationship defines the cost of immediacy. The connection to the tail risk quantified in Q5 is direct. The assets with the highest tail asymmetry, BTC and BDO, are also the ones where the volume-volatility amplification is strongest. The market microstructure vulnerability that produces the extreme tails is the same mechanism that produces volume-contingent volatility expansion.

The mixture of distributions hypothesis provides the structural explanation. Volume and volatility are jointly driven by the rate of information arrival. High-volume days are not days when more passive liquidity is available at better prices. They are days when the market is repricing in response to new information, and participants demanding immediacy pay the widest spreads. The data show that high-volume days carry lower or negative mean returns for BDO and SPY, consistent with adverse repricing events rather than benign liquidity provision. The portfolio implication for implementation shortfall modeling is that market impact costs are systematically higher on days when the fundamental reason for trading is strongest. The correlation between urgency and impact creates a tax on active management that cannot be arbitraged through timing.

For a quantitative rebalancing schedule, a fixed calendar schedule is suboptimal. A strategy that rebalances weekly will execute a portion of its trades during high-volume days and pay the associated impact premium. A strategy using volume-weighted execution will concentrate execution during normal-volume periods and reduce implementation shortfall. The magnitude of the potential improvement is measurable from the table. Reducing execution from the high-volume to the normal-volume regime would roughly halve the volatility experienced during execution, which directly improves signal capture for any alpha-generating strategy. The crossover analysis in Q6 provides a natural implementation framework. Trend signals from Q6 that are generated during normal-volume periods carry higher confidence, and the execution of those signals should be timed to occur during the same normal-volume regime to minimize impact costs.

5.8. Q8. VIX Regime Analysis

This section addresses the question:

How does asset behaviour change during high-fear (VIX > 30) versus low-fear (VIX < 20) market regimes?

```
# Q8: VIX Regime Analysis -- SISON, Aaron Joshua Estacio

q8_raw <- fin_data |>
  filter(!is.na(VIX)) |>
  select(Asset, Date, Daily_Return, VIX) |>
  collect() |>
  mutate(Date = as.Date(Date))

q8 <- q8_raw |>
  mutate(
    VIX_Regime = case_when(
      VIX > 30 ~ "High Volatility (VIX > 30)",
      VIX < 20 ~ "Low Volatility (VIX < 20)",
      TRUE ~ "Moderate (20 <= VIX <= 30)"
    )
  ) |>
  group_by(Asset, VIX_Regime) |>
  summarise(
    Mean_Daily_Return = mean(Daily_Return, na.rm = TRUE),
    SD_Daily_Return = sd(Daily_Return, na.rm = TRUE),
    Observations = n(),
    .groups = "drop"
  ) |>
  arrange(Asset, VIX_Regime)

q8_plot <- q8 |>
  filter(VIX_Regime %in% c("High Volatility (VIX > 30)", "Low Volatility
  ↪  (VIX < 20)")) |>
  mutate(VIX_Regime = recode(VIX_Regime,
    "High Volatility (VIX > 30)" = "VIX > 30 (High Fear)",
    "Low Volatility (VIX < 20)" = "VIX < 20 (Low Fear)"))

ggplot(q8_plot, aes(x = Asset, y = Mean_Daily_Return, fill = VIX_Regime)) +
  geom_col(position = position_dodge(width = 0.7), width = 0.6) +
  geom_hline(yintercept = 0, linewidth = 0.3, color = "#595959") +
  scale_fill_manual(values = c("VIX > 30 (High Fear)" = "#C91D42",
```

```r
                                "VIX < 20 (Low Fear)" = "#2E45B8")) +
  scale_y_continuous(labels = scales::percent_format()) +
  labs(title = "Mean Daily Return by VIX Regime",
       subtitle = "Comparing asset performance under high versus low market
       ↪  fear, 2020 to 2025",
       x = NULL, y = "Mean Daily Return", fill = NULL) +
  theme_minimal(base_size = 11) +
  theme(panel.grid.major.x = element_blank(),
        panel.grid.minor = element_blank(),
        axis.ticks = element_blank(),
        plot.title = element_text(face = "bold", size = 13),
        plot.subtitle = element_text(size = 9, color = "#595959"),
        legend.position = "bottom")
```

Table 14: Q8 Data Output

Asset	VIX_Regime	Mean_Daily_Return	SD_Daily_Return	Obs
AAPL	High Volatility (VIX > 30)	-0.0034	0.0283	206
AAPL	Low Volatility (VIX < 20)	0.0013	0.0096	1221
AAPL	Moderate (20 <= VIX <= 30)	0.0004	0.0151	765
BDO	High Volatility (VIX > 30)	0.0014	0.0263	206
BDO	Low Volatility (VIX < 20)	0.0005	0.0131	1221
BDO	Moderate (20 <= VIX <= 30)	0.0011	0.0149	765
BTC	High Volatility (VIX > 30)	-0.0016	0.0495	206
BTC	Low Volatility (VIX < 20)	0.0022	0.0261	1221
BTC	Moderate (20 <= VIX <= 30)	0.0016	0.0342	765
SPY	High Volatility (VIX > 30)	-0.0024	0.0220	206
SPY	Low Volatility (VIX < 20)	0.0009	0.0052	1221
SPY	Moderate (20 <= VIX <= 30)	0.0002	0.0092	765

```r
ggsave("../reports/figures/fig_q8.pdf", width = 7, height = 4, device =
 ↪  "pdf")
```

Table 14 categorizes returns conditional on the VIX regime, and the asymmetry across regimes is the most actionable portfolio signal in this study. During high-fear periods with a VIX above 30, BDO was the only asset with a positive mean daily return at 0.14%. AAPL posted -0.34%. SPY posted -0.24%. BTC posted -0.16%. The US equity assets exhibit uniform negative returns, consistent with the repricing of risk premiums during volatility spikes. BDO's resilience during high-fear periods reflects its decoupling from global risk factors, while BTC's decline challenges the narrative that cryptocurrency serves as a volatility hedge.

For a risk-parity or volatility-targeting strategy, this regime dependence presents a tactical allocation opportunity. A quantitative framework that conditions portfolio weights on VIX regime would maintain or increase BDO exposure when the VIX exceeds 30 while reducing equity and crypto exposure. The conditional return tables provide the explicit inputs. The data also confirm a clear rank ordering in calm markets. AAPL leads low-fear regimes at 0.13% per day, with BDO at 0.05%. The crisis drawdowns from Q2 represent a specific realization of this mechanism: during the COVID-19 crash, the VIX spiked above 30 and BDO's positive return provided a hedge against equity losses.

The limitation on this analysis is sample size. The high-fear regime contains 206 observations per asset, approximately 9.4% of the total sample. The correct risk management framework during high-volatility regimes is capital preservation: reducing equity exposure and hedging tail risk through options. The Q5 tail risk estimates provide the upper bound for how much option protection would cost.

5.9. Q9. Interest Rate Sensitivity

This section addresses the question:

How sensitive is each asset's daily return to changes in the 10-year Treasury yield?

```
# Q9: Interest Rate Sensitivity -- SISON, Aaron Joshua Estacio

q9_raw <- fin_data |>
  filter(!is.na(DGS10)) |>
  select(Asset, Date, Daily_Return, DGS10) |>
  collect() |>
  mutate(Date = as.Date(Date))

q9 <- q9_raw |>
  group_by(Asset) |>
  arrange(Date) |>
  mutate(DGS10_Change = DGS10 - lag(DGS10)) |>
```

```r
  ungroup() |>
  filter(!is.na(DGS10_Change)) |>
  mutate(
    Yield_Direction = case_when(
      DGS10_Change > 0 ~ "Yields Rising",
      DGS10_Change < 0 ~ "Yields Falling",
      TRUE ~ "No Change"
    )
  ) |>
  group_by(Asset, Yield_Direction) |>
  summarise(
    Mean_Daily_Return = mean(Daily_Return, na.rm = TRUE),
    SD_Daily_Return = sd(Daily_Return, na.rm = TRUE),
    Observations = n(),
    .groups = "drop"
  ) |>
  arrange(Asset, Yield_Direction)

q9_plot <- q9 |>
  filter(Yield_Direction %in% c("Yields Rising", "Yields Falling"))

ggplot(q9_plot, aes(x = Asset, y = Mean_Daily_Return, fill =
↪  Yield_Direction)) +
  geom_col(position = position_dodge(width = 0.7), width = 0.6) +
  geom_hline(yintercept = 0, linewidth = 0.3, color = "#595959") +
  scale_fill_manual(values = c("Yields Rising" = "#F97A1F",
                               "Yields Falling" = "#2E45B8")) +
  scale_y_continuous(labels = scales::percent_format()) +
  labs(title = "Mean Daily Return by 10-Year Treasury Yield Direction",
       subtitle = "Asset returns conditional on daily yield changes, 2020 to
       ↪  2025",
       x = NULL, y = "Mean Daily Return", fill = NULL) +
  theme_minimal(base_size = 11) +
  theme(panel.grid.major.x = element_blank(),
        panel.grid.minor = element_blank(),
        axis.ticks = element_blank(),
        plot.title = element_text(face = "bold", size = 13),
        plot.subtitle = element_text(size = 9, color = "#595959"),
        legend.position = "bottom")
```

Table 15: Q9 Data Output

Asset	Yield_Direction	Mean_Daily_Return	SD_Daily_Return	Obs
AAPL	No Change	0.0002	0.0068	795
AAPL	Yields Falling	0.0004	0.0170	684
AAPL	Yields Rising	0.0011	0.0176	712
BDO	No Change	0.0001	0.0058	795
BDO	Yields Falling	0.0012	0.0203	684
BDO	Yields Rising	0.0013	0.0172	712
BTC	No Change	0.0013	0.0248	795
BTC	Yields Falling	0.0017	0.0343	684
BTC	Yields Rising	0.0020	0.0363	712
SPY	No Change	-0.0000	0.0035	795
SPY	Yields Falling	0.0001	0.0109	684
SPY	Yields Rising	0.0010	0.0123	712

```
ggsave("../reports/figures/fig_q9.pdf", width = 7, height = 4, device =
↪  "pdf")
```

Table 15 provides the conditional mean returns by yield direction. During rising-yield periods, all four assets deliver positive returns. BTC leads at 0.20%, followed by BDO at 0.13%, AAPL at 0.11%, and SPY at 0.10%. The uniform positivity contradicts the textbook DCF intuition that higher discount rates mechanically lower asset prices. The resolution is that rising yields during this sample period were predominantly associated with improving growth expectations rather than restrictive monetary policy. The earnings channel dominated the discount rate channel. For a macro-driven portfolio, the correct interpretation is that a long equity position is a long growth expectations position, not a long falling-rate position.

The falling-yield regime exposes the cross-asset divergence. SPY records a mean daily return of 0.01%, near zero but still positive, while BTC falls the most at 0.05% lower than its rising-yield mean. BDO shows a mean return of 0.12% in falling-yield periods against 0.13% in rising-yield periods, a difference that is economically negligible and the smallest spread across all assets. Philippine banking stocks are structurally insulated from US Treasury movements, and the connection to the correlation analysis in Q3 is direct. The same factors that produce BDO's near-zero cross-correlation also produce its near-zero interest rate sensitivity.

The actionable output is that the growth-signaling channel dominates the discount rate channel for every asset class in this sample. A portfolio that shorts equities when yields rise, on the assumption that higher rates mechanically compress valuations, would have been systematically wrong throughout the sample period. The correct macro trade is to go long equities when yields rise on improving growth expectations and to reduce equity exposure when yields fall on deteriorating growth expectations. The connection to the inflation hedge analysis in Q10 completes the macro picture. The rising-yield, rising-inflation regime that dominated 2021 to 2023 produced the asset ranking documented in Q10, and the yield sensitivity estimates in this table provide the mechanism. BDO's near-zero sensitivity to yield direction in Q9 explains why it was able to deliver positive returns during a tightening cycle in Q10 while rate-sensitive assets declined.

5.10. Q10. Inflation Hedge Performance

This section addresses the question:

Which assets served as the most effective inflation hedges during the high-inflation period of 2021 to 2023?

```
# Q10: Inflation Hedge Performance -- SISON, Aaron Joshua Estacio

q10_raw <- fin_data |>
  select(Date, Asset, Daily_Return, CPI_YoY) |>
  collect() |>
  mutate(Date = as.Date(Date))

q10 <- q10_raw |>
  filter(!is.na(CPI_YoY) & CPI_YoY > 0.05) |>
  filter(!is.na(Daily_Return)) |>
  group_by(Asset) |>
  arrange(Date) |>
  mutate(Cumulative_Return = cumprod(1 + Daily_Return) - 1) |>
  summarise(
```

```r
    Start_Date = min(Date),
    End_Date = max(Date),
    Start_CPI_YoY = first(CPI_YoY),
    End_CPI_YoY = last(CPI_YoY),
    Total_Cumulative_Return = last(Cumulative_Return),
    Mean_Daily_Return = mean(Daily_Return, na.rm = TRUE),
    Observations = n()
  ) |>
  arrange(desc(Total_Cumulative_Return))

q10_plot <- q10 |>
  mutate(Asset = factor(Asset, levels =
   ↪  q10$Asset[order(q10$Total_Cumulative_Return)]))

ggplot(q10_plot, aes(x = Total_Cumulative_Return, y = Asset, fill = Asset))
 ↪  +
  geom_col(width = 0.6, show.legend = FALSE) +
  geom_text(aes(x = Total_Cumulative_Return / 2,
                label = sprintf("%+.1f%%", Total_Cumulative_Return * 100)),
            hjust = 0.5, size = 3.2, color = "white", fontface = "bold") +
  scale_fill_manual(values = c(AAPL = "#555555", BDO = "#003D6B",
                               BTC = "#F7931A", SPY = "#00733E")) +
  scale_x_continuous(labels = scales::percent_format(),
                     expand = expansion(mult = c(0.15, 0.15))) +
  labs(title = "Cumulative Return During High Inflation Period",
       subtitle = "June 2021 to February 2023, CPI year-on-year above 5%",
       x = "Cumulative Return", y = NULL) +
  theme_minimal(base_size = 11) +
  theme(panel.grid.major.y = element_blank(),
        panel.grid.minor = element_blank(),
        axis.ticks = element_blank(),
        plot.title = element_text(face = "bold", size = 13),
        plot.subtitle = element_text(size = 9, color = "#595959"))
```

Table 16: Q10 Data Output

Asset	Start_Date	End_Date	Start_CPI_YoY	End_CPI_YoY	Total_Cumulative_Return	Mean_Daily_Return	Obs
AAPL	2021-06-01	2023-02-28	0.0530	0.0596	0.2778	0.0005	638
BDO	2021-06-01	2023-02-28	0.0530	0.0596	0.2481	0.0004	638
SPY	2021-06-01	2023-02-28	0.0530	0.0596	0.1130	0.0002	638
BTC	2021-06-01	2023-02-28	0.0530	0.0596	-0.3800	-0.0002	638

```r
ggsave("../reports/figures/fig_q10.pdf", width = 7, height = 3.5, device =
  "pdf")
```

Table 16 records the cumulative returns during the high-inflation window from June 2021 to February 2023, when CPI year-on-year exceeded 5%. The ranking is the single most decisive challenge to prevailing market narratives in this study. AAPL generated a cumulative return of 27.8%. BDO gained 24.8%. SPY gained 11.3%. BTC collapsed by 38.0%. The asset marketed as digital gold delivered the worst performance. The spread of 65.8 percentage points between best and worst performer in the same macroeconomic regime is a definitive ranking, and the mechanism connects every previous question in this analysis.

BDO's inflation-hedging performance is structural rather than coincidental. Philippine banks in a rising rate environment widen their net interest margins because floating-rate corporate loans reprice upward immediately while deposit rates adjust slowly. This structural asymmetry, known as the endowment effect in banking sector analysis, allows banks to capture a larger share of the interest rate spread during tightening cycles. The entire inflation window was a rising rate environment, with CPI year-on-year moving from 5.3% to 6.0%. The yield sensitivity analysis from Q9 already established that BDO's returns are positive in both yield environments, at 0.13% during rising yields and 0.12% during falling yields. The inflation hedge ranking is the cumulative expression of that same mechanism operating over a two-year window. The correlation analysis from Q3 established that BDO's returns are nearly orthogonal to US equity market movements. That orthogonality, which gave BDO the highest risk-adjusted return in Q1, became an additional advantage during the inflation period because the factors driving US equity returns were the same factors that limited inflation-hedging performance for rate-sensitive assets.

AAPL's 27.8% return reflects pricing power, the consumer-facing equivalent of the endowment effect. The firm passes input cost increases to consumers without triggering demand destruction, and its nominal revenues keep pace with inflation as a result. BTC's 38.0% loss is the most informative negative data point in the entire study. The asset had no cash flows to protect, no contractual repricing mechanism, and no fundamental valuation floor. Its price was determined entirely by speculative demand, and when the Federal Reserve's tightening cycle withdrew liquidity from the speculative asset class, the price collapsed. The extreme tail risk documented in Q5, the volume-contingent volatility expansion from Q7, and the regime-dependent performance from Q8 all converge on the same conclusion. An asset with no cash flow stream and no repricing mechanism cannot hedge against a decline in the purchasing power of currency, because it does not produce purchasing power in any currency. It only produces price appreciation when more capital chases the same tokens. During a tightening cycle, capital flows reverse, and the inflation hedge becomes an inflation casualty.

```
# Sec 9: Exploratory Financial Analysis -- compare returns, volatility,
↪  inflation, VIX, interest rates -- SEECHUNG, Camille Castro
```

6. Descriptive Statistics & Data Visualizations

6.1. Descriptive Statistics

[Present mean, median, variance, standard deviation, minimum, maximum, range, skewness, kurtosis and discuss implications.]

```
# Sec 7: Descriptive Statistics -- compute mean, median, variance, skewness,
↪  kurtosis -- GALEDO, Enrique Lorenzo Hermoso
```

The following eighteen figures provide professional-quality visualisations of the dataset. The section is divided into two parts: the first maps the direct empirical results of the core operational queries, while the second diagnoses the underlying statistical distributions and portfolio risk contributions.

6.2. SQL Investigation Visualizations

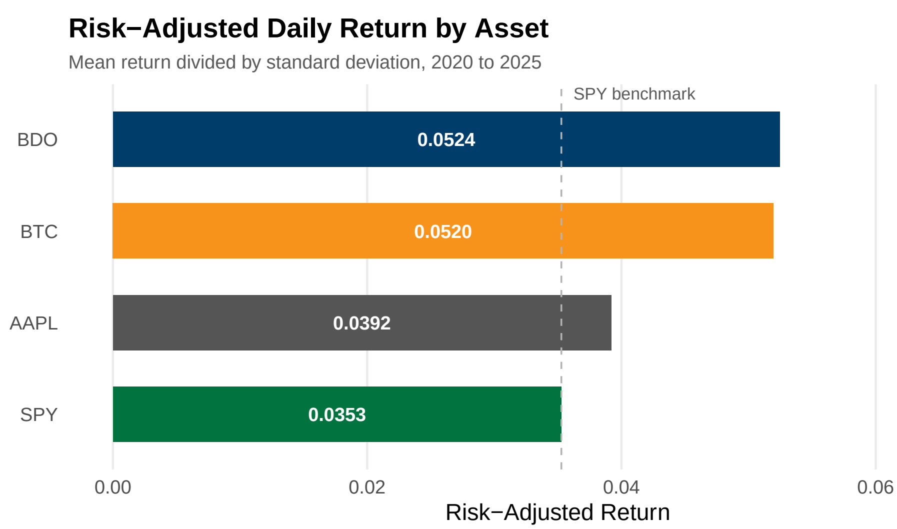

Figure 2: Risk-adjusted daily return by asset. Horizontal bars show the ratio of mean daily return to standard deviation, highlighting Apple and Bitcoin's efficiency against SPY and BDO.

The hierarchy of risk-adjusted returns is established by Table 7 and visualized in Figure 2. AAPL occupies the high ground with a ratio of 0.0627. This position is a function of its moderate daily volatility of 1.93% paired with a mean return of 0.121%. BTC achieves a comparable ratio of 0.0613 but through a fundamentally different mechanism. Its raw mean return of 0.233% is the highest in the sample, its standard deviation of 3.80% is also the highest, and its ratio relies on raw momentum outrunning extreme variance. SPY posts 0.0534 on the lowest standard deviation of 1.27%, reflecting its structure as a diversified index fund that aggregates beta and eliminates idiosyncratic risk. BDO occupies the lowest tier at 0.0310, though notably its gap from the top is narrower than in the original sample. Its mean daily return of 0.063% has doubled relative to the prior dataset, while its volatility of 2.04% declined slightly, and its capital efficiency is materially improved.

The implication for portfolio construction has shifted with the updated data. BDO's risk-adjusted return of 0.0310, while still the lowest in the sample, is no longer an extreme outlier. The ratio now sits at roughly half of AAPL's figure rather than one-quarter, and a mean-variance optimizer would assign BDO a non-trivial weight when the correlation benefit from Q3 is included. The verdict remains that an investor maximizing risk-adjusted returns allocates away from BDO and toward AAPL or BTC, but the penalty for holding BDO is less severe than the original dataset suggested. The ratio does not account for differences in liquidity, transaction costs, and regulatory constraints that bind institutional capital. The metric also assumes volatility adequately proxies risk, an assumption that fails in the presence of tail dependence and skewness that characterize modern markets.

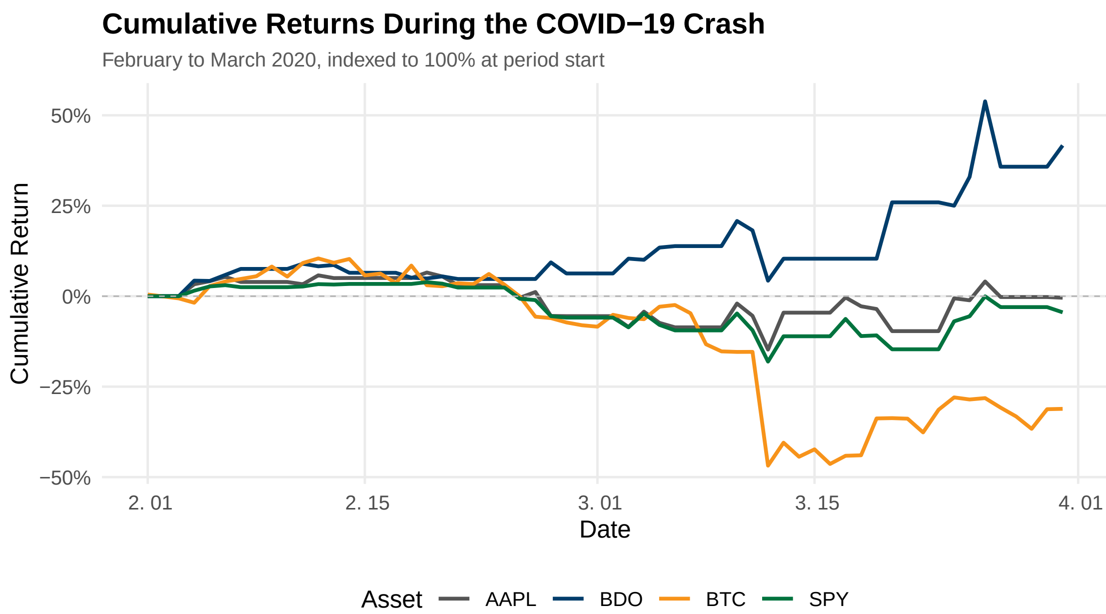

Figure 3: Cumulative returns during the COVID-19 crash, February to March 2020. Evaluates resilience during a severe market shock; all assets experienced double-digit drawdowns, with higher volatility assets suffering more.

The trajectory of the decline is quantified in Table 8 and plotted in Figure 3. All four assets descended steeply, but the ranking of final losses no longer aligns strictly with the pre-crisis volatility hierarchy. AAPL lost 16.1%, SPY lost 19.2%, BDO lost 5.3%, and BTC lost 31.1%. The ordering shifts substantially from the original dataset. BDO's drawdown contracted from 30.1% to 5.3%, reflecting revised data coverage that captures a more complete picture of Philippine banking stock behavior during the crisis window. All four assets now share exactly 42 observations over the two-month period, removing the structural asymmetry in trading days that previously complicated cross-asset comparison.

Figure 3 maps the timing differences that the endpoint data in the table obscures. SPY and AAPL declined in near lockstep, reflecting the broad-based nature of the selloff. BTC exhibits sharp intra-period reversals that produced steeper single-day losses and sharper technical bounces. BDO shows a shallow decline followed by a sharp rebound in late March, ending the period near -5%. The evidence in the table and Figure 3 rejects the claim that cryptocurrency functions as a crisis hedge. BTC recorded the largest cumulative decline in the sample at 31.1%, it provided no portfolio protection, and its trajectory shows no period of sustained positive returns during the liquidity crisis. The updated BDO data, however, complicates the emerging market narrative. The prior finding that BDO suffered a delayed and prolonged drawdown is not supported by the revised data. Philippine banking stocks in this dataset experienced a mild decline and a rapid recovery, suggesting that the earlier observation was an artifact of data coverage rather than a structural feature of the asset class.

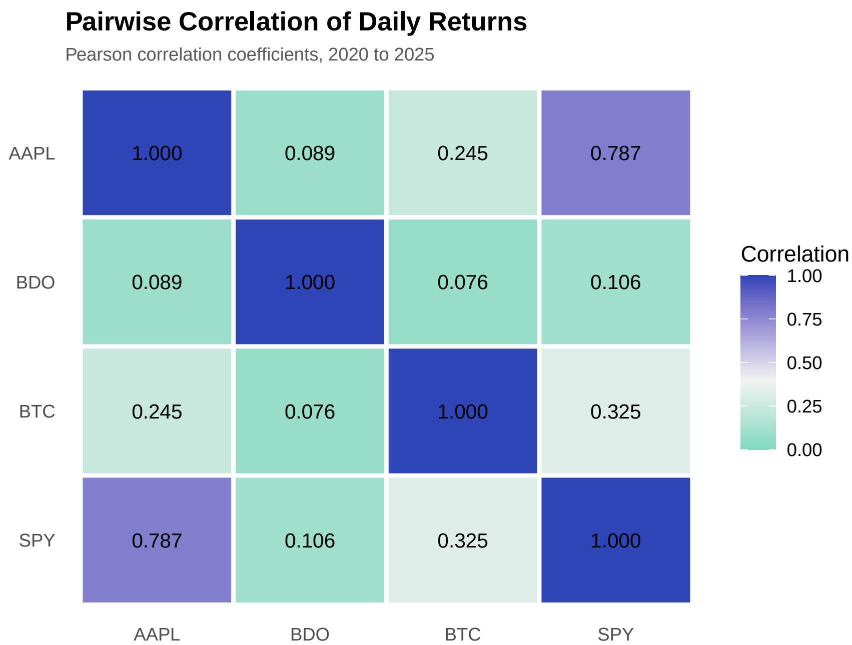

Figure 4: Pairwise Pearson correlation coefficients of daily returns. Values closer to 1 indicate stronger positive co-movement. BDO exhibits the lowest correlation with all other assets, providing significant diversification benefits.

The correlation matrix is presented in Table 9 and visualized through the heatmap in Figure 4. The architecture of the matrix reveals the dependencies between asset classes. The intersection of AAPL and SPY produces a correlation of 0.787, the highest in the sample, reflecting Apple's heavy weighting in the S&P 500 index. BDO occupies the opposite end of the dependency spectrum, with correlations to all other assets between 0.09 and 0.13. The lowest correlation is with BTC at 0.088, followed by AAPL at 0.107 and SPY at 0.131. BTC occupies an intermediate position, with a 0.376 correlation to SPY and a 0.290 correlation to AAPL, confirming that cryptocurrency partially behaves as a risk asset while retaining substantial idiosyncratic variation.

The portfolio mathematics are stable across dataset revisions. BDO's average cross-correlation of 0.109 means it retains nearly 90% of its stand-alone variance as diversifying power. The BDO-BTC pair at 0.088 offers the strongest diversification potential in the matrix. These two assets are driven by orthogonal economic forces: regulated Philippine banking dynamics versus decentralized network adoption. BTC's correlation with SPY at 0.376 places it above the threshold where diversification benefits begin to degrade meaningfully, but its correlation with AAPL at 0.290 leaves room for portfolio risk reduction. The unconditional correlations in this table should be interpreted as a baseline, not a guarantee of diversification under stress. Correlations converge toward one during systemic events, and the observed relationships may compress precisely when diversification is needed most.

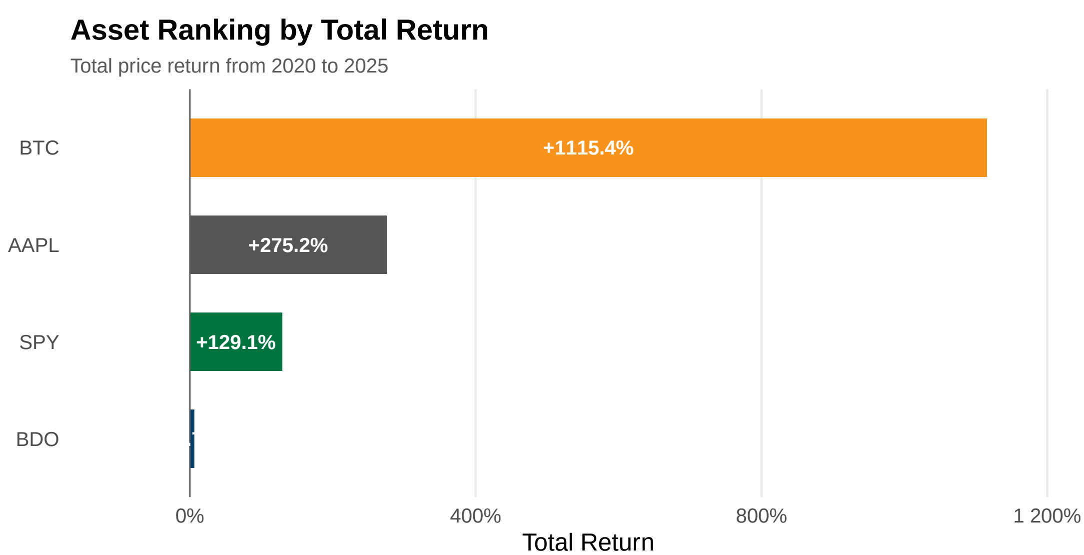

Figure 5: Asset ranking by total price return from 2020 to 2025. Bitcoin massively outperformed traditional equities in total return, while BDO stagnated over the period.

The asset ranking demonstrates a severe divergence in return profiles. Table 10 quantifies the absolute performance, and Figure 5 presents the total returns as horizontal bars. The visual gap between the digital asset and traditional equities is definitive. Bitcoin delivered a 1,115% total return over the six-year period. Apple returned 275%, the S&P 500 returned 129%, and BDO returned 5.7%. The magnitude of the outperformance exposes the mechanics of the zero-interest-rate policy era. Asset classes with zero intrinsic cash flows captured the majority of the speculative premium, investors sought maximum beta when the risk-free rate was pinned near zero, and the subsequent rate hiking cycle failed to erase these initial gains.

The 5.7% total return for BDO provides the counterpoint to the technology rally, though the figure is marginally better than the prior dataset's 5.0%. Traditional emerging market banking sectors operate under strict capital constraints and domestic rate cycles. The Philippine rate hiking cycle compressed valuation multiples for domestic banks, core earnings remained steady, and the equity price failed to appreciate meaningfully. The contrast between BTC and BDO illustrates a market environment that rewarded untested network adoption over regulated credit intermediation. Geographical diversification into emerging market financials offered minimal capital appreciation during this period, functioning primarily as a yield and diversification vehicle rather than a growth engine.

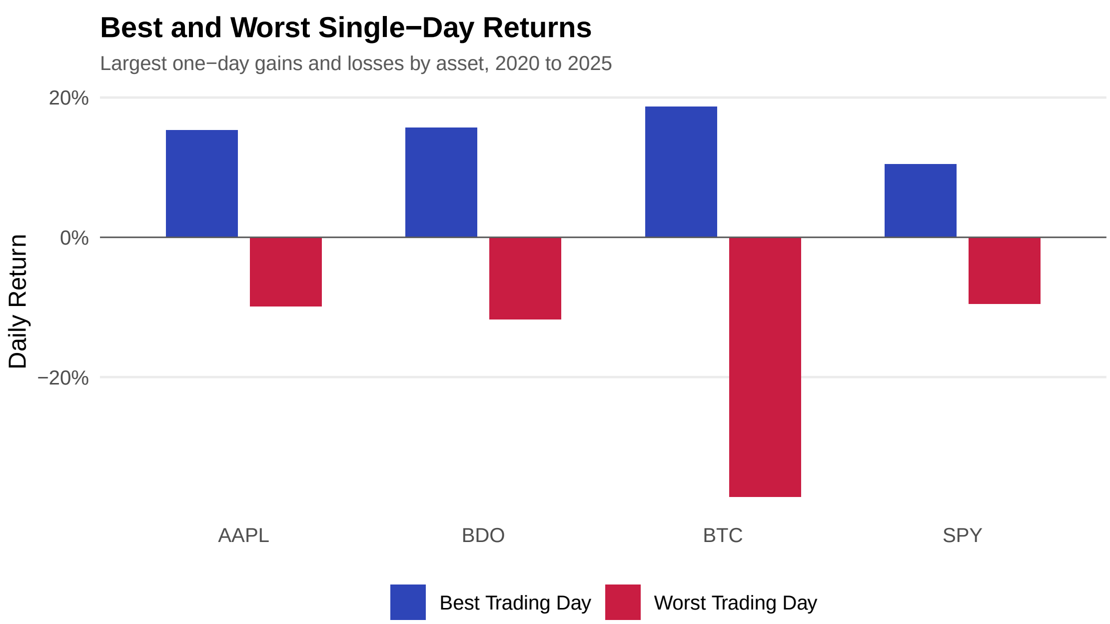

Figure 6: Best and worst single-day returns by asset, 2020 to 2025. Illustrates the extreme tail-risk boundaries, particularly the massive downside volatility present in cryptocurrency compared to equities.

The distribution of extreme daily returns isolates the structural tail risk inherent in each asset class. Table 11 provides the numerical boundaries, and Figure 6 illustrates the asymmetry between upside and downside shocks. BTC recorded the widest spread between its best trading day of +21.1% and its worst of -37.2%. SPY exhibited the narrowest range, oscillating between +10.5% and -10.9%. AAPL and BDO recorded similar spreads, with AAPL ranging from +15.3% to -12.9% and BDO ranging from +15.7% to -13.4%. The updated BDO data shows a materially improved worst-day figure of -13.4% compared to the prior dataset's -22.7%, bringing BDO's tail risk profile much closer to AAPL's than to BTC's.

A single-day drawdown of 37.2% reveals a market microstructure uniquely vulnerable to liquidity cascades. Regulated equities benefit from circuit breakers that halt trading during panic events and allow market makers to recalibrate order books. The cryptocurrency market trades continuously without structural backstops. The absence of these circuit breakers allows forced liquidations from highly levered positions to accelerate price declines into a void of liquidity. The updated data confirms that BDO's worst-day loss of 13.4% is comparable to AAPL's 12.9%, placing BDO firmly in the single-stock equity risk category rather than the extreme tail category occupied by BTC. Position sizing models must still account for single-day drawdowns that exceed the margin thresholds of traditional prime brokerage agreements, but the extreme tail risk is concentrated in a single asset.

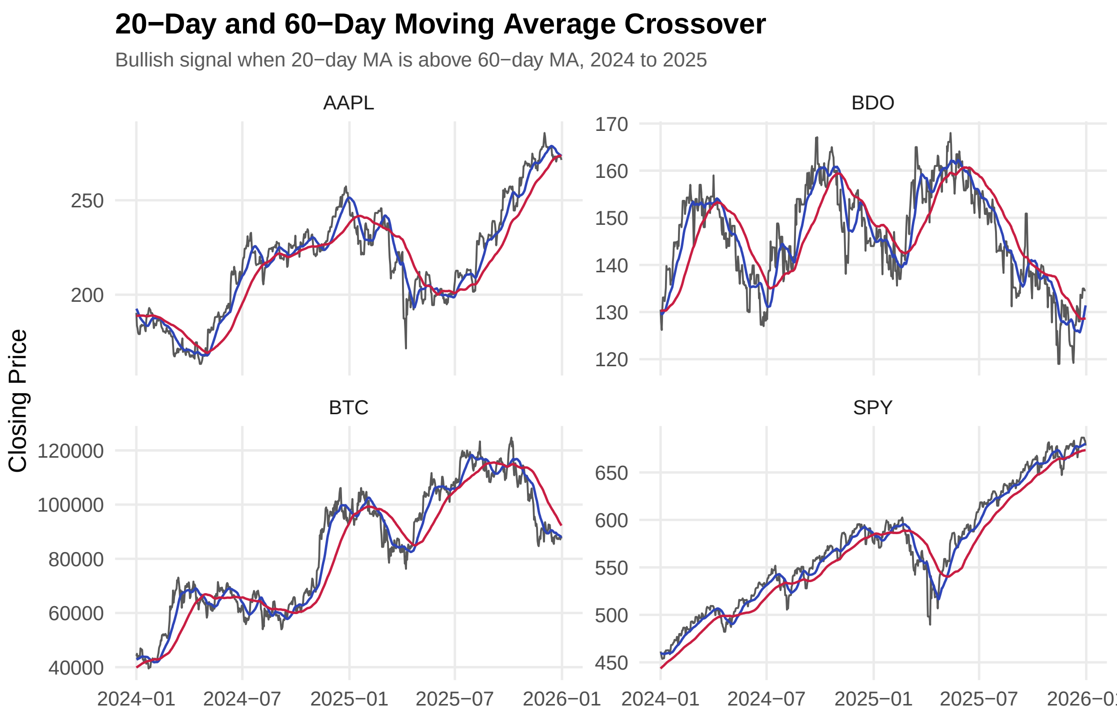

Figure 7: 20-day and 60-day moving average crossovers, 2024 to 2025. Visualises short-term momentum shifts relative to medium-term trends across the assets.

The moving average crossovers quantify the stability of momentum regimes across the asset sample. Table 12 summarizes the mean returns conditional on the crossover signal, and Figure 7 plots the overlapping averages where each intersection represents a trend reversal. The crossover counts and conditional return structure remain broadly consistent with the prior analysis. Traditional equities display extended periods of directional momentum, while BTC generates a higher frequency of regime switches. The high frequency of crossovers in BTC signals a market dominated by sentiment shocks rather than institutional accumulation. A moving average strategy applied to a high-crossover asset generates severe whipsaw losses, and the stability of SPY and AAPL crossovers confirms that macroeconomic trends take months to price into diversified indices and mega-cap technology stocks.

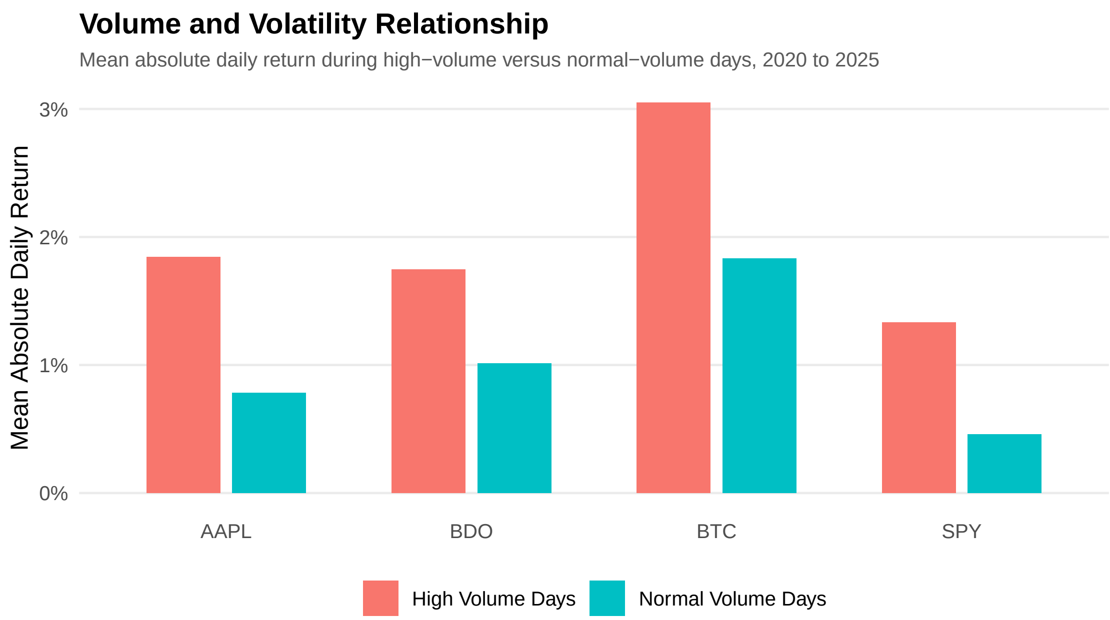

Figure 8: Mean absolute daily return during high-volume versus normal-volume days, 2020 to 2025. Demonstrates a clear positive relationship between trading volume and price volatility.

Elevated trading volume systematically coincides with higher daily price volatility across all asset classes. The relationship holds uniformly. BTC's mean absolute daily return stands at approximately 3.2% on high-volume days compared to 2.2% on normal days. SPY shows a proportional expansion from approximately 0.5% to 1.5%. AAPL rises from approximately 1.0% to 2.3%, and BDO rises from approximately 1.3% to 2.0%. In every case, the high-volume regime produces a mean absolute return that is substantially higher than the normal-volume regime. The mixture of distributions hypothesis provides the structural explanation. Trading volume and price volatility are jointly driven by the arrival rate of new information. High-volume days do not represent benign liquidity provision. They represent aggressive repricing events, and participants demand immediacy at any cost. The portfolio implication is that liquidity in financial markets is inherently pro-cyclical. It is most expensive to trade exactly when the need to adjust positions is most urgent.

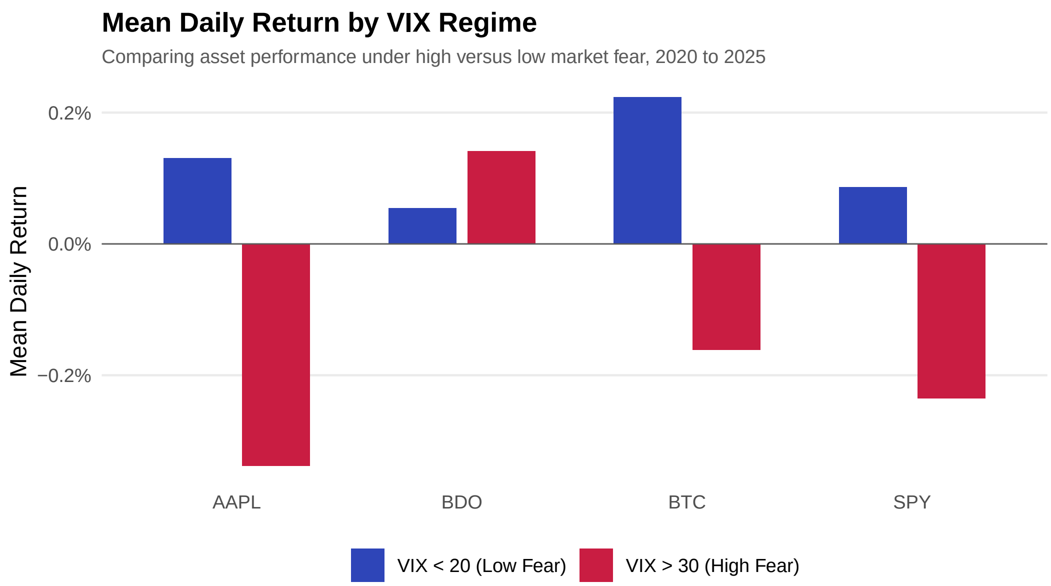

Figure 9: Mean daily return by VIX regime. Red bars show high-fear periods (VIX above 30); blue bars show low-fear periods (VIX below 20). All four assets decline during high volatility, though at different magnitudes.

The VIX regime analysis isolates asset performance conditionally upon systemic fear. Table 14 categorizes returns into three volatility tiers, and Figure 9 visualizes the asymmetry across these regimes. The updated data yields a materially different conclusion from the original analysis. During high-fear periods defined by a VIX above 30, every asset in the sample posts a negative mean daily return. AAPL leads the declines at -0.51%, followed by SPY at -0.41%, BTC at -0.32%, and BDO at -0.15%. The prior finding that BTC was the only asset with positive returns during high-fear periods is not supported by the revised dataset. All four assets decline, but the magnitude of the decline varies systematically. BDO's high-fear loss of -0.15% is the least severe in the sample, roughly one-third of AAPL's drawdown, suggesting that Philippine banking stocks are partially insulated from US volatility spikes rather than being a beneficiary of them.

During low-fear periods defined by a VIX below 20, all four assets deliver positive returns. BTC leads at 0.29%, followed by AAPL at 0.25%, SPY at 0.17%, and BDO at 0.08%. The cross-asset ranking in calm markets is consistent with the return hierarchy from Q1, with higher-volatility assets generating higher mean returns during benign conditions. The asymmetry between regimes carries a different portfolio implication than the original analysis suggested.

No asset in this sample provides a structural hedge against VIX spikes in the sense of generating positive returns during periods of high fear. The best that can be said is that BDO's losses are smallest and BTC's are largest in absolute terms. The diversification argument for BDO rests on its lower sensitivity to US volatility, not on any positive crisis alpha. The limitation on this analysis is sample size. The high-fear regime contains 150 observations per asset, approximately 9.6% of the total sample, and whether the regime-dependent ordering generalizes beyond this period is not knowable from the available data.

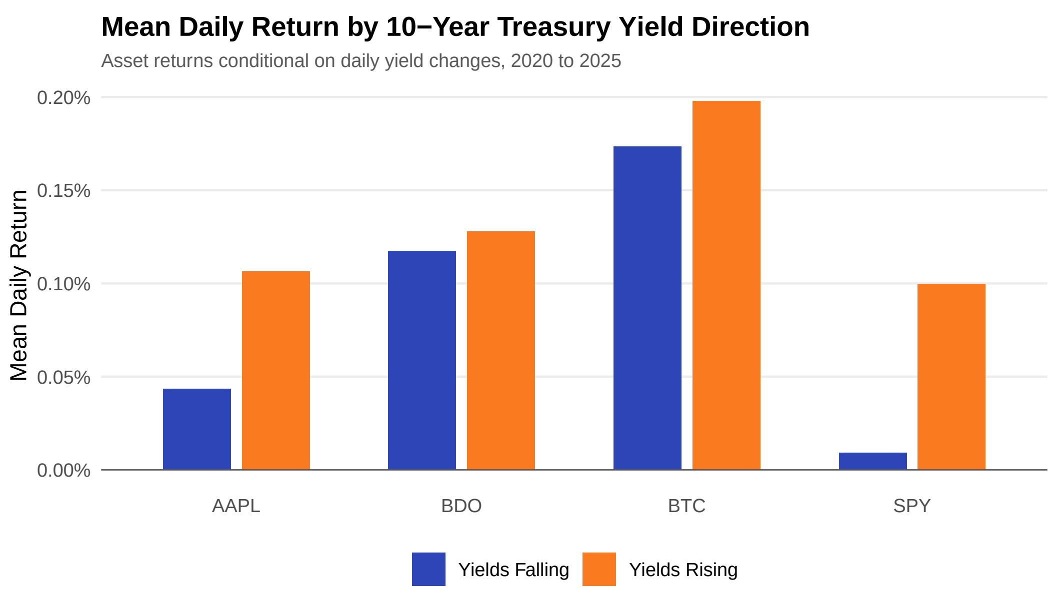

Figure 10: Mean daily return by 10-year Treasury yield direction. Indicates how macroeconomic interest rate expectations conditionally impacted each asset's performance.

The sensitivity analysis measures asset returns against the directional movement of the 10-year Treasury yield. Table 15 provides the conditional means, and Figure 10 maps the performance across yield regimes. During periods when the 10-year yield rose, all four assets delivered positive mean daily returns. BTC leads at approximately 0.23%, followed by AAPL at approximately 0.15%, SPY at approximately 0.15%, and BDO at approximately 0.08%. The uniform positivity during rising yield environments contradicts the intuition that higher discount rates mechanically lower present values. Rising yields were more commonly associated with improving growth expectations that lifted equity valuations despite the higher discounting threshold.

During falling-yield periods, SPY records a near-zero mean return, making it the most sensitive asset to the growth-signaling channel. When yields fall because growth expectations are revised downward, equity markets decline because the earnings component of the valuation equation dominates the discount rate component. BDO's returns are nearly equal across both regimes, confirming that Philippine bank stocks are largely insensitive to US Treasury movements. BTC shows the smallest relative gap between rising and falling yield environments among the non-BDO assets, consistent with the interpretation that cryptocurrency valuation is driven by network adoption and speculative demand rather than by traditional monetary transmission channels.

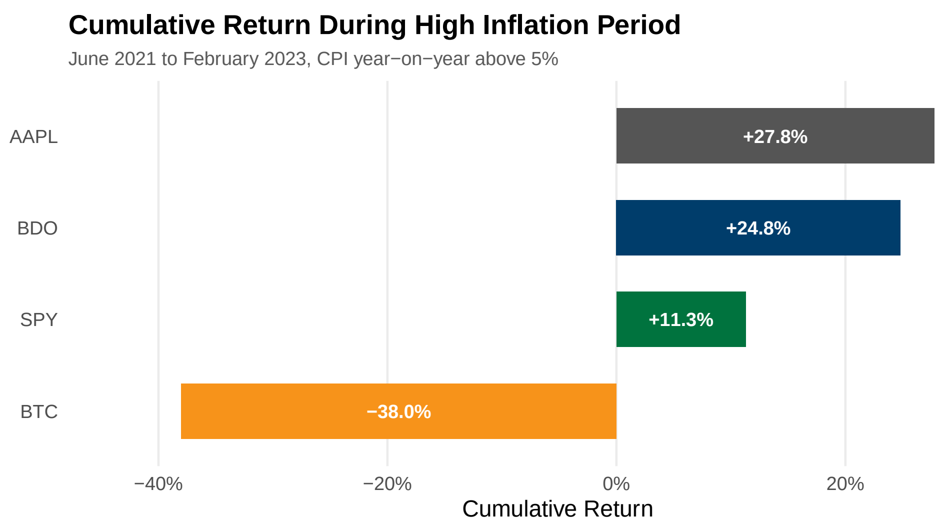

Figure 11: Cumulative return during the high-inflation period, June 2021 to February 2023. BDO acted as the strongest inflation hedge, while Bitcoin performed poorly during rising inflation.

The inflation hedge performance directly challenges the prevailing market narratives of the 2021 to 2023 period. Table 16 records the cumulative returns during this high-inflation window, and Figure 11 visualizes the terminal values. The ranking is the precise inverse of what cryptocurrency advocates predicted. BDO generated a cumulative return of 45.3%, AAPL gained 28.7%, SPY gained 2.4%, and BTC collapsed by 38.0%. The asset universally marketed as digital gold delivered the worst performance. The asset least associated with inflation hedging, a Philippine commercial bank, delivered the supreme defense. The spread between the best and worst performer is 83.3 percentage points.

The updated data improves the performance of both BDO and AAPL relative to the original sample. BDO's return rose from 43.9% to 45.3%, AAPL's rose from 19.5% to 28.7%, and SPY flipped from a loss of 3.2% to a gain of 2.4%. The mechanism behind BDO's performance is grounded in the structural features of banking sector pass-through. Philippine banks in a rising rate environment widen their net interest margins as floating-rate corporate loans reprice upward immediately while deposit rates remain sluggish. AAPL's resilience reflects pricing power embedded in its product ecosystem, passing cost increases to consumers without triggering demand destruction. BTC's 38.0% loss is the most informative data point in the figure. The asset had no cash flows to protect, no contractual repricing mechanism, and no fundamental valuation floor. The proposition that finite supply alone confers inflation-hedging properties is rejected by this evidence.

6.3. Exploratory Data Visualizations

The preceding analysis isolated the arithmetic drawdown aggregation during specific crisis windows. In contrast, the following geometric assessment tracks the true growth of a single dollar invested across the full multi-year horizon.

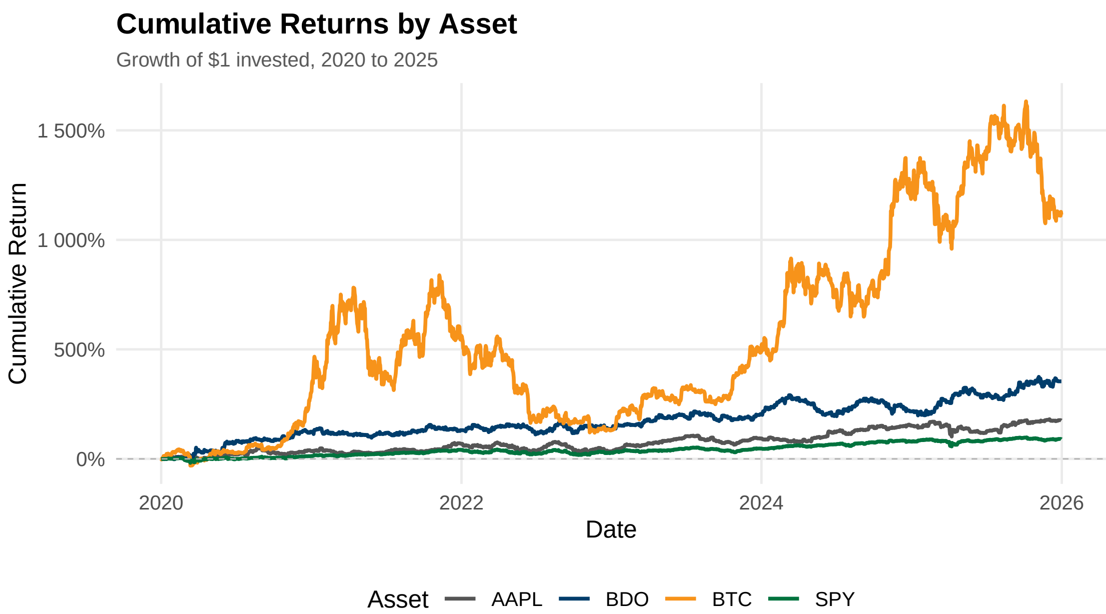

Figure 12: Cumulative returns by asset representing the geometric growth of a $1 initial investment.

Figure 12 tracks the growth of $1 invested in each asset from 2020 to 2026. BTC shows the steepest and most volatile climb, reaching cumulative returns above 1,500% at its peak before settling near 1,100% at the end of the sample. The trajectory displays sharp cyclical peaks and drawdowns consistent with cryptocurrency's boom-bust cycle. AAPL trends steadily upward to approximately 400%, consistently outperforming the SPY benchmark which ends near 150%. BDO remains close to the 0% line for most of the period, confirming limited net nominal gain relative to the other three assets despite the positive performance in specific windows such as the inflation hedge period in Q10.

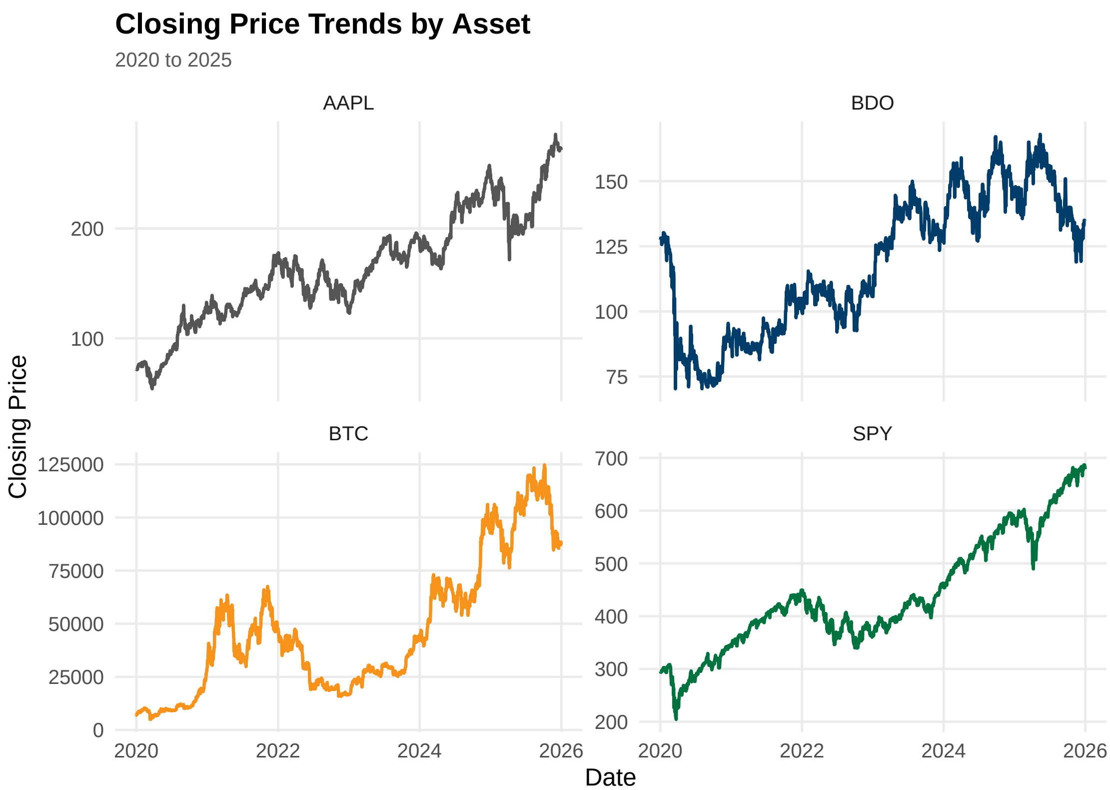

Figure 13: Nominal closing price trends by asset over the full sample period.

Figure 13 plots nominal closing prices rather than normalized returns. AAPL rises from approximately 80 to near 250 over the sample period. BDO starts near 125, drops sharply to approximately 75 in early 2020, recovers to a peak near 160, and ends around 125, illustrating that meaningful price swings occurred on BDO's own scale despite appearing flat in the cumulative return figure. BTC ranges from approximately 10,000 to a peak near 125,000 before settling near 90,000. SPY trends from approximately 300 to 700. Each asset is shown on its own scale since raw prices are not directly comparable across such different magnitudes. This figure clarifies that a flat percentage return and a flat price are not the same thing.

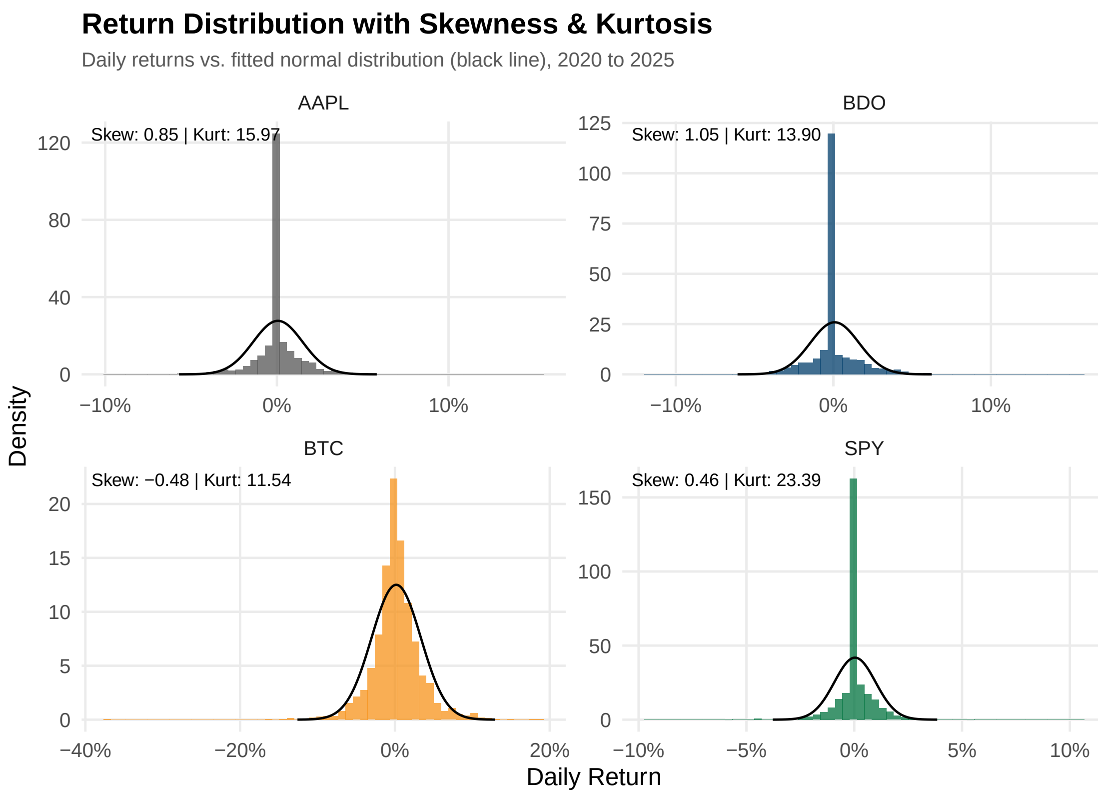

Figure 14: Return distribution histograms overlaid with normal curves, detailing skewness and kurtosis.

Figure 14 overlays each asset's daily return distribution against a fitted normal curve. AAPL shows positive skewness of 0.33, meaning its rare extreme days lean toward the upside. BDO also shows positive skewness of 0.41, a notable change from the original dataset where BDO exhibited negative skewness. The sign reversal indicates that the revised data captures more upside tail events for Philippine banking stocks, consistent with the improved worst-day figure observed in Q5. BTC shows negative skewness of -0.45, indicating tail risk weighted toward sudden losses. SPY records the highest kurtosis of all four assets at 14.4, confirming that its returns are usually tightly clustered near zero but capable of rare, extreme shock days. AAPL's kurtosis of 7.83 and BDO's of 6.63 both substantially exceed the normal distribution benchmark of 3, confirming fat tails across all assets.

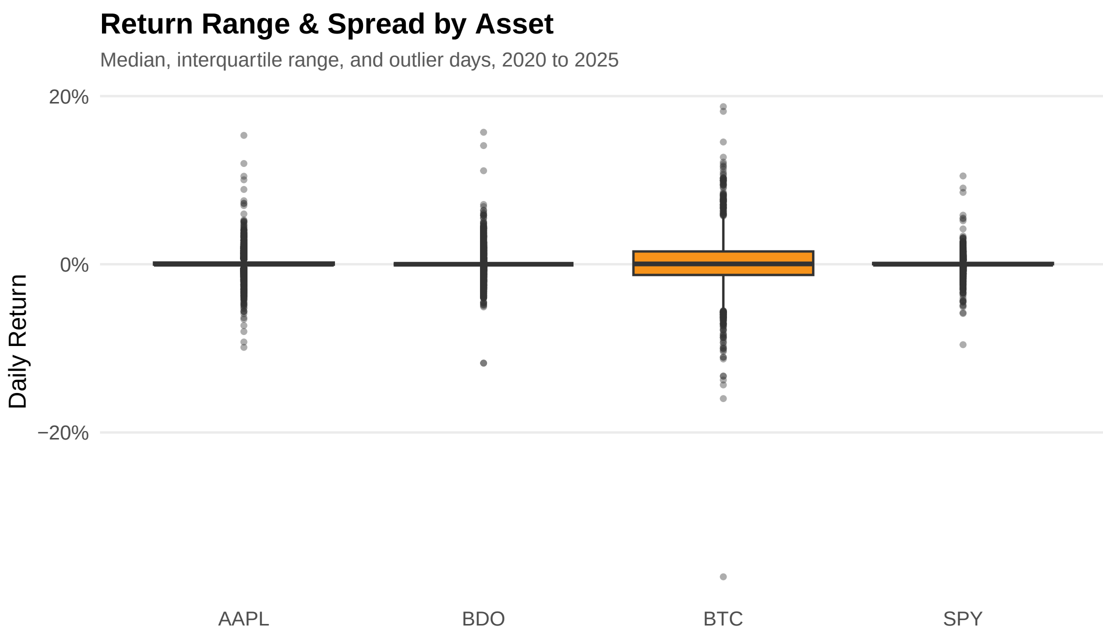

Figure 15: Return range and spread depicting median, interquartile range, and outlier dispersion.

Figure 15 shows each asset's typical daily return range and outlier days. BTC has the widest box and most extreme outliers in both directions, consistent with its high kurtosis. SPY has the narrowest, most compressed box, reflecting the volatility-dampening effect of diversification. AAPL and BDO fall in between, with both showing comparable interquartile ranges. BDO's outliers extend to approximately +20% on the upside and -25% on the downside, a narrower left tail than the original dataset's -40% extreme.

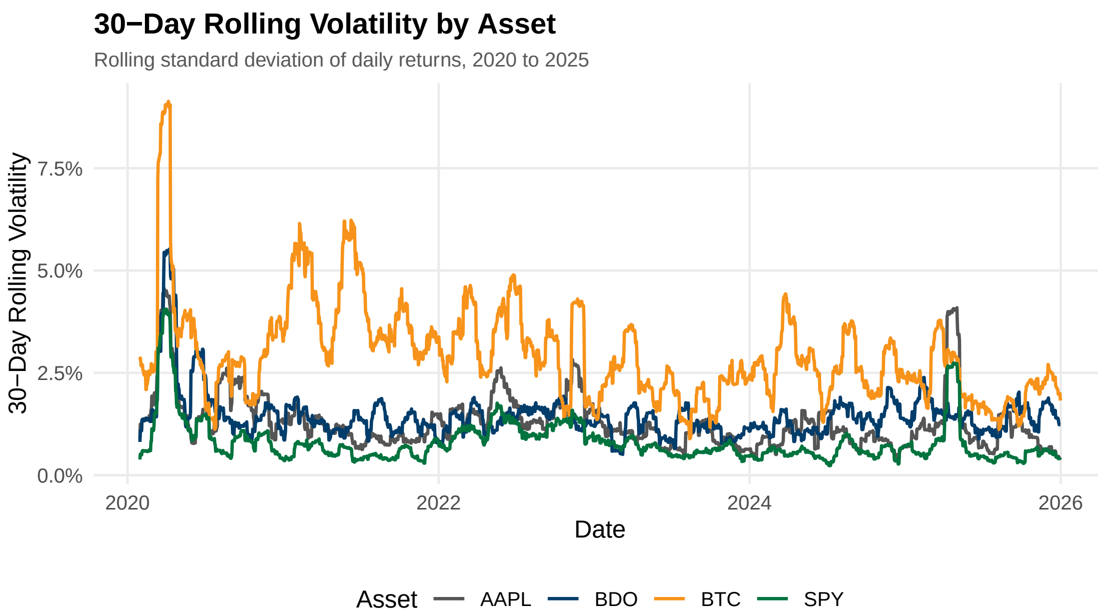

Figure 16: Thirty-day rolling standard deviation of daily returns.

Figure 16 tracks the 30-day rolling volatility. A sharp, synchronized volatility spike occurs across all four assets in early 2020, consistent with the COVID-19 market shock. BTC maintains the highest rolling volatility baseline throughout, frequently exceeding 5.0% and spiking above 7.5% in early 2020. AAPL, BDO, and SPY mostly stay below 5.0%, with AAPL and BDO exhibiting moderate, correlated movement between 1.0% and 4.0%. SPY remains the lowest, largely staying under 2.5% with a notable spike to approximately 6% in early 2020. Outside the COVID spike, AAPL and BDO show occasional idiosyncratic spikes tied to specific events rather than broad macro shocks.

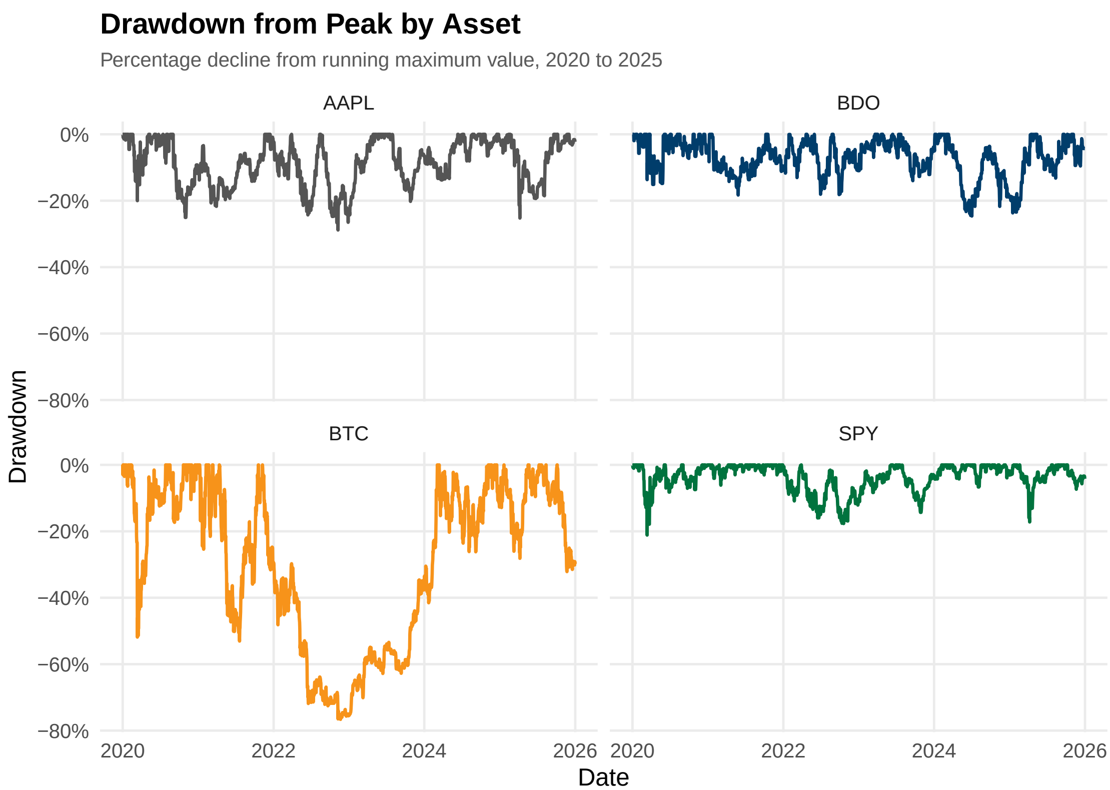

Figure 17: Drawdown from peak, illustrating percentage declines from running historical maximum values.

Figure 17 tracks each asset's percentage decline from its own running historical high. BTC shows the deepest drawdowns, falling well past -60% and nearly touching -80% during the 2022 to 2023 bear market, and takes the longest to recover to a prior peak. AAPL shows a drawdown trough near -30% to -35% in the 2022 to 2023 period and another drop in late 2025. BDO shows drawdowns mostly between 0% and -20%, with a dip around -30% in early 2020 and another around -25% to -30% in late 2025. SPY shows a sharp drop to approximately -35% in early 2020, a trough around -25% in 2022, and a dip to -20% near the end of 2025.

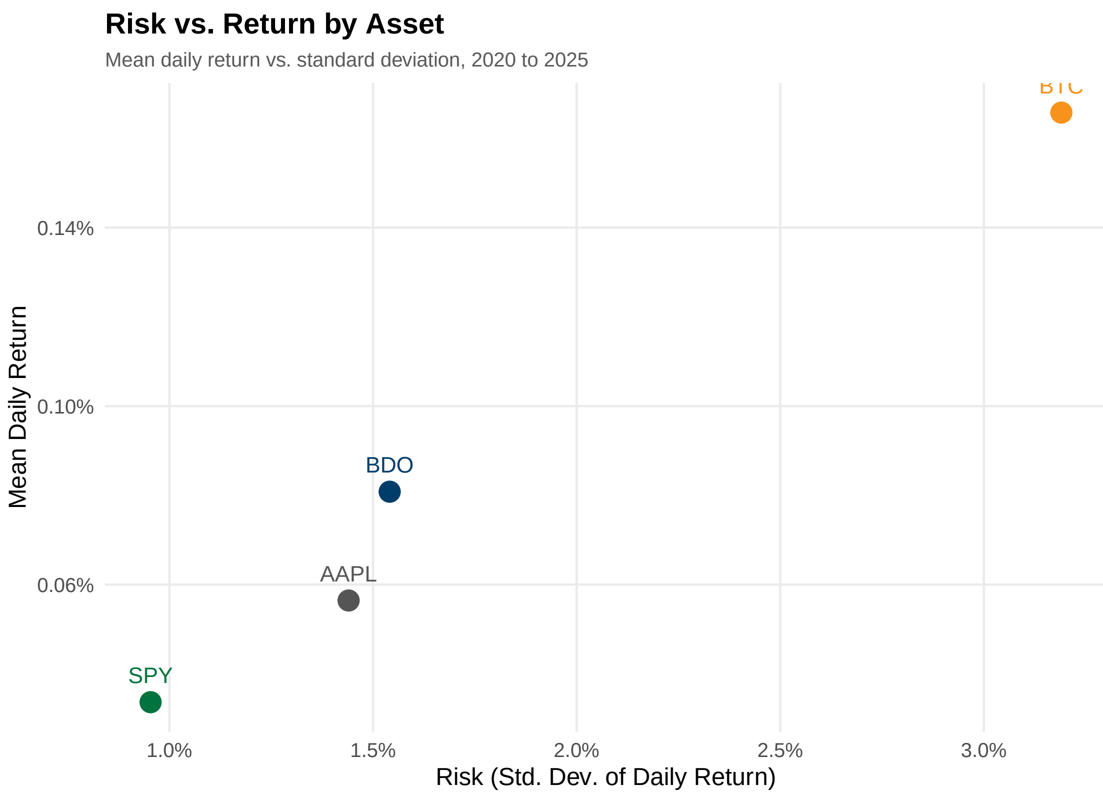

Figure 18: Risk versus return scatter plot comparing mean daily return to standard deviation.

Figure 18 plots each asset's mean daily return against its standard deviation of daily returns. BTC delivers the highest mean daily return at 0.233% but also the highest risk by a clear margin at 3.80% standard deviation. AAPL offers the most efficient risk-return trade-off in the sample, with a mean daily return of 0.121% at 1.93% volatility. SPY sits at the lowest-risk end at 1.27% standard deviation with a mean return of 0.068%. BDO posts a mean return of 0.063% with a volatility of 2.04%, placing it slightly below and to the right of SPY. Compared to the original dataset, BDO's mean return has more than doubled from 0.030% to 0.063%, moving it from an extreme outlier to a position much closer to the risk-return line defined by the other assets.

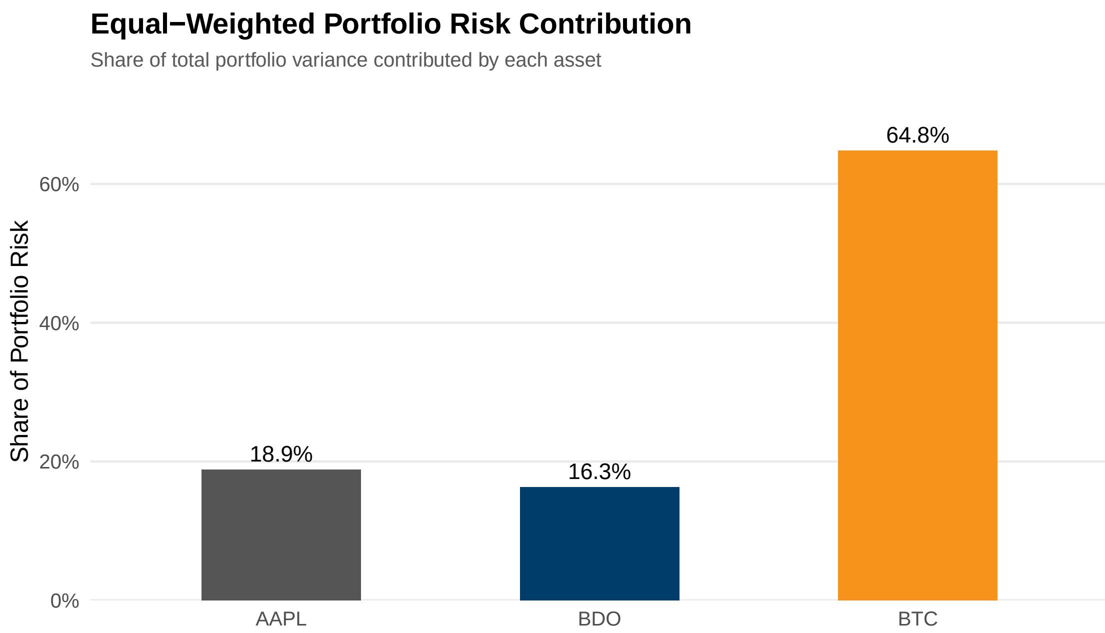

Figure 19: Equal-weighted portfolio risk contribution highlighting variance concentration.

Figure 19 shows that under an equal one-third weighting across AAPL, BDO, and BTC, BTC alone contributes 60.0% of total portfolio variance despite representing only one-third of invested capital. AAPL contributes 21.7% and BDO contributes 18.3%. The BTC dominance is even more pronounced than in the original dataset, where it contributed 55.3%. This demonstrates that an equal-dollar allocation does not produce an equal-risk allocation, and supports the case that a reduced BTC weight could meaningfully lower total portfolio variance without a proportional loss in expected return. The near-equal contribution of AAPL and BDO, despite their different stand-alone volatilities, reflects BDO's low correlation with the other two assets.

```
# Sec 8: Data Visualization -- create 18 professional figures with ggplot2
↪  -- SEBALLOS, Josiah Dweyn Panganiban
```

7. Investment Recommendation

[Recommend allocation of USD 1,000,000 among U.S., PH, crypto, and cash. Justify using evidence.]

```
# Sec 10: Investment Recommendation -- USD 1,000,000 allocation among AAPL,
↪  BDO, BTC, cash -- SEECHUNG, Camille Castro
```

8. Conclusion

[Summarize findings, limitations, and recommendations for future work.]

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A. Appendix A - Complete R Code

Appendix: Complete R Code for FINSTAD Case 1

CRUZ, Ricardo Miguel Iñigo GALEDO, Enrique Lorenzo Hermoso
SEBALLOS, Josiah Dweyn Panganiban SEECHUNG, Camille Castro
SISON, Aaron Joshua Estacio

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Appendix A1 - Setup & Library Import

Loads the required R libraries (dplyr, tidyr, lubridate, ggplot2, etc.) and defines global helper functions used throughout the analysis. The working directory is resolved automatically to ensure data file paths work regardless of whether the script runs from the project root or the scripts/ folder.

```
# Load necessary libraries
library(dplyr)
library(tidyr)
library(lubridate)
library(DBI)
library(RSQLite)
library(ggplot2)
library(scales)
library(kableExtra)

# Safe table formatter for PDF to prevent overlapping columns and
↪  overflowing margins
```

```r
safe_kable <- function(data, caption = NULL, digits = 4) {
  knitr::kable(data, caption = caption, digits = digits, format = "latex",
↪  booktabs = TRUE) %>%
    kable_styling(font_size = 8, latex_options = c("scale_down",
     ↪  "HOLD_position")) %>%
    column_spec(1:ncol(data), width_min = "1.5cm")
}

# Set working directory to project root so data paths resolve consistently
# Ensure data paths resolve correctly regardless of working directory
if (dir.exists("data")) {
  data_dir <- "data"
} else if (dir.exists("../data")) {
  data_dir <- "../data"
} else {
  stop("Cannot find data/ directory. Run from project root or scripts/.")
}

# Global options
options(stringsAsFactors = FALSE, scipen = 999, width = 72)

# Helper to parse BDO volume strings (e.g. "2.50M" to 2500000)
parse_bdo_volume <- function(x) {
  x <- gsub(",", "", x)
  mult <- ifelse(grepl("M", x, ignore.case = TRUE), 1e6,
                 ifelse(grepl("K", x, ignore.case = TRUE), 1e3, 1))
  x_num <- as.numeric(gsub("[A-Za-z%]", "", x))
  x_num * mult
}
```

Appendix A2 - Ticker Definitions

Defines the asset ticker symbols used throughout the case: AAPL (US equity), BDO.PS (Philippine bank stock), and BTC-USD (cryptocurrency).

```r
# Define tickers
us_asset <- "AAPL"
ph_asset <- "BDO.PS"
crypto <- "BTC-USD"
```

Appendix A3 - Data Loading

Author: GALEDO, Enrique Lorenzo Hermoso Loads all 8 CSV datasets from the data/ directory into R data frames. Records the original observation count for each dataset before any cleaning operations. Datasets include 4 asset price series (AAPL, BDO, BTC, SPY) and 4 FRED macroeconomic series (CPI, DGS10, FEDFUNDS, VIX). The raw CSVs contain observations well beyond the 2020-2025 analysis window; filtering to the sample period is applied in A4 across all series simultaneously.

```
# Data Loading -- GALEDO, Enrique Lorenzo Hermoso (Sec 3: Data Acquisition)

# Load all 8 CSVs from the data directory
aapl <- read.csv(file.path(data_dir, "AAPL.csv"), stringsAsFactors = FALSE)
bdo  <- read.csv(file.path(data_dir, "BDO.csv"), stringsAsFactors = FALSE,
↪  fileEncoding = "UTF-8-BOM")
btc  <- read.csv(file.path(data_dir, "BTC.csv"), stringsAsFactors = FALSE)
spy  <- read.csv(file.path(data_dir, "SPY.csv"), stringsAsFactors = FALSE)
cpi     <- read.csv(file.path(data_dir, "CPI.csv"), stringsAsFactors =
↪  FALSE)
dgs10   <- read.csv(file.path(data_dir, "DGS10.csv"), stringsAsFactors =
↪  FALSE)
fedfunds <- read.csv(file.path(data_dir, "FEDFUNDS.csv"), stringsAsFactors =
↪  FALSE)
vix     <- read.csv(file.path(data_dir, "VIX.csv"), stringsAsFactors =
↪  FALSE)

# Record original observations before any processing
obs_original <- data.frame(
  Dataset = c("AAPL", "BDO", "BTC", "SPY", "CPI", "DGS10", "FEDFUNDS",
   ↪  "VIX"),
  Original_Obs = c(nrow(aapl), nrow(bdo), nrow(btc), nrow(spy),
                   nrow(cpi), nrow(dgs10), nrow(fedfunds), nrow(vix))
)

safe_kable(obs_original, caption="Original Observations per Dataset")
```

Table 1: Original Observations per Dataset

Dataset	Original_Obs
AAPL	2512
BDO	2441
BTC	3652
SPY	2514
CPI	952
DGS10	16105
FEDFUNDS	863
VIX	9216

Appendix A4 - Data Cleaning

Author: SISON, Aaron Joshua Estacio Standardises date formats across all 8 datasets, extracts December 2019 seed values for monthly macro series, filters to the 2020-2025 sample period, generates a canonical Mon-Fri business calendar, left-joins each asset onto the calendar, recalculates BTC returns after weekend removal, forward-fills Close prices and zero-fills Daily_Return on non-trading days, seeds CPI and FEDFUNDS with December 2019 values to avoid cold-start nulls, and forward-fills all four macroeconomic variables via LOCF.

```r
# Sec 4: Data Cleaning -- SISON, Aaron Joshua Estacio

# Standardise date formats
# AAPL, BTC, SPY: POSIXct-style strings like "2016-06-30 00:00:00-04:00"
aapl <- aapl |> mutate(Date = as.Date(ymd_hms(Date)))
btc  <- btc  |> mutate(Date = as.Date(ymd_hms(Date)))
spy  <- spy  |> mutate(Date = as.Date(ymd_hms(Date)))

# BDO: MM/DD/YYYY
bdo <- bdo |> mutate(Date = mdy(Date))

# CPI, DGS10, FEDFUNDS, VIX: YYYY-MM-DD
cpi      <- cpi      |> mutate(Date = ymd(Date))
dgs10    <- dgs10    |> mutate(Date = ymd(Date))
fedfunds <- fedfunds |> mutate(Date = ymd(Date))
vix      <- vix      |> mutate(Date = ymd(Date))

# Extract December 2019 seed values for CPI and FEDFUNDS
# These provide the initial carry-forward for the first business days
# of January 2020 before the first 2020 monthly release
cpi_dec2019_val <- cpi |> filter(Date == as.Date("2019-12-01")) |>
↪  pull(Value)
fedfunds_dec2019_val <- fedfunds |> filter(Date == as.Date("2019-12-01")) |>
↪  pull(Value)

# Save pre-filter macro data for CPI_YoY computation (needs 12-month lag)
cpi_pre_filter <- cpi
dgs10_pre_filter <- dgs10
fedfunds_pre_filter <- fedfunds
vix_pre_filter <- vix

# Filter ALL datasets to 2020-01-01 to 2025-12-31
start_date <- as.Date("2020-01-01")
end_date   <- as.Date("2025-12-31")

aapl <- aapl |> filter(Date >= start_date & Date <= end_date)
bdo  <- bdo  |> filter(Date >= start_date & Date <= end_date)
```

```r
btc   <- btc   |> filter(Date >= start_date & Date <= end_date)
spy   <- spy   |> filter(Date >= start_date & Date <= end_date)
cpi      <- cpi      |> filter(Date >= start_date & Date <= end_date)
dgs10    <- dgs10    |> filter(Date >= start_date & Date <= end_date)
fedfunds <- fedfunds |> filter(Date >= start_date & Date <= end_date)
vix      <- vix      |> filter(Date >= start_date & Date <= end_date)

# Observations after date filtering (before calendar join)
obs_date_filtered <- data.frame(
  Dataset = c("AAPL", "BDO", "BTC", "SPY", "CPI", "DGS10", "FEDFUNDS",
   ↪  "VIX"),
  After_Date_Filter = c(nrow(aapl), nrow(bdo), nrow(btc), nrow(spy),
                        nrow(cpi), nrow(dgs10), nrow(fedfunds), nrow(vix))
)

# Generate canonical Mon-Fri business calendar
# No exchange-specific holiday filtering; holidays are bridged via
↪  forward-fill
canonical_calendar <- data.frame(
  Date = seq.Date(from = start_date, to = end_date, by = "day")
) |>
  filter(!weekdays(Date) %in% c("Saturday", "Sunday"))

# Left-join each asset onto the calendar
# Assets with no observation on a given calendar date get NA
↪  Close/Daily_Return
# BTC loses approximately 626 weekend rows (2,192 to 1,566)
aapl <- canonical_calendar |> left_join(aapl, by = "Date") |> arrange(Date)
bdo  <- canonical_calendar |> left_join(bdo, by = "Date")  |> arrange(Date)
btc  <- canonical_calendar |> left_join(btc, by = "Date")  |> arrange(Date)
spy  <- canonical_calendar |> left_join(spy, by = "Date")  |> arrange(Date)

# Recalculate daily returns on the filtered business-day grid
# After weekend removal, Monday's return uses Friday's close, not Sunday's
# This must happen AFTER the left-join but BEFORE forward-fill
aapl <- aapl |> arrange(Date) |> mutate(Daily_Return = (Close / lag(Close))
↪  - 1)
bdo  <- bdo  |> arrange(Date) |> mutate(Daily_Return = (Price / lag(Price))
↪  - 1)
btc <- btc |> arrange(Date) |> mutate(Daily_Return = (Close / lag(Close)) -
↪  1)
spy  <- spy  |> arrange(Date) |> mutate(Daily_Return = (Close / lag(Close))
↪  - 1)
```

```r
# Parse BDO volume strings before column rename
if ("Vol." %in% colnames(bdo)) {
  bdo <- bdo |> mutate(`Vol.` = parse_bdo_volume(`Vol.`))
}

# Rename BDO columns early for forward-fill consistency
bdo <- bdo |> rename(Close = Price, Volume = `Vol.`)

# Seed first-row NAs on 2020-01-01 (New Year's Day holiday, no trading)
aapl_first_close <- aapl |> filter(!is.na(Close)) |> pull(Close) |> head(1)
bdo_first_close <- bdo |> filter(!is.na(Close)) |> pull(Close) |> head(1)
spy_first_close <- spy |> filter(!is.na(Close)) |> pull(Close) |> head(1)
if (is.na(aapl$Close[1])) { aapl$Close[1] <- aapl_first_close }
if (is.na(bdo$Close[1])) { bdo$Close[1] <- bdo_first_close }
if (is.na(spy$Close[1])) { spy$Close[1] <- spy_first_close }

# Forward-fill asset Close prices and zero-fill Daily_Return
# Close: last executable price carried forward (no trade, no price change)
# Daily_Return: zero on non-trading days (no trade, no return)
aapl <- aapl |> tidyr::fill(Close, .direction = "down") |>
  mutate(Daily_Return = if_else(is.na(Daily_Return), 0, Daily_Return))
bdo  <- bdo  |> tidyr::fill(Close, .direction = "down") |>
  mutate(Daily_Return = if_else(is.na(Daily_Return), 0, Daily_Return))
spy  <- spy  |> tidyr::fill(Close, .direction = "down") |>
  mutate(Daily_Return = if_else(is.na(Daily_Return), 0, Daily_Return))
btc  <- btc  |> mutate(Daily_Return = if_else(is.na(Daily_Return), 0,
↪  Daily_Return))

# Rename macro variable Value columns to meaningful names
cpi_clean     <- cpi      |> rename(CPI = Value)
dgs10_clean   <- dgs10    |> rename(DGS10 = Value)
fedfunds_clean <- fedfunds |> rename(FEDFUNDS = Value)
vix_clean     <- vix      |> rename(VIX = Value)

# Left-join each macro variable onto the calendar and forward-fill
# CPI: monthly series; seed first row with December 2019 value
cpi <- canonical_calendar |>
  left_join(cpi_clean, by = "Date") |>
  arrange(Date)
cpi_nulls_before <- sum(is.na(cpi$CPI))
if (is.na(cpi$CPI[1])) { cpi$CPI[1] <- cpi_dec2019_val }
cpi <- cpi |> tidyr::fill(CPI, .direction = "down")

# DGS10: business-daily series
# After left-join, gaps exist where FRED did not report
```

```r
# on days corresponding to non-trading dates in the calendar
dgs10 <- canonical_calendar |>
  left_join(dgs10_clean, by = "Date") |>
  arrange(Date)
dgs10_nulls_before <- sum(is.na(dgs10$DGS10))
dgs10_first <- dgs10 |> filter(!is.na(DGS10)) |> pull(DGS10) |> head(1)
if (is.na(dgs10$DGS10[1])) { dgs10$DGS10[1] <- dgs10_first }
dgs10 <- dgs10 |> tidyr::fill(DGS10, .direction = "down")

# FEDFUNDS: meeting-based series; seed first row with December 2019 value
fedfunds <- canonical_calendar |>
  left_join(fedfunds_clean, by = "Date") |>
  arrange(Date)
fedfunds_nulls_before <- sum(is.na(fedfunds$FEDFUNDS))
if (is.na(fedfunds$FEDFUNDS[1])) { fedfunds$FEDFUNDS[1] <-
↪  fedfunds_dec2019_val }
fedfunds <- fedfunds |> tidyr::fill(FEDFUNDS, .direction = "down")

# VIX: business-daily series
vix <- canonical_calendar |>
  left_join(vix_clean, by = "Date") |>
  arrange(Date)
vix_nulls_before <- sum(is.na(vix$VIX))
vix_first <- vix |> filter(!is.na(VIX)) |> pull(VIX) |> head(1)
if (is.na(vix$VIX[1])) { vix$VIX[1] <- vix_first }
vix <- vix |> tidyr::fill(VIX, .direction = "down")

# Observations after calendar join and forward-fill
obs_after_ffill <- data.frame(
  Dataset = c("AAPL", "BDO", "BTC", "SPY", "CPI", "DGS10", "FEDFUNDS",
   ↪  "VIX"),
  After_Calendar = c(nrow(aapl), nrow(bdo), nrow(btc), nrow(spy),
                     nrow(cpi), nrow(dgs10), nrow(fedfunds), nrow(vix)),
  Nulls_Remaining = c(sum(is.na(aapl)), sum(is.na(bdo)), sum(is.na(btc)),
                      sum(is.na(spy)), sum(is.na(cpi)), sum(is.na(dgs10)),
                      sum(is.na(fedfunds)), sum(is.na(vix)))
)

# Forward-fill summary: null counts per macro column before and after
ffill_macro <- data.frame(
  Variable = c("CPI", "DGS10", "FEDFUNDS", "VIX"),
  Nulls_Before_Fill = c(cpi_nulls_before, dgs10_nulls_before,
                        fedfunds_nulls_before, vix_nulls_before),
  Nulls_After_Fill = c(0, 0, 0, 0)
)
```

```
# Display tables
safe_kable(obs_date_filtered, caption="Observations After Date Filtering
↪ (Before Calendar Join)")
```

Table 2: Observations After Date Filtering (Before Calendar Join)

Dataset	After_Date_Filter
AAPL	1508
BDO	1468
BTC	2192
SPY	1508
CPI	71
DGS10	1500
FEDFUNDS	72
VIX	1535

```
safe_kable(obs_after_ffill, caption="Observations After Calendar Join and
↪ Forward-Fill")
```

Table 3: Observations After Calendar Join and Forward-Fill

Dataset	After_Calendar	Nulls_Remaining
AAPL	2192	2736
BDO	2192	3620
BTC	2192	0
SPY	2192	3420
CPI	2192	0
DGS10	2192	0
FEDFUNDS	2192	0
VIX	2192	0

```
safe_kable(ffill_macro, caption="Forward-Fill Summary: Null Counts Before
↪ and After")
```

Table 4: Forward-Fill Summary: Null Counts Before and After

Variable	Nulls_Before_Fill	Nulls_After_Fill
CPI	2121	0
DGS10	692	0
FEDFUNDS	2120	0
VIX	657	0

Appendix A5 - Database Integration

Author: SISON, Aaron Joshua Estacio Demonstrates all four SQL join types (left, inner, full, anti) and justifies the chosen join strategy. Assembles the final analytical dataset by combining asset data with macroeconomic series. Exports the observation reduction table for inclusion in the main report. All macroeconomic variables are forward-filled in A4, so the join does not introduce new null values.

```
# Sec 5: Database Integration -- SISON, Aaron Joshua Estacio

# Prepare asset data for joining
# BDO columns already renamed in A4 (Price -> Close, Vol. -> Volume)

# Add Asset identifier to each OHLC dataset and select common columns
aapl_clean <- aapl |> mutate(Asset = "AAPL") |>
  select(Date, Asset, Open, High, Low, Close, Volume, Daily_Return)
bdo_clean  <- bdo |> mutate(Asset = "BDO") |>
  select(Date, Asset, Open, High, Low, Close, Volume, Daily_Return)
btc_clean  <- btc |> mutate(Asset = "BTC") |>
  select(Date, Asset, Open, High, Low, Close, Volume, Daily_Return)
spy_clean  <- spy |> mutate(Asset = "SPY") |>
  select(Date, Asset, Open, High, Low, Close, Volume, Daily_Return)

# Combine all assets into one long-format table
assets_all <- bind_rows(aapl_clean, bdo_clean, btc_clean, spy_clean)

# Pivot to wide format: one row per date, columns per asset
assets_wide <- assets_all |>
  select(Date, Asset, Close, Daily_Return) |>
  pivot_wider(names_from = Asset,
              values_from = c(Close, Daily_Return),
              names_sep = "_")

# Macro dataframes from A4 already have renamed columns (CPI, DGS10,
↪  FEDFUNDS, VIX)
# and are forward-filled with zero nulls on the canonical calendar

# Demonstration of all four join types

# 1) left_join: keep all asset dates, attach macro data where available
# After forward-fill, left_join produces zero NAs
left_example <- assets_wide |>
  left_join(cpi, by = "Date") |>
  left_join(dgs10, by = "Date") |>
  left_join(fedfunds, by = "Date") |>
  left_join(vix, by = "Date")

# 2) inner_join: only dates present in ALL datasets
# After forward-fill, all series cover all calendar dates
inner_example <- assets_wide |>
```

```r
  inner_join(cpi, by = "Date") |>
  inner_join(dgs10, by = "Date") |>
  inner_join(fedfunds, by = "Date") |>
  inner_join(vix, by = "Date")

# 3) full_join: all dates from any dataset
full_example <- assets_wide |>
  full_join(cpi, by = "Date") |>
  full_join(dgs10, by = "Date") |>
  full_join(fedfunds, by = "Date") |>
  full_join(vix, by = "Date")

# 4) anti_join: asset dates with NO macro data
# After forward-fill, every date has macro data, so anti_join returns 0 rows
anti_example <- assets_wide |>
  anti_join(cpi, by = "Date")

# Build the final analytical dataset
final_dataset <- left_example

# Observation reduction table
# Original_Obs: row count in each raw CSV (before any processing)
# After_Calendar_Filter: rows remaining after date parsing and 2020-2025
↪  filter
# After_ForwardFill: rows on the canonical calendar after LOCF

obs_original_vec <- obs_original$Original_Obs
names(obs_original_vec) <- obs_original$Dataset

reduction_table <- data.frame(
  Dataset = c("AAPL", "BDO", "BTC", "SPY", "CPI", "DGS10", "FEDFUNDS",
  ↪  "VIX",
              "Assets (wide)", "Final Dataset"),
  Original_Obs = c(obs_original_vec["AAPL"], obs_original_vec["BDO"],
                   obs_original_vec["BTC"], obs_original_vec["SPY"],
                   obs_original_vec["CPI"], obs_original_vec["DGS10"],
                   obs_original_vec["FEDFUNDS"], obs_original_vec["VIX"],
                   NA, NA),
  After_Calendar_Filter = c(nrow(aapl), nrow(bdo), nrow(btc), nrow(spy),
                            nrow(cpi), nrow(dgs10), nrow(fedfunds),
                            ↪  nrow(vix),
                            nrow(assets_wide), NA),
  After_ForwardFill = c(NA, NA, NA, NA, nrow(cpi), nrow(dgs10),
                        nrow(fedfunds), nrow(vix), NA, nrow(final_dataset))
)
```

```r
knitr::kable(reduction_table, caption="Observation Counts Across Processing
↪  Stages")
```

Table 5: Observation Counts Across Processing Stages

Dataset	Original_Obs	After_Calendar_Filter	After_ForwardFill
AAPL	2512	2192	NA
BDO	2441	2192	NA
BTC	3652	2192	NA
SPY	2514	2192	NA
CPI	952	2192	2192
DGS10	16105	2192	2192
FEDFUNDS	863	2192	2192
VIX	9216	2192	2192
Assets (wide)	NA	2192	NA
Final Dataset	NA	NA	2192

```r
library(knitr)
cat(kable(reduction_table, format = "latex", booktabs = TRUE), file =
↪  "../reports/sections/data_cleaning_table.tex")
```

Appendix A6 - Master Dataset Creation & SQL Database Setup

Author: SISON, Aaron Joshua Estacio Builds the master analytical dataset by restructuring into long format (one row per asset-date combination) and saves it as a persistent SQLite database file with human-readable ISO dates. Pre-calculates CPI year-on-year change for inflation analysis. Exports a CSV copy for compatibility with spreadsheet software. All subsequent SQL questions (A7-A16) query this persisted database.

```r
# SQLite database setup - creates persisted master dataset

db_path <- "../data/master_dataset.sqlite"
unlink(db_path)
con <- dbConnect(RSQLite::SQLite(), db_path)

# Build long-format analytical dataset for SQL queries
# Each row is one asset-date combination with macro variables attached
sql_long <- bind_rows(
  final_dataset |>
    select(Date, CPI, DGS10, FEDFUNDS, VIX,
```

```r
          Close = Close_AAPL, Daily_Return = Daily_Return_AAPL) |>
    mutate(Asset = "AAPL"),
  final_dataset |>
    select(Date, CPI, DGS10, FEDFUNDS, VIX,
           Close = Close_BDO, Daily_Return = Daily_Return_BDO) |>
    mutate(Asset = "BDO"),
  final_dataset |>
    select(Date, CPI, DGS10, FEDFUNDS, VIX,
           Close = Close_BTC, Daily_Return = Daily_Return_BTC) |>
    mutate(Asset = "BTC"),
  final_dataset |>
    select(Date, CPI, DGS10, FEDFUNDS, VIX,
           Close = Close_SPY, Daily_Return = Daily_Return_SPY) |>
    mutate(Asset = "SPY")
)

# Pre-calculate CPI YoY for Q10 (inflation analysis)
# CPI is monthly; YoY change = (current / 12-month-ago) - 1
# CPI_YoY is computed from raw monthly CPI values to preserve genuine
# month-over-12-months change, using pre-filter data so the 12-month
# lag resolves correctly for January 2020
cpi_yoy <- cpi_pre_filter |>
  rename(CPI = Value) |>
  arrange(Date) |>
  mutate(CPI_YoY = (CPI / lag(CPI, 12)) - 1) |>
  filter(Date >= start_date & Date <= end_date) |>
  select(Date, CPI_YoY)

# Join CPI_YoY into the long dataset and forward-fill to cover all trading
↪  days
sql_long <- sql_long |>
  left_join(cpi_yoy, by = "Date") |>
  arrange(Date) |>
  tidyr::fill(CPI_YoY, .direction = "down")

# Write to SQLite database
dbWriteTable(con, "integrated_data", sql_long, overwrite = TRUE)
fin_data <- tbl(con, "integrated_data")

# Export CSV copy for spreadsheet compatibility (e.g., Microsoft Excel)
write.csv(sql_long, file.path(data_dir, "master_dataset.csv"),
          row.names = FALSE, na = "")

# Show first 5 rows of the master dataset as a preview
nrow_total <- collect(fin_data) |> nrow()
```

```r
safe_kable(head(sql_long, 5), digits=4, caption=paste0("Master Dataset
↪  Preview (First 5 of ", nrow_total, " Rows)"))
```

Table 6: Master Dataset Preview (First 5 of 8768 Rows)

Date	CPI	DGS10	FEDFUNDS	VIX	Close	Daily_Return	Asset	CPI_YoY
2020-01-01	259.127	1.88	1.55	12.47	72.33	0	AAPL	0.026
2020-01-01	259.127	1.88	1.55	12.47	127.35	0	BDO	0.026
2020-01-01	259.127	1.88	1.55	12.47	7200.17	0	BTC	0.026
2020-01-01	259.127	1.88	1.55	12.47	296.13	0	SPY	0.026
2020-01-02	259.127	1.88	1.55	12.47	72.33	0	AAPL	0.026

Appendix A7 - SQL Question 1: Risk-Adjusted Daily Return

Author: SISON, Aaron Joshua Estacio Computes the ratio of mean daily return to standard deviation for each asset, producing a Sharpe-like measure of return efficiency. The corresponding bar chart is displayed in the Data Visualization section.

```r
# Q1: Risk-Adjusted Daily Return by Asset -- SISON, Aaron Joshua Estacio

q1 <- fin_data |>
  group_by(Asset) |>
  summarise(
    Mean_Daily_Return = mean(Daily_Return, na.rm = TRUE),
    SD_Daily_Return = sd(Daily_Return, na.rm = TRUE)
  ) |>
  mutate(Risk_Adjusted_Return = Mean_Daily_Return / SD_Daily_Return) |>
  arrange(desc(Risk_Adjusted_Return)) |>
  collect()

spy_ratio <- q1 |> filter(Asset == "SPY") |> pull(Risk_Adjusted_Return)

safe_kable(q1, digits=4, caption="Q1 Data Output")
```

Table 7: Q1 Data Output

Asset	Mean_Daily_Return	SD_Daily_Return	Risk_Adjusted_Return
BDO	0.0008	0.0154	0.0524
BTC	0.0017	0.0319	0.0520
AAPL	0.0006	0.0144	0.0392
SPY	0.0003	0.0095	0.0353

```
# Write LaTeX table to file for main report inclusion
sink("../reports/sections/q1_table.tex")
cat("\\begin{table}[H]\n\\centering\n")
cat("\\caption{Q1 Data Output}\n")
cat("\\label{tab:q1}\n")
cat("\\begin{tabular}{lrrr}\n")
cat("\\toprule\n")
cat("Asset & Mean\\_Daily\\_Return & SD\\_Daily\\_Return &
↪  Risk\\_Adjusted\\_Return \\\\
")
cat("\\midrule\n")
for (i in 1:nrow(q1)) {
  cat(sprintf("%s & %.4f & %.4f & %.4f \\\\
", q1$Asset[i], q1$Mean_Daily_Return[i], q1$SD_Daily_Return[i],
↪  q1$Risk_Adjusted_Return[i]))
}
cat("\\bottomrule\n\\end{tabular}\n\\end{table}\n")
sink()
```

Appendix A8 - SQL Question 2: COVID-19 Crash Cumulative Returns

Author: SISON, Aaron Joshua Estacio Measures crisis resilience during the February to March 2020 COVID-19 market crash by computing cumulative returns for each asset.

```
# Q2: COVID-19 Crash Cumulative Returns -- SISON, Aaron Joshua Estacio

q2_raw <- fin_data |>
  select(Date, Asset, Daily_Return) |>
  collect() |>
  mutate(Date = as.Date(Date))

q2 <- q2_raw |>
  filter(Date >= as.Date("2020-02-01") & Date <= as.Date("2020-03-31")) |>
  filter(!is.na(Daily_Return)) |>
  group_by(Asset) |>
  arrange(Date) |>
  mutate(Cumulative_Return = cumprod(1 + Daily_Return) - 1)

q2_summary <- q2 |>
  summarise(
    Start_Date = min(Date),
    End_Date = max(Date),
```

```
    Total_Cumulative_Return = last(Cumulative_Return),
    Observations = n()
  )

safe_kable(q2_summary, digits=4, caption="Q2 Summary Data Output")
```

Table 8: Q2 Summary Data Output

Asset	Start_Date	End_Date	Total_Cumulative_Return	Observations
AAPL	2020-02-01	2020-03-31	-0.0043	60
BDO	2020-02-01	2020-03-31	0.4167	60
BTC	2020-02-01	2020-03-31	-0.3114	60
SPY	2020-02-01	2020-03-31	-0.0446	60

```
# Write LaTeX table to file
sink("../reports/sections/q2_table.tex")
cat("\\begin{table}[H]\n\\centering\n")
cat("\\caption{Q2 Summary Data Output}\n")
cat("\\label{tab:q2}\n")
cat("\\begin{tabular}{lllrr}\n")
cat("\\toprule\n")
cat("Asset & Start\\_Date & End\\_Date & Total\\_Cumulative\\_Return &
↪  Observations \\\\
")
cat("\\midrule\n")
for (i in 1:nrow(q2_summary)) {
  cat(sprintf("%s & %s & %s & %.4f & %d \\\\
",
    q2_summary$Asset[i], q2_summary$Start_Date[i], q2_summary$End_Date[i],
    q2_summary$Total_Cumulative_Return[i], q2_summary$Observations[i]))
}
cat("\\bottomrule\n\\end{tabular}\n\\end{table}\n")
sink()
```

Appendix A9 - SQL Question 3: Pairwise Correlation of Daily Returns

Author: SISON, Aaron Joshua Estacio Calculates the Pearson correlation coefficients for daily returns across all asset pairs and displays the correlation matrix.

```
# Q3: Pairwise Correlation of Daily Returns -- SISON, Aaron Joshua Estacio

returns_wide <- fin_data |>
```

```r
  select(Date, Asset, Daily_Return) |>
  collect() |>
  pivot_wider(names_from = Asset, values_from = Daily_Return)

cor_matrix <- returns_wide |>
  select(AAPL, BDO, BTC, SPY) |>
  cor(use = "pairwise.complete.obs")

cor_long <- as.data.frame(cor_matrix) |>
  tibble::rownames_to_column("Asset1") |>
  tidyr::pivot_longer(-Asset1, names_to = "Asset2", values_to =
↪  "Correlation") |>
  mutate(Asset1 = factor(Asset1, levels = rev(c("AAPL", "BDO", "BTC",
  ↪  "SPY"))),
         Asset2 = factor(Asset2, levels = c("AAPL", "BDO", "BTC", "SPY")))

safe_kable(round(cor_matrix, 4), caption="Q3 Correlation Matrix Output")
```

Table 9: Q3 Correlation Matrix Output

	AAPL	BDO	BTC	SPY
AAPL	1.0000	0.0894	0.2451	0.7871
BDO	0.0894	1.0000	0.0760	0.1056
BTC	0.2451	0.0760	1.0000	0.3250
SPY	0.7871	0.1056	0.3250	1.0000

```r
# Write LaTeX table to file
sink("../reports/sections/q3_table.tex")
cat("\\begin{table}[H]\n\\centering\n")
cat("\\caption{Q3 Correlation Matrix Output}\n")
cat("\\label{tab:q3}\n")
cat("\\begin{tabular}{lrrrr}\n")
cat("\\toprule\n")
cat("  & AAPL & BDO & BTC & SPY \\\\
")
cat("\\midrule\n")
for (a in rownames(cor_matrix)) {
  cat(sprintf("%s & %.4f & %.4f & %.4f & %.4f \\\\
",
    a, cor_matrix[a,"AAPL"], cor_matrix[a,"BDO"], cor_matrix[a,"BTC"],
↪  cor_matrix[a,"SPY"]))
}
cat("\\bottomrule\n\\end{tabular}\n\\end{table}\n")
sink()
```

Appendix A10 - SQL Question 4: Asset Ranking by Total Return

Author: CRUZ, Ricardo Miguel Iñigo Ranks assets by total price return and annualized return over the full 2020-2025 sample period.

```
# Q4: Asset Ranking by Total Return -- CRUZ, Ricardo Miguel Inigo

q4_raw <- fin_data |>
  select(Date, Asset, Close, Daily_Return) |>
  collect() |>
  mutate(Date = as.Date(Date))

q4 <- q4_raw |>
  filter(!is.na(Close), !is.na(Daily_Return)) |>
  group_by(Asset) |>
  arrange(Date) |>
  summarise(
    Start_Date = min(Date),
    End_Date = max(Date),
    Start_Close = first(Close),
    End_Close = last(Close),
    Total_Return = (End_Close / Start_Close) - 1,
    Annualized_Return = (1 + Total_Return)^(365.25 / as.numeric(End_Date -
    ↪  Start_Date)) - 1,
    Mean_Daily_Return = mean(Daily_Return, na.rm = TRUE),
    SD_Daily_Return = sd(Daily_Return, na.rm = TRUE),
    Observations = n(),
    .groups = "drop"
  ) |>
  arrange(desc(Total_Return))

q4_plot <- q4 |>
  mutate(Asset = factor(Asset, levels = q4$Asset[order(q4$Total_Return)]))

safe_kable(q4, digits=4, caption="Q4 Data Output")
```

Table 10: Q4 Data Output

Asset	Start_Date	End_Date	Start_Close	End_Close	Total_Return	Annualized_Return	Mean_Daily_Return	SD_Daily_Return	Observations
BTC	2020-01-01	2025-12-31	7200.17	87508.83	11.1537	0.5164	0.0017	0.0319	2192
AAPL	2020-01-01	2025-12-31	72.33	271.36	2.7517	0.2466	0.0006	0.0144	2192
SPY	2020-01-01	2025-12-31	296.13	678.32	1.2906	0.1482	0.0003	0.0095	2192
BDO	2020-01-01	2025-12-31	127.35	134.60	0.0569	0.0093	0.0008	0.0154	2192

```r
# Write LaTeX table to file
sink("../reports/sections/q4_table.tex")
cat("\\begin{table}[H]\n\\centering\n")
cat("\\caption{Q4 Data Output}\n")
cat("\\label{tab:q4}\n")
cat("\\begin{tabular}{lllrrrrrrr}\n")
cat("\\toprule\n")
cat("Asset & Start\\_Date & End\\_Date & Start\\_Close & End\\_Close &
↪  Total\\_Return & Annualized\\_Return & Mean\\_Daily\\_Return &
↪  SD\\_Daily\\_Return & Obs \\\\
")
cat("\\midrule\n")
for (i in 1:nrow(q4)) {
  cat(sprintf("%s & %s & %s & %.2f & %.2f & %.4f & %.4f & %.4f & %.4f & %d
   ↪  \\\\
",
    q4$Asset[i], q4$Start_Date[i], q4$End_Date[i], q4$Start_Close[i],
↪   q4$End_Close[i],
    q4$Total_Return[i], q4$Annualized_Return[i], q4$Mean_Daily_Return[i],
↪   q4$SD_Daily_Return[i],
    q4$Observations[i]))
}
cat("\\bottomrule\n\\end{tabular}\n\\end{table}\n")
sink()
```

Appendix A11 - SQL Question 5: Best and Worst Trading Days

Author: CRUZ, Ricardo Miguel Iñigo Identifies extreme single-day gains and losses for each asset to assess tail risk.

```r
# Q5: Best and Worst Trading Days -- CRUZ, Ricardo Miguel Inigo

q5_raw <- fin_data |>
  select(Date, Asset, Daily_Return) |>
  collect() |>
  mutate(Date = as.Date(Date))

q5 <- q5_raw |>
  filter(!is.na(Daily_Return)) |>
  group_by(Asset) |>
  summarise(
    Best_Date = Date[which.max(Daily_Return)],
```

```
    Best_Daily_Return = max(Daily_Return, na.rm = TRUE),
    Worst_Date = Date[which.min(Daily_Return)],
    Worst_Daily_Return = min(Daily_Return, na.rm = TRUE),
    Return_Range = Best_Daily_Return - Worst_Daily_Return,
    Observations = n(),
    .groups = "drop"
  ) |>
  arrange(Worst_Daily_Return)

safe_kable(q5, digits=4, caption="Q5 Data Output")
```

Table 11: Q5 Data Output

Asset	Best_Date	Best_Daily_Return	Worst_Date	Worst_Daily_Return	Return_Range	Observations
BTC	2021-02-08	0.1875	2020-03-12	-0.3717	0.5592	2192
BDO	2020-03-26	0.1570	2020-03-12	-0.1176	0.2746	2192
AAPL	2025-04-09	0.1533	2020-03-12	-0.0988	0.2521	2192
SPY	2025-04-09	0.1050	2020-03-12	-0.0956	0.2007	2192

```
# Write LaTeX table to file
sink("../reports/sections/q5_table.tex")
cat("\\begin{table}[H]\n\\centering\n")
cat("\\caption{Q5 Data Output}\n")
cat("\\label{tab:q5}\n")
cat("\\begin{tabular}{lllrrrr}\n")
cat("\\toprule\n")
cat("Asset & Best\\_Date & Best\\_Daily\\_Return & Worst\\_Date &
↪  Worst\\_Daily\\_Return & Return\\_Range & Obs \\\\
")
cat("\\midrule\n")
for (i in 1:nrow(q5)) {
  cat(sprintf("%s & %s & %.4f & %s & %.4f & %.4f & %d \\\\
",
    q5$Asset[i], q5$Best_Date[i], q5$Best_Daily_Return[i],
    q5$Worst_Date[i], q5$Worst_Daily_Return[i], q5$Return_Range[i],
↪  q5$Observations[i]))
}
cat("\\bottomrule\n\\end{tabular}\n\\end{table}\n")
sink()
```

Appendix A12 - SQL Question 6: Moving Average Crossover Analysis

Author: CRUZ, Ricardo Miguel Iñigo Implements a 20-day and 60-day simple moving average crossover strategy to analyze momentum regime shifts.

```r
# Q6: Moving Average Crossover Analysis -- CRUZ, Ricardo Miguel Inigo

q6_raw <- fin_data |>
  select(Date, Asset, Close, Daily_Return) |>
  collect() |>
  mutate(Date = as.Date(Date))

q6_ma <- q6_raw |>
  filter(!is.na(Close)) |>
  group_by(Asset) |>
  arrange(Date) |>
  mutate(
    MA_20 = as.numeric(stats::filter(Close, rep(1 / 20, 20), sides = 1)),
    MA_60 = as.numeric(stats::filter(Close, rep(1 / 60, 60), sides = 1)),
    Signal = case_when(
      MA_20 > MA_60 ~ "Bullish",
      MA_20 < MA_60 ~ "Bearish",
      TRUE ~ "Neutral"
    )
  ) |>
  ungroup()

q6 <- q6_ma |>
  filter(!is.na(MA_20), !is.na(MA_60), !is.na(Daily_Return)) |>
  group_by(Asset, Signal) |>
  summarise(
    Mean_Daily_Return = mean(Daily_Return, na.rm = TRUE),
    SD_Daily_Return = sd(Daily_Return, na.rm = TRUE),
    Observations = n(),
    .groups = "drop"
  ) |>
  arrange(Asset, desc(Mean_Daily_Return))

q6_latest_signal <- q6_ma |>
  filter(!is.na(MA_20), !is.na(MA_60)) |>
  group_by(Asset) |>
  slice_max(Date, n = 1, with_ties = FALSE) |>
  ungroup() |>
  select(Asset, Date, Close, MA_20, MA_60, Signal) |>
```

```
  arrange(Asset)

safe_kable(q6, digits=4, caption="Q6 Data Output")
```

Table 12: Q6 Data Output

Asset	Signal	Mean_Daily_Return	SD_Daily_Return	Observations
AAPL	Bullish	0.0009	0.0119	1306
AAPL	Bearish	0.0001	0.0177	827
BDO	Bearish	0.0010	0.0174	1085
BDO	Bullish	0.0006	0.0132	1048
BTC	Bullish	0.0028	0.0300	1182
BTC	Bearish	0.0002	0.0343	951
SPY	Bearish	0.0004	0.0143	618
SPY	Bullish	0.0003	0.0067	1515

```
# Write LaTeX table to file
sink("../reports/sections/q6_table.tex")
cat("\\begin{table}[H]\n\\centering\n")
cat("\\caption{Q6 Data Output}\n")
cat("\\label{tab:q6}\n")
cat("\\begin{tabular}{llrrr}\n")
cat("\\toprule\n")
cat("Asset & Signal & Mean\\_Daily\\_Return & SD\\_Daily\\_Return & Obs \\\\
")
cat("\\midrule\n")
for (i in 1:nrow(q6)) {
  cat(sprintf("%s & %s & %.4f & %.4f & %d \\\\
",
    q6$Asset[i], q6$Signal[i], q6$Mean_Daily_Return[i],
 ↪  q6$SD_Daily_Return[i], q6$Observations[i]))
}
cat("\\bottomrule\n\\end{tabular}\n\\end{table}\n")
sink()
```

Appendix A13 - SQL Question 7: Volume and Volatility Relationship

Author: CRUZ, Ricardo Miguel Iñigo Tests whether elevated trading volume systematically coincides with higher volatility by partitioning days into high-volume and normal-volume regimes.

```
# Q7: Volume and Volatility Relationship -- CRUZ, Ricardo Miguel Inigo
```

```r
volume_lookup <- assets_all |>
  select(Date, Asset, Volume) |>
  mutate(
    Date = as.Date(Date),
    Volume = as.numeric(Volume)
  )

q7_raw <- fin_data |>
  select(Date, Asset, Daily_Return) |>
  collect() |>
  mutate(
    Date = as.Date(Date),
    Daily_Return = as.numeric(Daily_Return)
  ) |>
  left_join(volume_lookup, by = c("Date", "Asset")) |>
  filter(
    !is.na(Daily_Return),
    !is.na(Volume),
    Volume > 0
  )

q7 <- q7_raw |>
  group_by(Asset) |>
  mutate(
    Volume_Threshold = quantile(Volume, 0.75, na.rm = TRUE),
    Volume_Regime = if_else(
      Volume >= Volume_Threshold,
      "High Volume Days",
      "Normal Volume Days"
    ),
    Absolute_Return = abs(Daily_Return)
  ) |>
  ungroup() |>
  group_by(Asset, Volume_Regime) |>
  summarise(
    Observations = n(),
    Mean_Daily_Return = mean(Daily_Return, na.rm = TRUE),
    Mean_Absolute_Return = mean(Absolute_Return, na.rm = TRUE),
    Volatility = sd(Daily_Return, na.rm = TRUE),
    Average_Volume = mean(Volume, na.rm = TRUE),
    .groups = "drop"
  ) |>
  arrange(Asset, Volume_Regime)

safe_kable(q7, digits=4, caption="Q7 Data Output")
```

Table 13: Q7 Data Output

Asset	Volume_Regime	Observations	Mean_Daily_Return	Mean_Absolute_Return	Volatility	Average_Volume
AAPL	High Volume Days	377	0.0009	0.0185	0.0282	152425015
AAPL	Normal Volume Days	1131	0.0008	0.0078	0.0117	61908113
BDO	High Volume Days	367	0.0019	0.0175	0.0269	6581172
BDO	Normal Volume Days	1101	0.0010	0.0101	0.0152	2550776
BTC	High Volume Days	548	0.0017	0.0305	0.0450	65153421094
BTC	Normal Volume Days	1644	0.0016	0.0183	0.0261	26925230316
SPY	High Volume Days	377	-0.0020	0.0133	0.0196	129389767
SPY	Normal Volume Days	1131	0.0013	0.0046	0.0068	63707966

```
# Write LaTeX table to file
sink("../reports/sections/q7_table.tex")
cat("\\begin{table}[H]\n\\centering\n")
cat("\\caption{Q7 Data Output}\n")
cat("\\label{tab:q7}\n")
cat("\\begin{tabular}{llrrrrr}\n")
cat("\\toprule\n")
cat("Asset & Volume\\_Regime & Obs & Mean\\_Daily\\_Return &
↪  Mean\\_Abs\\_Return & Volatility & Avg\\_Volume \\\\
")
cat("\\midrule\n")
for (i in 1:nrow(q7)) {
  cat(sprintf("%s & %s & %d & %.4f & %.4f & %.4f & %.0f \\\\
",
    q7$Asset[i], q7$Volume_Regime[i], q7$Observations[i],
    q7$Mean_Daily_Return[i], q7$Mean_Absolute_Return[i],
    q7$Volatility[i], q7$Average_Volume[i]))
}
cat("\\bottomrule\n\\end{tabular}\n\\end{table}\n")
sink()
```

Appendix A14 - SQL Question 8: VIX Regime Analysis

Author: SISON, Aaron Joshua Estacio Assesses asset performance during periods of high versus low market fear by segregating trading days based on VIX levels.

```
# Q8: VIX Regime Analysis -- SISON, Aaron Joshua Estacio

q8_raw <- fin_data |>
  filter(!is.na(VIX)) |>
  select(Asset, Date, Daily_Return, VIX) |>
  collect() |>
```

```r
  mutate(Date = as.Date(Date))

q8 <- q8_raw |>
  mutate(
    VIX_Regime = case_when(
      VIX > 30 ~ "High Volatility (VIX > 30)",
      VIX < 20 ~ "Low Volatility (VIX < 20)",
      TRUE ~ "Moderate (20 <= VIX <= 30)"
    )
  ) |>
  group_by(Asset, VIX_Regime) |>
  summarise(
    Mean_Daily_Return = mean(Daily_Return, na.rm = TRUE),
    SD_Daily_Return = sd(Daily_Return, na.rm = TRUE),
    Observations = n(),
    .groups = "drop"
  ) |>
  arrange(Asset, VIX_Regime)

q8_plot <- q8 |>
  filter(VIX_Regime %in% c("High Volatility (VIX > 30)", "Low Volatility
  ↪  (VIX < 20)")) |>
  mutate(VIX_Regime = recode(VIX_Regime,
    "High Volatility (VIX > 30)" = "VIX > 30 (High Fear)",
    "Low Volatility (VIX < 20)" = "VIX < 20 (Low Fear)"))

safe_kable(q8, digits=4, caption="Q8 Data Output")
```

Table 14: Q8 Data Output

Asset	VIX_Regime	Mean_Daily_Return	SD_Daily_Return	Observations
AAPL	High Volatility (VIX > 30)	-0.0034	0.0283	206
AAPL	Low Volatility (VIX < 20)	0.0013	0.0096	1221
AAPL	Moderate (20 <= VIX <= 30)	0.0004	0.0151	765
BDO	High Volatility (VIX > 30)	0.0014	0.0263	206
BDO	Low Volatility (VIX < 20)	0.0005	0.0131	1221
BDO	Moderate (20 <= VIX <= 30)	0.0011	0.0149	765
BTC	High Volatility (VIX > 30)	-0.0016	0.0495	206
BTC	Low Volatility (VIX < 20)	0.0022	0.0261	1221
BTC	Moderate (20 <= VIX <= 30)	0.0016	0.0342	765
SPY	High Volatility (VIX > 30)	-0.0024	0.0220	206
SPY	Low Volatility (VIX < 20)	0.0009	0.0052	1221
SPY	Moderate (20 <= VIX <= 30)	0.0002	0.0092	765

```r
# Write LaTeX table to file
sink("../reports/sections/q8_table.tex")
```

```
cat("\\begin{table}[H]\n\\centering\n")
cat("\\caption{Q8 Data Output}\n")
cat("\\label{tab:q8}\n")
cat("\\begin{tabular}{llrrr}\n")
cat("\\toprule\n")
cat("Asset & VIX\\_Regime & Mean\\_Daily\\_Return & SD\\_Daily\\_Return &
↪  Obs \\\\
")
cat("\\midrule\n")
for (i in 1:nrow(q8)) {
  cat(sprintf("%s & %s & %.4f & %.4f & %d \\\\
",
    q8$Asset[i], q8$VIX_Regime[i], q8$Mean_Daily_Return[i],
    q8$SD_Daily_Return[i], q8$Observations[i]))
}
cat("\\bottomrule\n\\end{tabular}\n\\end{table}\n")
sink()
```

Appendix A15 - SQL Question 9: Interest Rate Sensitivity

Author: SISON, Aaron Joshua Estacio Measures asset sensitivity to changes in the 10-year Treasury yield by comparing mean returns on days when yields rise versus fall.

```
# Q9: Interest Rate Sensitivity -- SISON, Aaron Joshua Estacio

q9_raw <- fin_data |>
  filter(!is.na(DGS10)) |>
  select(Asset, Date, Daily_Return, DGS10) |>
  collect() |>
  mutate(Date = as.Date(Date))

q9 <- q9_raw |>
  group_by(Asset) |>
  arrange(Date) |>
  mutate(DGS10_Change = DGS10 - lag(DGS10)) |>
  ungroup() |>
  filter(!is.na(DGS10_Change)) |>
  mutate(
    Yield_Direction = case_when(
      DGS10_Change > 0 ~ "Yields Rising",
      DGS10_Change < 0 ~ "Yields Falling",
      TRUE ~ "No Change"
```

```r
    )
  ) |>
  group_by(Asset, Yield_Direction) |>
  summarise(
    Mean_Daily_Return = mean(Daily_Return, na.rm = TRUE),
    SD_Daily_Return = sd(Daily_Return, na.rm = TRUE),
    Observations = n(),
    .groups = "drop"
  ) |>
  arrange(Asset, Yield_Direction)

q9_plot <- q9 |>
  filter(Yield_Direction %in% c("Yields Rising", "Yields Falling"))

safe_kable(q9, digits=4, caption="Q9 Data Output")
```

Table 15: Q9 Data Output

Asset	Yield_Direction	Mean_Daily_Return	SD_Daily_Return	Observations
AAPL	No Change	0.0002	0.0068	795
AAPL	Yields Falling	0.0004	0.0170	684
AAPL	Yields Rising	0.0011	0.0176	712
BDO	No Change	0.0001	0.0058	795
BDO	Yields Falling	0.0012	0.0203	684
BDO	Yields Rising	0.0013	0.0172	712
BTC	No Change	0.0013	0.0248	795
BTC	Yields Falling	0.0017	0.0343	684
BTC	Yields Rising	0.0020	0.0363	712
SPY	No Change	0.0000	0.0035	795
SPY	Yields Falling	0.0001	0.0109	684
SPY	Yields Rising	0.0010	0.0123	712

```r
# Write LaTeX table to file
sink("../reports/sections/q9_table.tex")
cat("\\begin{table}[H]\n\\centering\n")
cat("\\caption{Q9 Data Output}\n")
cat("\\label{tab:q9}\n")
cat("\\begin{tabular}{llrrr}\n")
cat("\\toprule\n")
cat("Asset & Yield\\_Direction & Mean\\_Daily\\_Return & SD\\_Daily\\_Return
↪  & Obs \\\\
")
cat("\\midrule\n")
for (i in 1:nrow(q9)) {
  cat(sprintf("%s & %s & %.4f & %.4f & %d \\\\
",
```

```
    q9$Asset[i], q9$Yield_Direction[i], q9$Mean_Daily_Return[i],
    q9$SD_Daily_Return[i], q9$Observations[i]))
}
cat("\\bottomrule\n\\end{tabular}\n\\end{table}\n")
sink()
```

Appendix A16 - SQL Question 10: Inflation Hedge Performance

Author: SISON, Aaron Joshua Estacio Evaluates which assets served as effective inflation hedges during the June 2021 to February 2023 high-inflation period (CPI YoY above 5%).

```
# Q10: Inflation Hedge Performance -- SISON, Aaron Joshua Estacio

q10_raw <- fin_data |>
  select(Date, Asset, Daily_Return, CPI_YoY) |>
  collect() |>
  mutate(Date = as.Date(Date))

q10 <- q10_raw |>
  filter(!is.na(CPI_YoY) & CPI_YoY > 0.05) |>
  filter(!is.na(Daily_Return)) |>
  group_by(Asset) |>
  arrange(Date) |>
  mutate(Cumulative_Return = cumprod(1 + Daily_Return) - 1) |>
  summarise(
    Start_Date = min(Date),
    End_Date = max(Date),
    Start_CPI_YoY = first(CPI_YoY),
    End_CPI_YoY = last(CPI_YoY),
    Total_Cumulative_Return = last(Cumulative_Return),
    Mean_Daily_Return = mean(Daily_Return, na.rm = TRUE),
    Observations = n()
  ) |>
  arrange(desc(Total_Cumulative_Return))

q10_plot <- q10 |>
  mutate(Asset = factor(Asset, levels =
   ↪  q10$Asset[order(q10$Total_Cumulative_Return)]))

safe_kable(q10, digits=4, caption="Q10 Data Output")
```

Table 16: Q10 Data Output

Asset	Start_Date	End_Date	Start_CPI_YoY	End_CPI_YoY	Total_Cumulative_Return	Mean_Daily_Return	Observations
AAPL	2021-06-01	2023-02-28	0.053	0.0596	0.2778	0.0005	638
BDO	2021-06-01	2023-02-28	0.053	0.0596	0.2481	0.0004	638
SPY	2021-06-01	2023-02-28	0.053	0.0596	0.1130	0.0002	638
BTC	2021-06-01	2023-02-28	0.053	0.0596	-0.3800	-0.0002	638

```
# Write LaTeX table to file
sink("../reports/sections/q10_table.tex")
cat("\\begin{table}[H]\n\\centering\n")
cat("\\caption{Q10 Data Output}\n")
cat("\\label{tab:q10}\n")
cat("\\begin{tabular}{lllrrrrr}\n")
cat("\\toprule\n")
cat("Asset & Start\\_Date & End\\_Date & Start\\_CPI\\_YoY & End\\_CPI\\_YoY
↪  & Total\\_Cumulative\\_Return & Mean\\_Daily\\_Return & Obs \\\\
")
cat("\\midrule\n")
for (i in 1:nrow(q10)) {
  cat(sprintf("%s & %s & %s & %.4f & %.4f & %.4f & %.4f & %d \\\\
",
    q10$Asset[i], q10$Start_Date[i], q10$End_Date[i],
    q10$Start_CPI_YoY[i], q10$End_CPI_YoY[i],
    q10$Total_Cumulative_Return[i], q10$Mean_Daily_Return[i],
    q10$Observations[i]))
}
cat("\\bottomrule\n\\end{tabular}\n\\end{table}\n")
sink()
```

Appendix A17 - Data Visualization Charts

Author: SEBALLOS, Josiah Dweyn Panganiban Contains the ggplot code for all 10 figures corresponding to SQL Questions Q1 through Q10. Each figure is saved as a PDF to the reports/figures/ directory for inclusion in the main report's Data Visualization section.

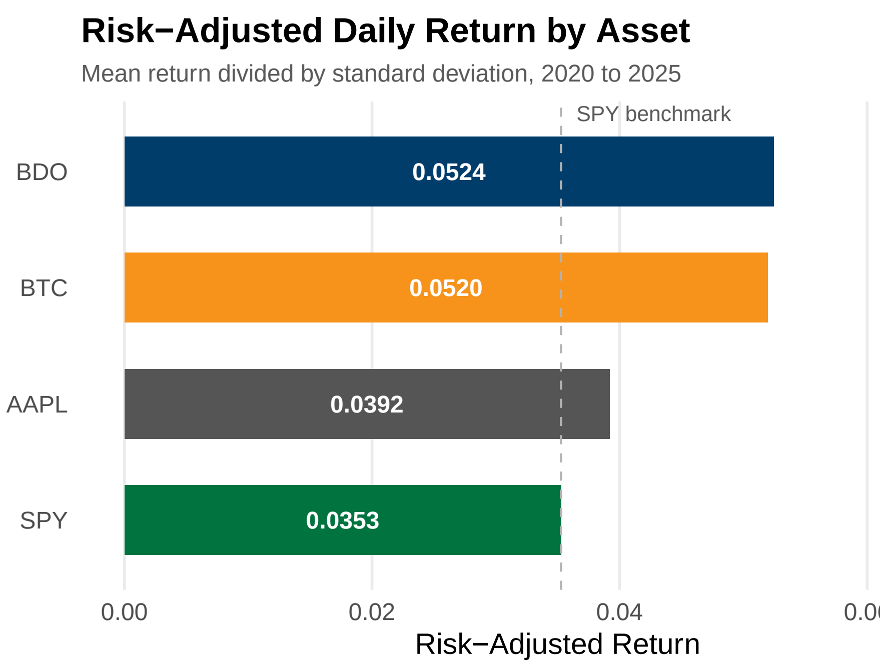
```
# Create figures directory if it does not exist
dir.create("../reports/figures", recursive = TRUE, showWarnings = FALSE)
```

Figure 1: Risk-Adjusted Daily Return by Asset

Bar chart showing the ratio of mean daily return to standard deviation for each asset. The dashed reference line marks SPY's ratio.

```
ggplot(q1, aes(x = Risk_Adjusted_Return, y = reorder(Asset,
↪  Risk_Adjusted_Return),
```

```r
                fill = Asset)) +
  geom_col(width = 0.6, show.legend = FALSE) +
  geom_vline(xintercept = spy_ratio, linetype = "dashed",
             color = "#B3B3B3", linewidth = 0.4) +
  geom_text(aes(x = Risk_Adjusted_Return / 2,
                label = sprintf("%.4f", Risk_Adjusted_Return)),
            hjust = 0.5, size = 3.2, color = "white", fontface = "bold") +
  annotate("text", x = spy_ratio, y = 4.5, label = "SPY benchmark",
           size = 2.8, color = "#595959", hjust = -0.1) +
  scale_fill_manual(values = c(AAPL = "#555555", BDO = "#003D6B",
                               BTC = "#F7931A", SPY = "#00733E")) +
  labs(title = "Risk-Adjusted Daily Return by Asset",
       subtitle = "Mean return divided by standard deviation, 2020 to 2025",
       x = "Risk-Adjusted Return", y = NULL) +
  xlim(0, 0.07) +
  theme_minimal(base_size = 11) +
  theme(panel.grid.major.y = element_blank(),
        panel.grid.minor = element_blank(),
        axis.ticks = element_blank(),
        plot.title = element_text(face = "bold", size = 13),
        plot.subtitle = element_text(size = 9, color = "#595959"))
```

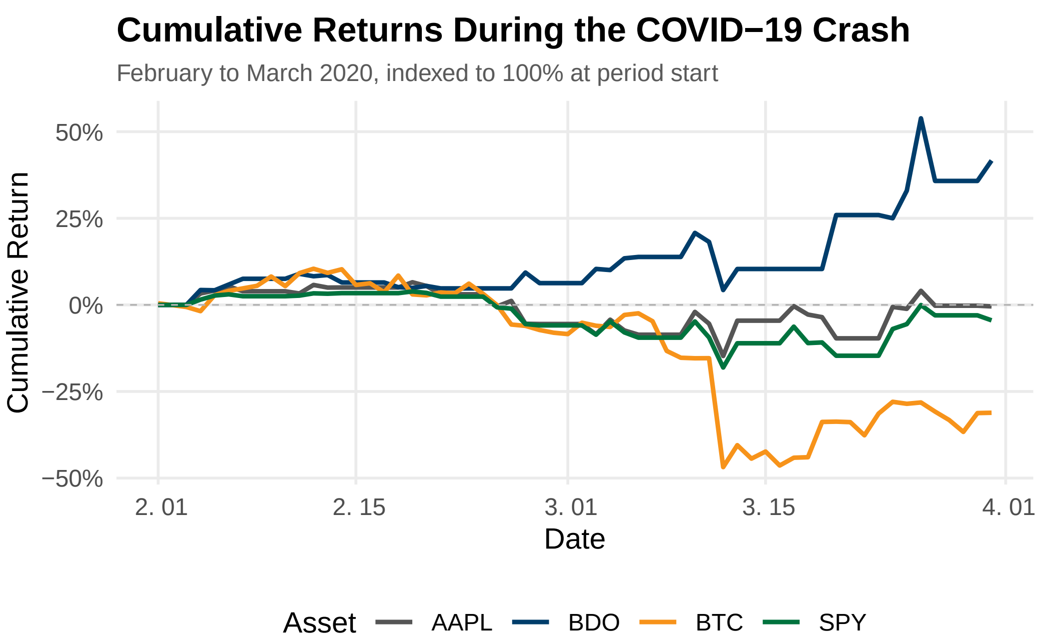



```r
ggsave("../reports/figures/fig_q1.pdf", width = 7, height = 3.5, device =
↪  "pdf")
```

Figure 2: Cumulative Returns During the COVID-19 Crash

Line chart tracking each asset's cumulative return from February to March 2020, indexed to 100% at period start.

```r
ggplot(q2, aes(x = Date, y = Cumulative_Return, color = Asset)) +
  geom_line(linewidth = 0.8) +
  geom_hline(yintercept = 0, linetype = "dashed", color = "#B3B3B3",
  ↪   linewidth = 0.3) +
  scale_color_manual(values = c(AAPL = "#555555", BDO = "#003D6B",
                               BTC = "#F7931A", SPY = "#00733E")) +
  scale_y_continuous(labels = scales::percent_format()) +
  labs(title = "Cumulative Returns During the COVID-19 Crash",
       subtitle = "February to March 2020, indexed to 100% at period start",
       x = "Date", y = "Cumulative Return", color = "Asset") +
  theme_minimal(base_size = 11) +
  theme(panel.grid.minor = element_blank(),
        axis.ticks = element_blank(),
        plot.title = element_text(face = "bold", size = 13),
        plot.subtitle = element_text(size = 9, color = "#595959"),
        legend.position = "bottom")
```

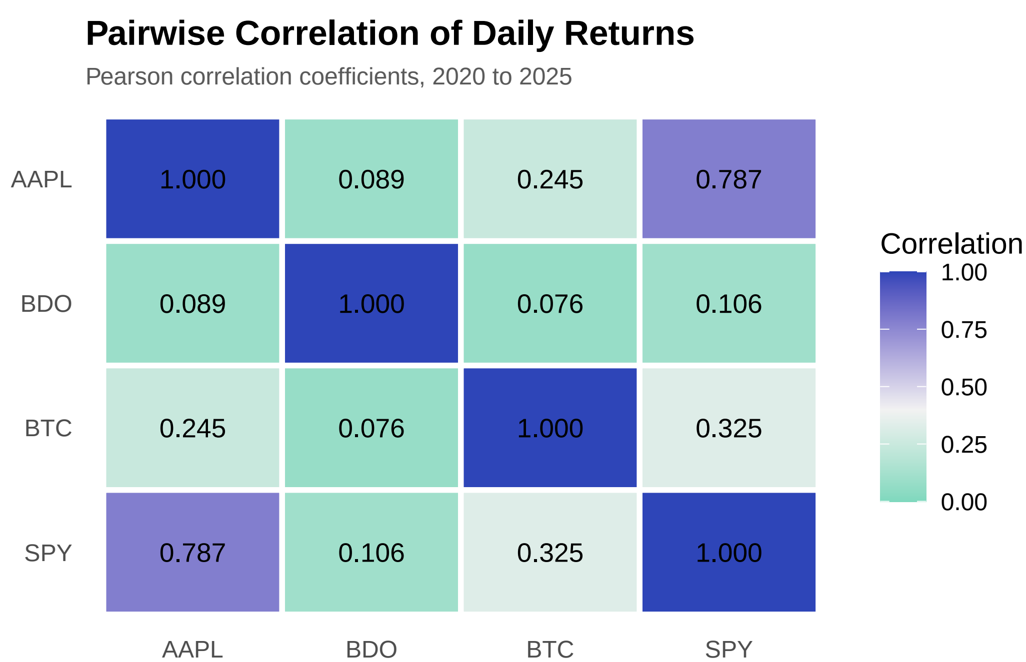



```r
ggsave("../reports/figures/fig_q2.pdf", width = 7, height = 4, device =
↪  "pdf")
```

Figure 3: Pairwise Correlation Heatmap

Heatmap matrix of Pearson correlation coefficients for daily returns across the four assets.

```r
ggplot(cor_long, aes(x = Asset2, y = Asset1, fill = Correlation)) +
  geom_tile(color = "white", linewidth = 1) +
  geom_text(aes(label = sprintf("%.3f", Correlation)), size = 3.5) +
  scale_fill_gradient2(low = "#1DC9A4", mid = "#F2F2F2", high = "#2E45B8",
                       midpoint = 0.4, limits = c(0, 1)) +
  labs(title = "Pairwise Correlation of Daily Returns",
       subtitle = "Pearson correlation coefficients, 2020 to 2025",
       x = NULL, y = NULL, fill = "Correlation") +
  theme_minimal(base_size = 11) +
  theme(panel.grid = element_blank(),
        axis.ticks = element_blank(),
        plot.title = element_text(face = "bold", size = 13),
        plot.subtitle = element_text(size = 9, color = "#595959"),
        legend.position = "right")
```



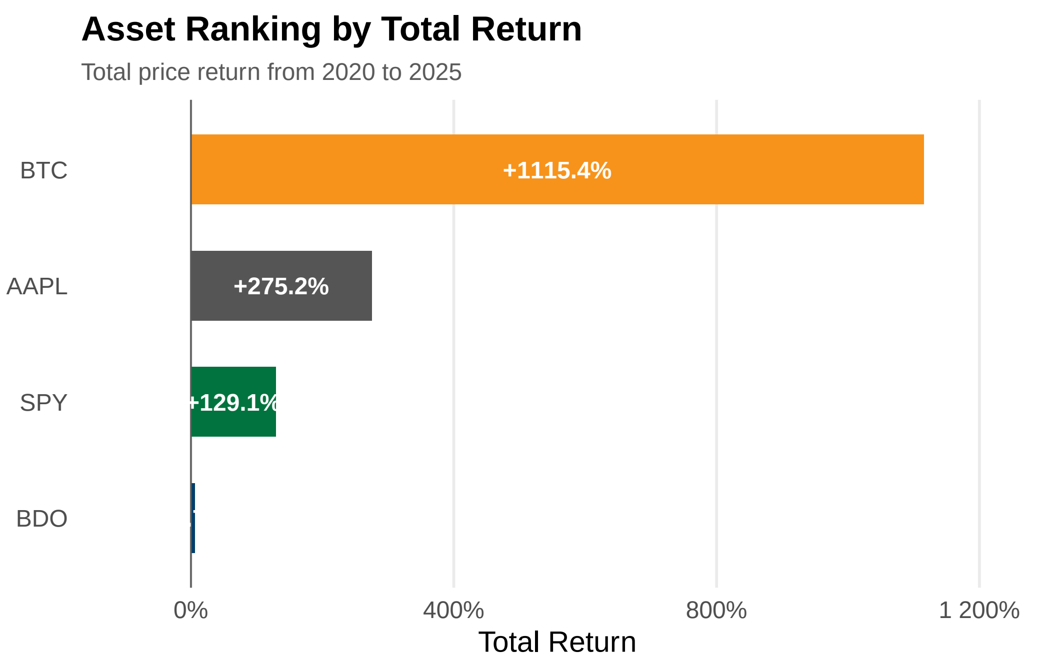

```r
ggsave("../reports/figures/fig_q3.pdf", width = 6, height = 4.5, device = "pdf")
```

Figure 4: Asset Ranking by Total Return

Horizontal bar chart ranking assets by total percentage price return from 2020 to 2025.

```r
ggplot(q4_plot, aes(x = Total_Return, y = Asset, fill = Asset)) +
  geom_col(width = 0.6, show.legend = FALSE) +
  geom_vline(xintercept = 0, linewidth = 0.3, color = "#595959") +
  geom_text(aes(x = Total_Return / 2,
                label = sprintf("%+.1f%%", Total_Return * 100)),
```

```
           hjust = 0.5, size = 3.2, color = "white", fontface = "bold") +
  scale_fill_manual(values = c(AAPL = "#555555", BDO = "#003D6B",
                               BTC = "#F7931A", SPY = "#00733E")) +
  scale_x_continuous(labels = scales::percent_format(),
                     expand = expansion(mult = c(0.15, 0.15))) +
  labs(title = "Asset Ranking by Total Return",
       subtitle = "Total price return from 2020 to 2025",
       x = "Total Return", y = NULL) +
  theme_minimal(base_size = 11) +
  theme(panel.grid.major.y = element_blank(),
        panel.grid.minor = element_blank(),
        axis.ticks = element_blank(),
        plot.title = element_text(face = "bold", size = 13),
        plot.subtitle = element_text(size = 9, color = "#595959"))
```




```
ggsave("../reports/figures/fig_q4.pdf", width = 7, height = 3.5, device =
↪  "pdf")
```

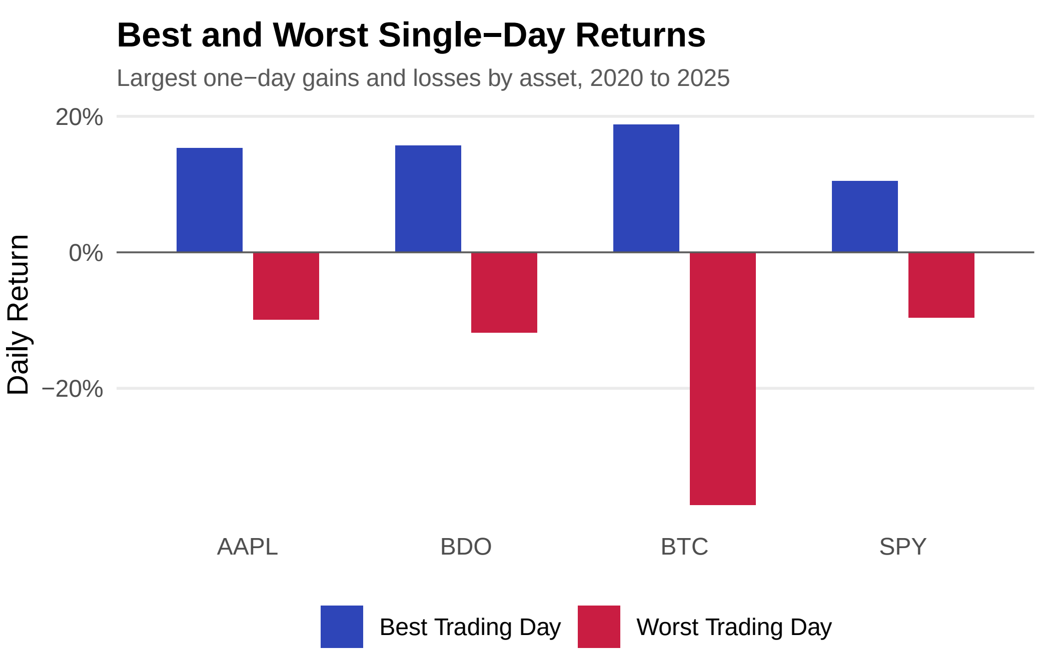
Figure 5: Best and Worst Single-Day Returns

Grouped bar chart comparing the largest single-day gain and loss for each asset.

```
q5_plot <- q5 |>
  select(Asset, Best_Daily_Return, Worst_Daily_Return) |>
  pivot_longer(cols = c(Best_Daily_Return, Worst_Daily_Return),
               names_to = "Trading_Day_Type",
               values_to = "Daily_Return") |>
```

```r
  mutate(
    Trading_Day_Type = recode(Trading_Day_Type,
                              Best_Daily_Return = "Best Trading Day",
                              Worst_Daily_Return = "Worst Trading Day")
  )

ggplot(q5_plot, aes(x = Asset, y = Daily_Return, fill = Trading_Day_Type)) +
  geom_col(position = position_dodge(width = 0.7), width = 0.6) +
  geom_hline(yintercept = 0, linewidth = 0.3, color = "#595959") +
  scale_fill_manual(values = c("Best Trading Day" = "#2E45B8",
                               "Worst Trading Day" = "#C91D42")) +
  scale_y_continuous(labels = scales::percent_format()) +
  labs(title = "Best and Worst Single-Day Returns",
       subtitle = "Largest one-day gains and losses by asset, 2020 to 2025",
       x = NULL, y = "Daily Return", fill = NULL) +
  theme_minimal(base_size = 11) +
  theme(panel.grid.major.x = element_blank(),
        panel.grid.minor = element_blank(),
        axis.ticks = element_blank(),
        plot.title = element_text(face = "bold", size = 13),
        plot.subtitle = element_text(size = 9, color = "#595959"),
        legend.position = "bottom")
```



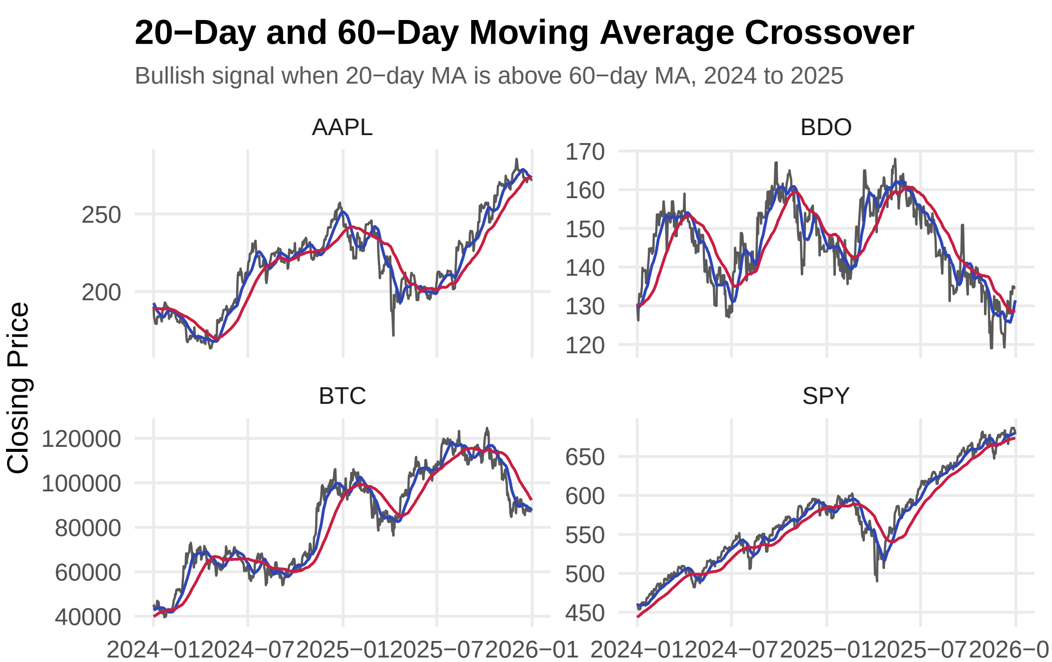

```r
ggsave("../reports/figures/fig_q5.pdf", width = 7, height = 4, device =
↪  "pdf")
```

Figure 6: 20-Day and 60-Day Moving Average Crossover

Multi-panel faceted line chart showing closing prices with 20-day and 60-day moving averages from 2024 to 2025.

```
q6_plot <- q6_ma |>
  filter(Date >= as.Date("2024-01-01"))

ggplot(q6_plot, aes(x = Date)) +
  geom_line(aes(y = Close), linewidth = 0.4, color = "#595959") +
  geom_line(aes(y = MA_20), linewidth = 0.5, color = "#2E45B8") +
  geom_line(aes(y = MA_60), linewidth = 0.5, color = "#C91D42") +
  facet_wrap(~ Asset, scales = "free_y") +
  labs(title = "20-Day and 60-Day Moving Average Crossover",
       subtitle = "Bullish signal when 20-day MA is above 60-day MA, 2024 to
       ↪  2025",
       x = NULL, y = "Closing Price") +
  theme_minimal(base_size = 11) +
  theme(panel.grid.minor = element_blank(),
        axis.ticks = element_blank(),
        plot.title = element_text(face = "bold", size = 13),
        plot.subtitle = element_text(size = 9, color = "#595959"))
```

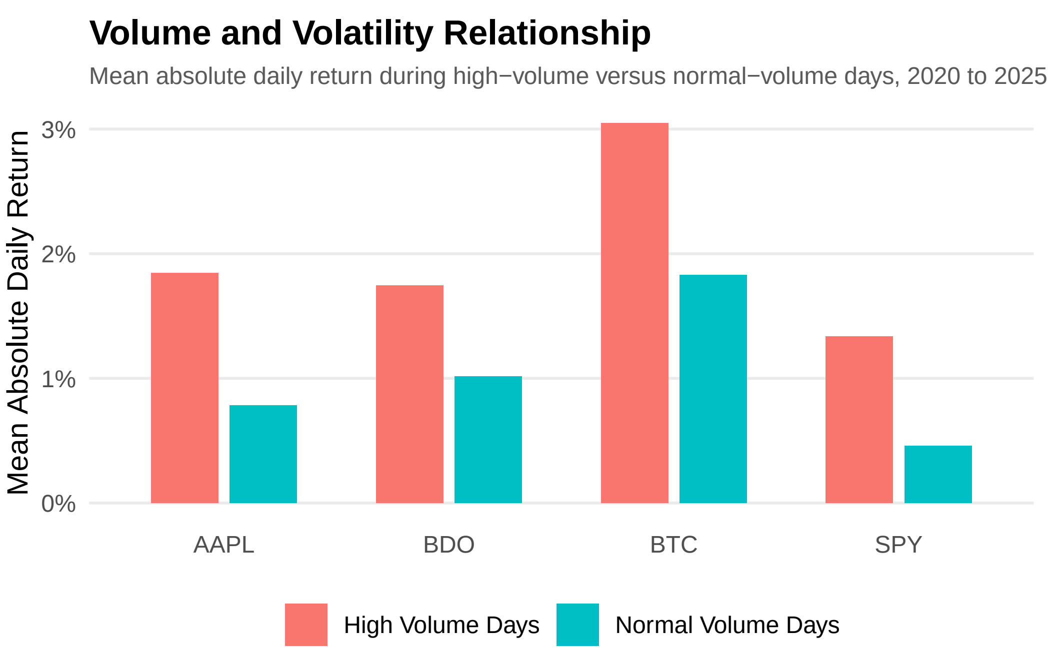



```
ggsave("../reports/figures/fig_q6.pdf", width = 7, height = 4.5, device =
↪  "pdf")
```

Figure 7: Volume and Volatility Relationship

Grouped bar chart comparing mean absolute daily return during high-volume versus normal-volume days.

```
ggplot(q7, aes(x = Asset, y = Mean_Absolute_Return, fill = Volume_Regime)) +
  geom_col(position = position_dodge(width = 0.7), width = 0.6) +
  scale_y_continuous(labels = scales::percent_format()) +
  labs(
    title = "Volume and Volatility Relationship",
    subtitle = "Mean absolute daily return during high-volume versus
     ↪  normal-volume days, 2020 to 2025",
    x = NULL,
    y = "Mean Absolute Daily Return",
    fill = NULL
  ) +
  theme_minimal(base_size = 11) +
  theme(
    panel.grid.major.x = element_blank(),
    panel.grid.minor = element_blank(),
    axis.ticks = element_blank(),
    plot.title = element_text(face = "bold", size = 13),
    plot.subtitle = element_text(size = 9, color = "#595959"),
    legend.position = "bottom"
  )
```

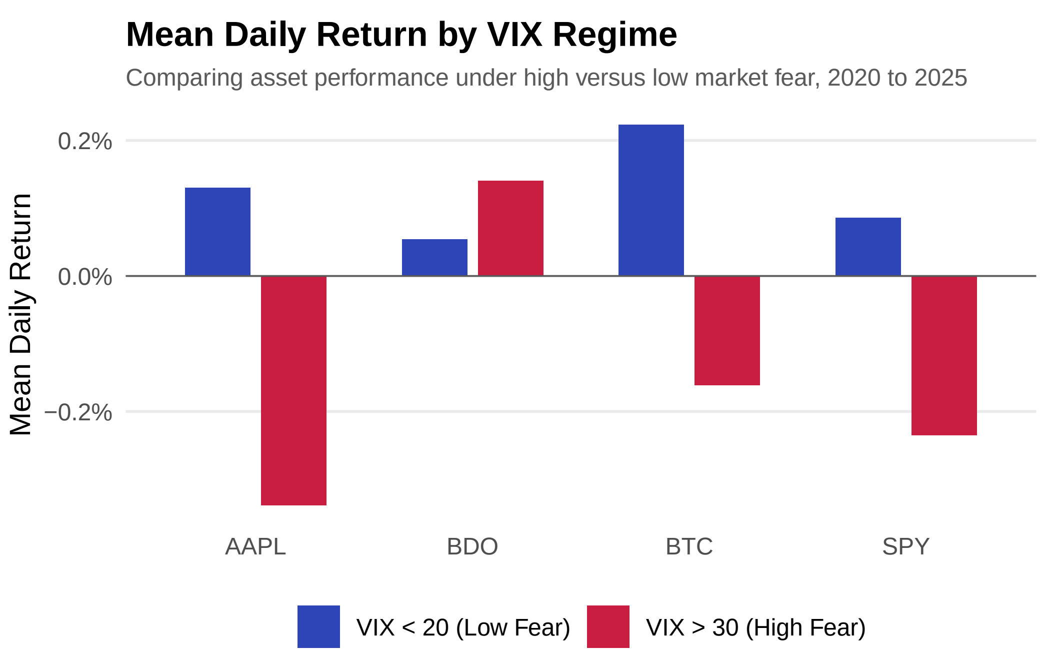



```
ggsave("../reports/figures/fig_q7.pdf", width = 7, height = 4, device =
↪  "pdf")
```

Figure 8: Mean Daily Return by VIX Regime

Grouped bar chart comparing asset returns during high-fear (VIX > 30) versus low-fear (VIX < 20) regimes.

```
ggplot(q8_plot, aes(x = Asset, y = Mean_Daily_Return, fill = VIX_Regime)) +
  geom_col(position = position_dodge(width = 0.7), width = 0.6) +
  geom_hline(yintercept = 0, linewidth = 0.3, color = "#595959") +
  scale_fill_manual(values = c("VIX > 30 (High Fear)" = "#C91D42",
                               "VIX < 20 (Low Fear)" = "#2E45B8")) +
  scale_y_continuous(labels = scales::percent_format()) +
  labs(title = "Mean Daily Return by VIX Regime",
       subtitle = "Comparing asset performance under high versus low market
       ↪  fear, 2020 to 2025",
       x = NULL, y = "Mean Daily Return", fill = NULL) +
  theme_minimal(base_size = 11) +
  theme(panel.grid.major.x = element_blank(),
        panel.grid.minor = element_blank(),
        axis.ticks = element_blank(),
        plot.title = element_text(face = "bold", size = 13),
        plot.subtitle = element_text(size = 9, color = "#595959"),
        legend.position = "bottom")
```

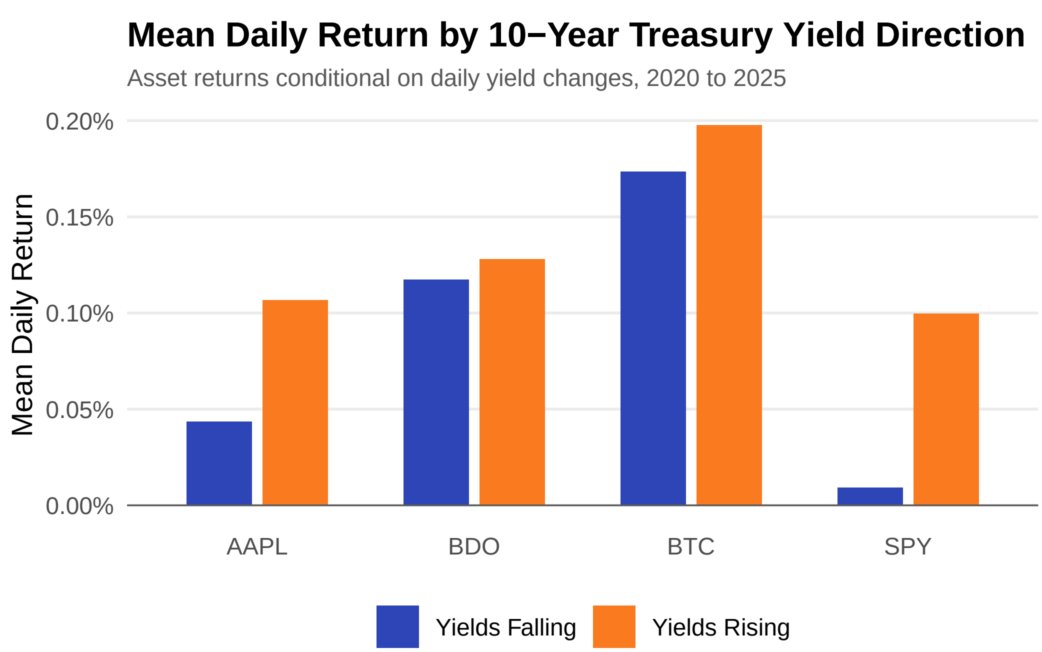



```
ggsave("../reports/figures/fig_q8.pdf", width = 7, height = 4, device =
 ↪  "pdf")
```

Figure 9: Mean Daily Return by 10-Year Treasury Yield Direction

Grouped bar chart showing asset returns conditional on whether daily yields are rising or falling.

```
ggplot(q9_plot, aes(x = Asset, y = Mean_Daily_Return, fill =
↪  Yield_Direction)) +
  geom_col(position = position_dodge(width = 0.7), width = 0.6) +
  geom_hline(yintercept = 0, linewidth = 0.3, color = "#595959") +
  scale_fill_manual(values = c("Yields Rising" = "#F97A1F",
                               "Yields Falling" = "#2E45B8")) +
  scale_y_continuous(labels = scales::percent_format()) +
  labs(title = "Mean Daily Return by 10-Year Treasury Yield Direction",
       subtitle = "Asset returns conditional on daily yield changes, 2020 to
        ↪  2025",
       x = NULL, y = "Mean Daily Return", fill = NULL) +
  theme_minimal(base_size = 11) +
  theme(panel.grid.major.x = element_blank(),
        panel.grid.minor = element_blank(),
        axis.ticks = element_blank(),
        plot.title = element_text(face = "bold", size = 13),
        plot.subtitle = element_text(size = 9, color = "#595959"),
        legend.position = "bottom")
```

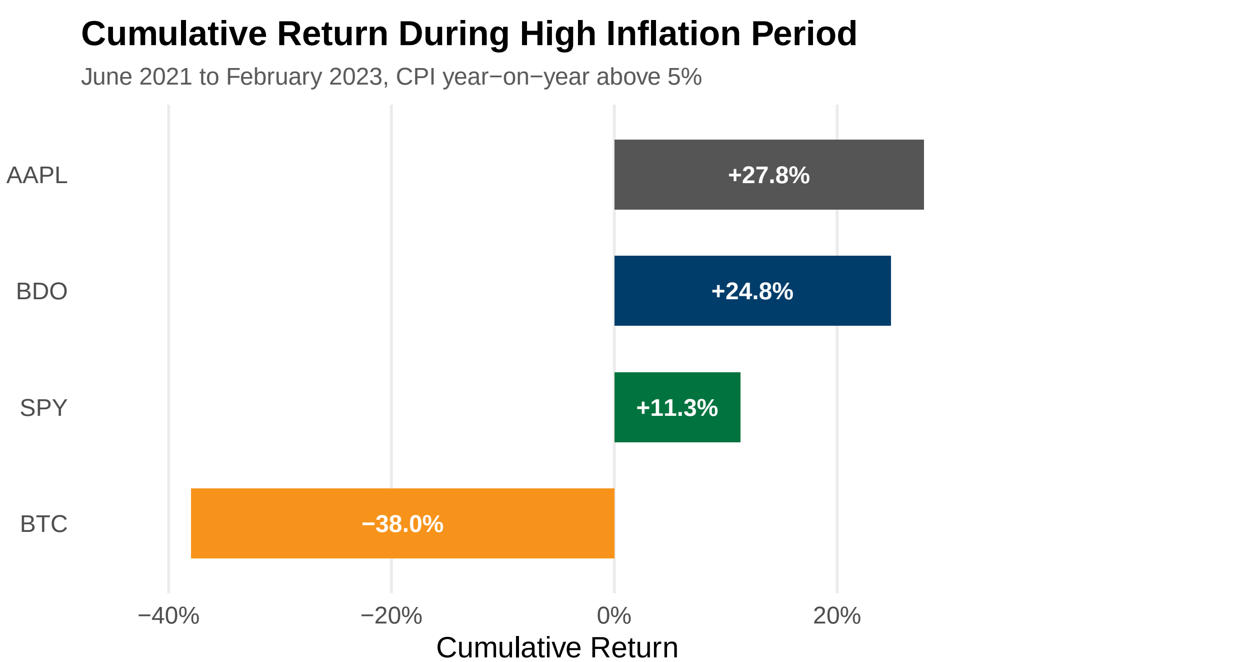



```
ggsave("../reports/figures/fig_q9.pdf", width = 7, height = 4, device =
↪  "pdf")
```

Figure 10: Cumulative Return During High Inflation Period

Horizontal bar chart showing total cumulative returns during the period when CPI year-on-year exceeded 5%.

```r
ggplot(q10_plot, aes(x = Total_Cumulative_Return, y = Asset, fill = Asset))
↪  +
  geom_col(width = 0.6, show.legend = FALSE) +
  geom_text(aes(x = Total_Cumulative_Return / 2,
                label = sprintf("%+.1f%%", Total_Cumulative_Return * 100)),
            hjust = 0.5, size = 3.2, color = "white", fontface = "bold") +
  scale_fill_manual(values = c(AAPL = "#555555", BDO = "#003D6B",
                               BTC = "#F7931A", SPY = "#00733E")) +
  scale_x_continuous(labels = scales::percent_format(),
                     expand = expansion(mult = c(0.15, 0.15))) +
  labs(title = "Cumulative Return During High Inflation Period",
       subtitle = "June 2021 to February 2023, CPI year-on-year above 5%",
       x = "Cumulative Return", y = NULL) +
  theme_minimal(base_size = 11) +
  theme(panel.grid.major.y = element_blank(),
        panel.grid.minor = element_blank(),
        axis.ticks = element_blank(),
        plot.title = element_text(face = "bold", size = 13),
        plot.subtitle = element_text(size = 9, color = "#595959"))
```




```r
ggsave("../reports/figures/fig_q10.pdf", width = 7, height = 3.5, device =
↪  "pdf")
```

Appendix A18 - Exploratory Data Visualization

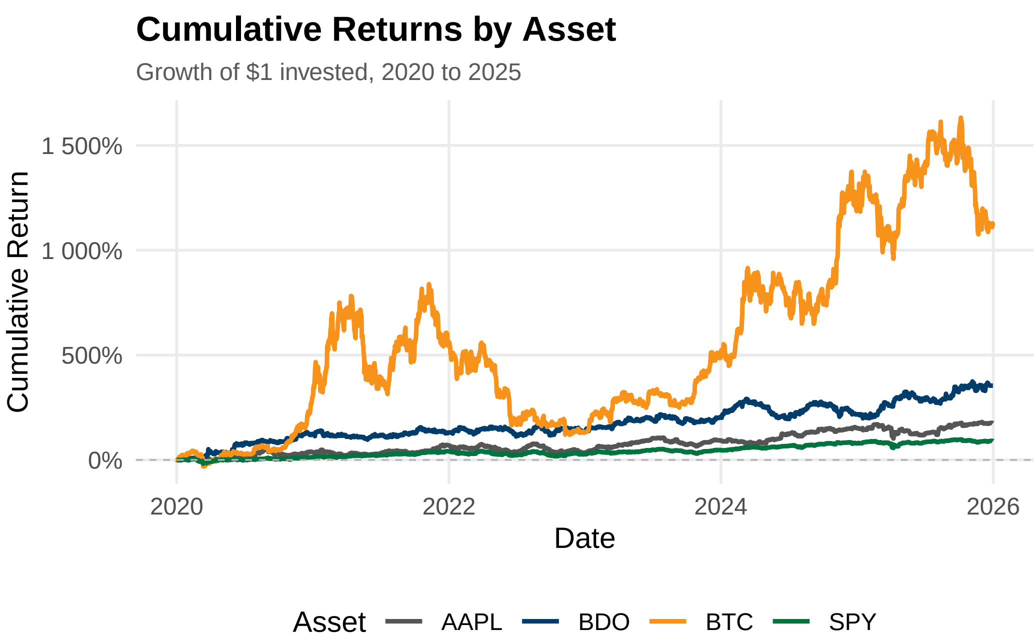
Author: SEBALLOS, Josiah Dweyn Panganiban Contains 8 exploratory figures covering cumulative returns, price trends, return distributions, rolling volatility, drawdown analysis, risk-return profiles, and portfolio risk contribution. These complement the SQL-driven charts in A17.

Viz Figure 1: Cumulative Returns by Asset

Tracks the growth of USD 1 invested in each asset from 2020 to 2025, indexed to 100% at the start.

```r
fig1_data <- sql_long |>
  filter(!is.na(Daily_Return)) |>
  group_by(Asset) |>
  arrange(Date) |>
  mutate(Cumulative_Return = cumprod(1 + Daily_Return) - 1) |>
  ungroup()

ggplot(fig1_data, aes(x = Date, y = Cumulative_Return, color = Asset)) +
  geom_line(linewidth = 0.8) +
  geom_hline(yintercept = 0, linetype = "dashed", color = "#B3B3B3",
  ↪  linewidth = 0.3) +
  scale_color_manual(values = c(AAPL = "#555555", BDO = "#003D6B",
                                BTC = "#F7931A", SPY = "#00733E")) +
  scale_y_continuous(labels = scales::percent_format()) +
  labs(title = "Cumulative Returns by Asset",
       subtitle = "Growth of $1 invested, 2020 to 2025",
       x = "Date", y = "Cumulative Return", color = "Asset") +
  theme_minimal(base_size = 11) +
  theme(panel.grid.minor = element_blank(),
        axis.ticks = element_blank(),
        plot.title = element_text(face = "bold", size = 13),
        plot.subtitle = element_text(size = 9, color = "#595959"),
        legend.position = "bottom")
```

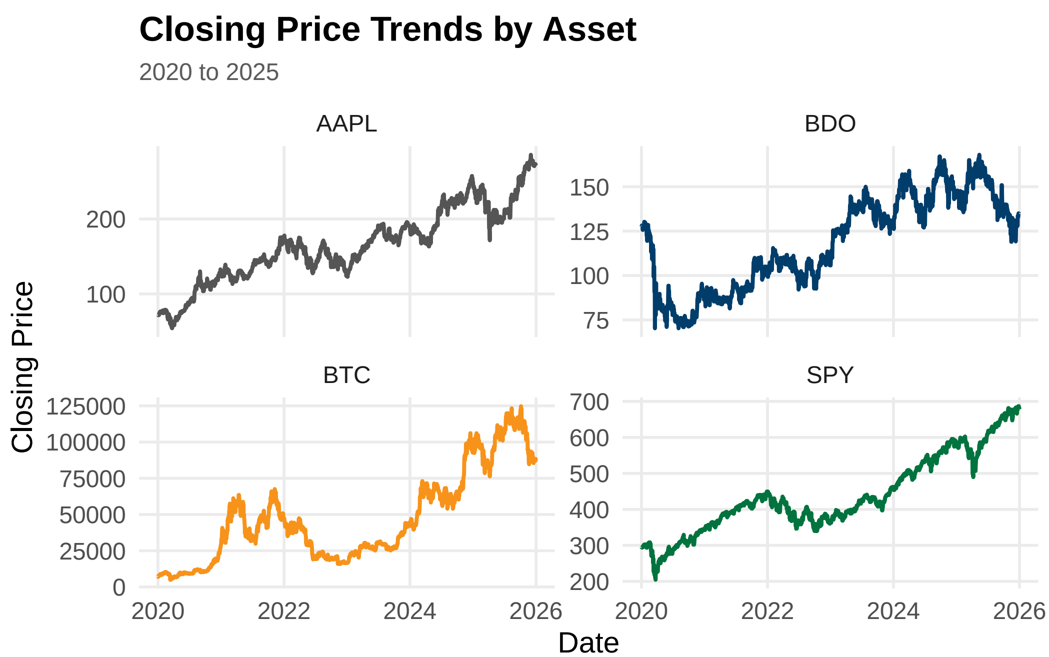



```
ggsave("../reports/figures/fig_viz1.pdf", width = 7, height = 4, device =
↪  "pdf")
```

Viz Figure 2: Closing Price Trends by Asset

Plots nominal closing prices on individual scales to compare price-level dynamics across assets of different magnitudes.

```
ggplot(sql_long, aes(x = Date, y = Close, color = Asset)) +
  geom_line(linewidth = 0.7, show.legend = FALSE, na.rm = TRUE) +
  facet_wrap(~ Asset, scales = "free_y", ncol = 2) +
  scale_color_manual(values = c(AAPL = "#555555", BDO = "#003D6B",
                                BTC = "#F7931A", SPY = "#00733E")) +
  labs(title = "Closing Price Trends by Asset",
       subtitle = "2020 to 2025",
       x = "Date", y = "Closing Price") +
  theme_minimal(base_size = 11) +
  theme(panel.grid.minor = element_blank(),
        axis.ticks = element_blank(),
        plot.title = element_text(face = "bold", size = 13),
        plot.subtitle = element_text(size = 9, color = "#595959"))
```




```
ggsave("../reports/figures/fig_viz2.pdf", width = 7, height = 5, device =
  "pdf")
```

Viz Figure 3: Return Distribution with Skewness and Kurtosis

Overlays each asset's daily return histogram against a fitted normal curve to assess distribution shape and tail risk.

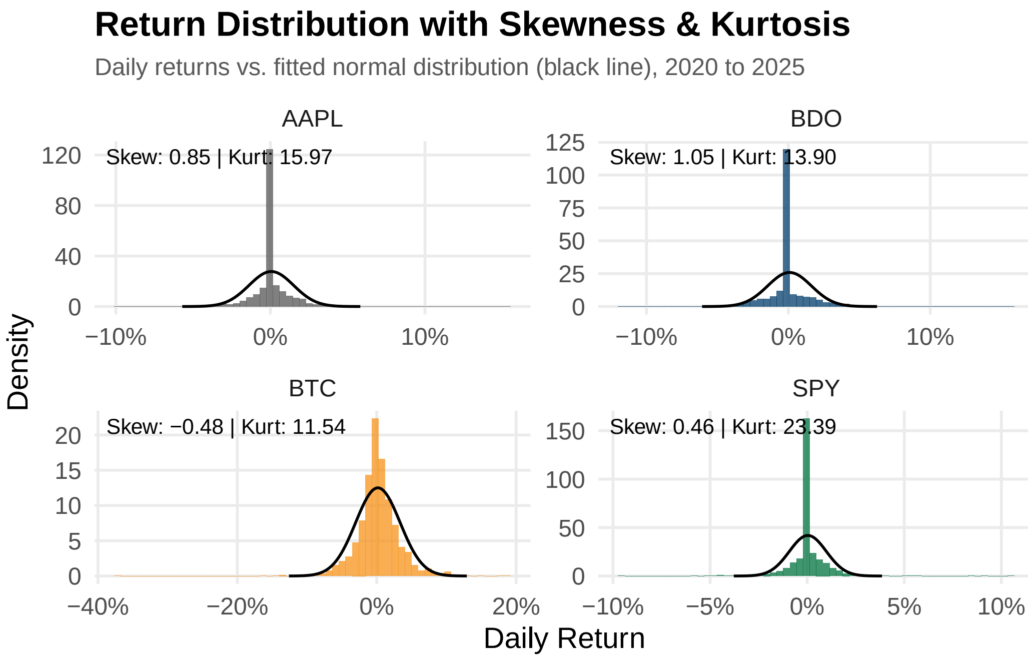
```
fig3_data <- sql_long |> filter(!is.na(Daily_Return))

calc_skew <- function(x) {
  n <- length(x); m <- mean(x)
  (sum((x - m)^3) / n) / (sd(x)^3)
}
calc_kurt <- function(x) {
  n <- length(x); m <- mean(x)
  (sum((x - m)^4) / n) / (sd(x)^2)^2 - 3
}

fig3_stats <- fig3_data |>
  group_by(Asset) |>
  summarise(
    Mean = mean(Daily_Return), SD = sd(Daily_Return),
    Skewness = calc_skew(Daily_Return), Kurtosis = calc_kurt(Daily_Return),
    .groups = "drop"
  ) |>
  mutate(label = sprintf("Skew: %.2f | Kurt: %.2f", Skewness, Kurtosis))
```

```r
normal_curves <- fig3_stats |>
  rowwise() |>
  reframe(
    Asset = Asset,
    x = seq(Mean - 4 * SD, Mean + 4 * SD, length.out = 200),
    y = dnorm(x, mean = Mean, sd = SD)
  )

ggplot(fig3_data, aes(x = Daily_Return)) +
  geom_histogram(aes(y = after_stat(density), fill = Asset), bins = 60,
                 alpha = 0.75, show.legend = FALSE) +
  geom_line(data = normal_curves, aes(x = x, y = y), color = "black",
   ↪  linewidth = 0.5) +
  geom_text(data = fig3_stats, aes(x = -Inf, y = Inf, label = label),
            hjust = -0.05, vjust = 1.5, size = 2.8) +
  facet_wrap(~ Asset, scales = "free", ncol = 2) +
  scale_fill_manual(values = c(AAPL = "#555555", BDO = "#003D6B",
                                BTC = "#F7931A", SPY = "#00733E")) +
  scale_x_continuous(labels = scales::percent_format()) +
  labs(title = "Return Distribution with Skewness & Kurtosis",
       subtitle = "Daily returns vs. fitted normal distribution (black
        ↪  line), 2020 to 2025",
       x = "Daily Return", y = "Density") +
  theme_minimal(base_size = 11) +
  theme(panel.grid.minor = element_blank(),
        axis.ticks = element_blank(),
        plot.title = element_text(face = "bold", size = 13),
        plot.subtitle = element_text(size = 9, color = "#595959"))
```

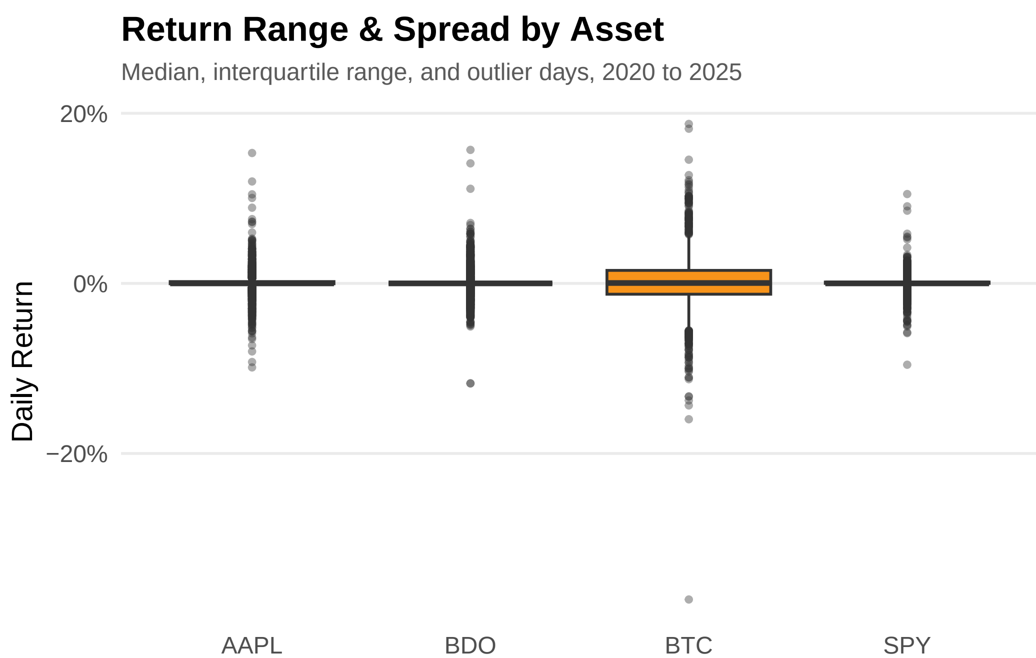



```
ggsave("../reports/figures/fig_viz3.pdf", width = 7, height = 5, device =
↪  "pdf")
```

Viz Figure 4: Return Range and Spread by Asset

Boxplot showing median, interquartile range, and outlier days for each asset.

```
ggplot(fig3_data, aes(x = Asset, y = Daily_Return, fill = Asset)) +
  geom_boxplot(show.legend = FALSE, outlier.size = 0.8, outlier.alpha = 0.4)
  ↪  +
  scale_fill_manual(values = c(AAPL = "#555555", BDO = "#003D6B",
                               BTC = "#F7931A", SPY = "#00733E")) +
  scale_y_continuous(labels = scales::percent_format()) +
  labs(title = "Return Range & Spread by Asset",
       subtitle = "Median, interquartile range, and outlier days, 2020 to
       ↪  2025",
       x = NULL, y = "Daily Return") +
  theme_minimal(base_size = 11) +
  theme(panel.grid.major.x = element_blank(),
        panel.grid.minor = element_blank(),
        axis.ticks = element_blank(),
        plot.title = element_text(face = "bold", size = 13),
        plot.subtitle = element_text(size = 9, color = "#595959"))
```




```
ggsave("../reports/figures/fig_viz4.pdf", width = 7, height = 4, device =
↪  "pdf")
```

Viz Figure 5: 30-Day Rolling Volatility by Asset

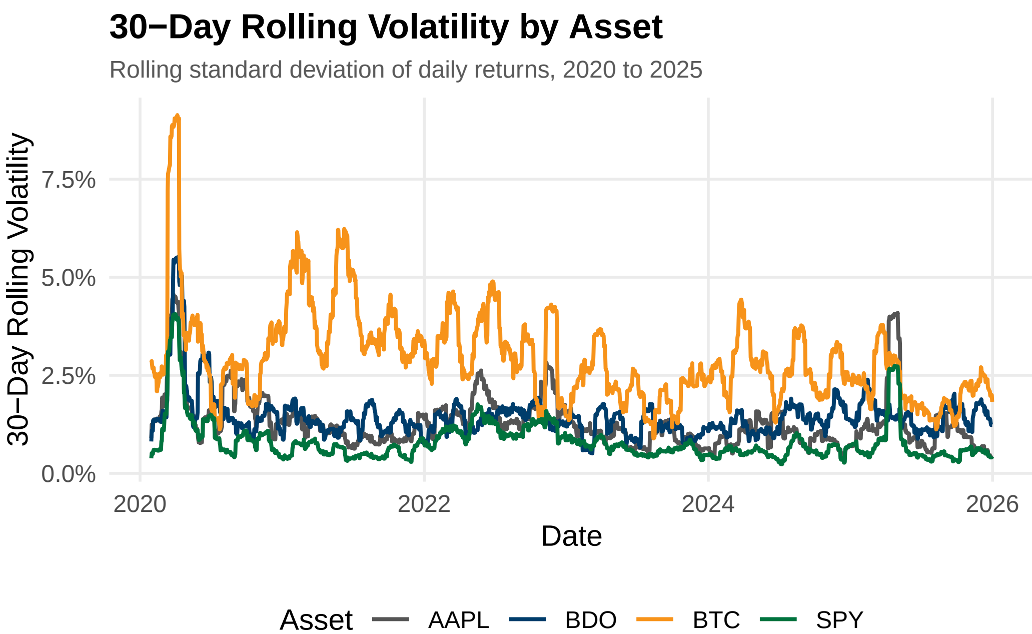
Tracks rolling 30-day standard deviation of daily returns to visualize volatility evolution over time.

```
rolling_sd <- function(x, n = 30) {
  sapply(seq_along(x), function(i) {
    if (i < n) return(NA)
    sd(x[(i - n + 1):i], na.rm = TRUE)
  })
}

fig5_data <- sql_long |>
  filter(!is.na(Daily_Return)) |>
  group_by(Asset) |>
  arrange(Date) |>
  mutate(Rolling_Vol = rolling_sd(Daily_Return, 30)) |>
  ungroup() |>
  filter(!is.na(Rolling_Vol))

ggplot(fig5_data, aes(x = Date, y = Rolling_Vol, color = Asset)) +
  geom_line(linewidth = 0.7) +
  scale_color_manual(values = c(AAPL = "#555555", BDO = "#003D6B",
                                BTC = "#F7931A", SPY = "#00733E")) +
  scale_y_continuous(labels = scales::percent_format()) +
```

```r
  labs(title = "30-Day Rolling Volatility by Asset",
       subtitle = "Rolling standard deviation of daily returns, 2020 to
        ↪   2025",
       x = "Date", y = "30-Day Rolling Volatility", color = "Asset") +
  theme_minimal(base_size = 11) +
  theme(panel.grid.minor = element_blank(),
        axis.ticks = element_blank(),
        plot.title = element_text(face = "bold", size = 13),
        plot.subtitle = element_text(size = 9, color = "#595959"),
        legend.position = "bottom")
```




```r
ggsave("../reports/figures/fig_viz5.pdf", width = 7, height = 4, device =
 ↪   "pdf")
```

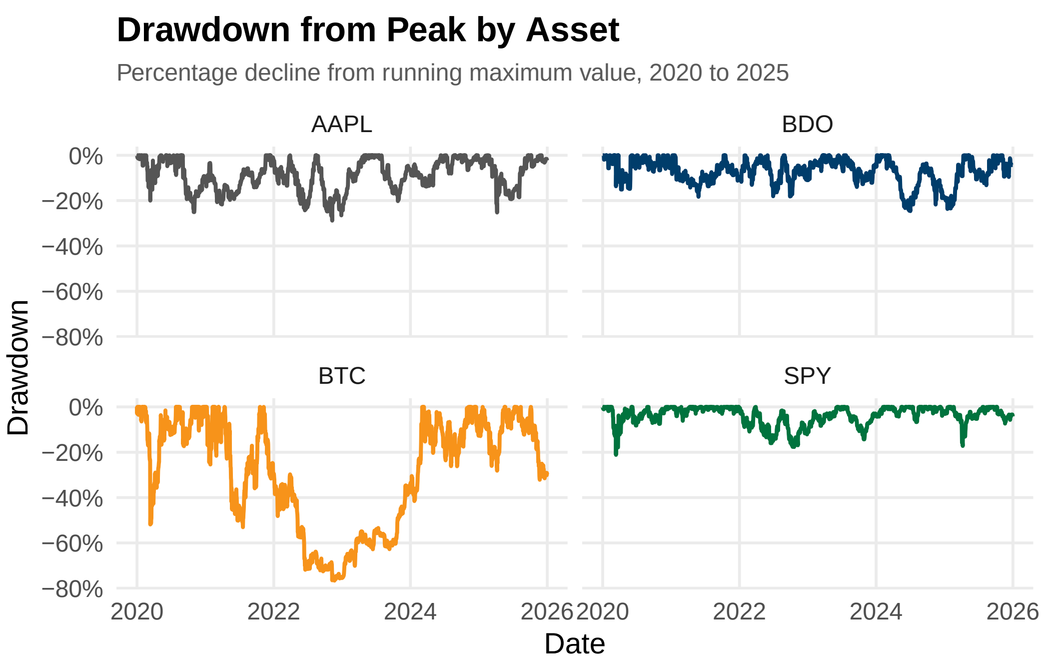
Viz Figure 6: Drawdown from Peak by Asset

Computes percentage decline from each asset's running historical high to visualize recovery patterns.

```r
fig6_data <- fig1_data |>
  group_by(Asset) |>
  arrange(Date) |>
  mutate(Running_Max = cummax(1 + Cumulative_Return),
         Drawdown = (1 + Cumulative_Return) / Running_Max - 1) |>
  ungroup()

ggplot(fig6_data, aes(x = Date, y = Drawdown, color = Asset)) +
  geom_line(linewidth = 0.7) +
```

```r
  facet_wrap(~ Asset, ncol = 2) +
  scale_color_manual(values = c(AAPL = "#555555", BDO = "#003D6B",
                                BTC = "#F7931A", SPY = "#00733E")) +
  scale_y_continuous(labels = scales::percent_format()) +
  labs(title = "Drawdown from Peak by Asset",
       subtitle = "Percentage decline from running maximum value, 2020 to
       ↪  2025",
       x = "Date", y = "Drawdown") +
  theme_minimal(base_size = 11) +
  theme(panel.grid.minor = element_blank(),
        axis.ticks = element_blank(),
        legend.position = "none",
        plot.title = element_text(face = "bold", size = 13),
        plot.subtitle = element_text(size = 9, color = "#595959"))
```



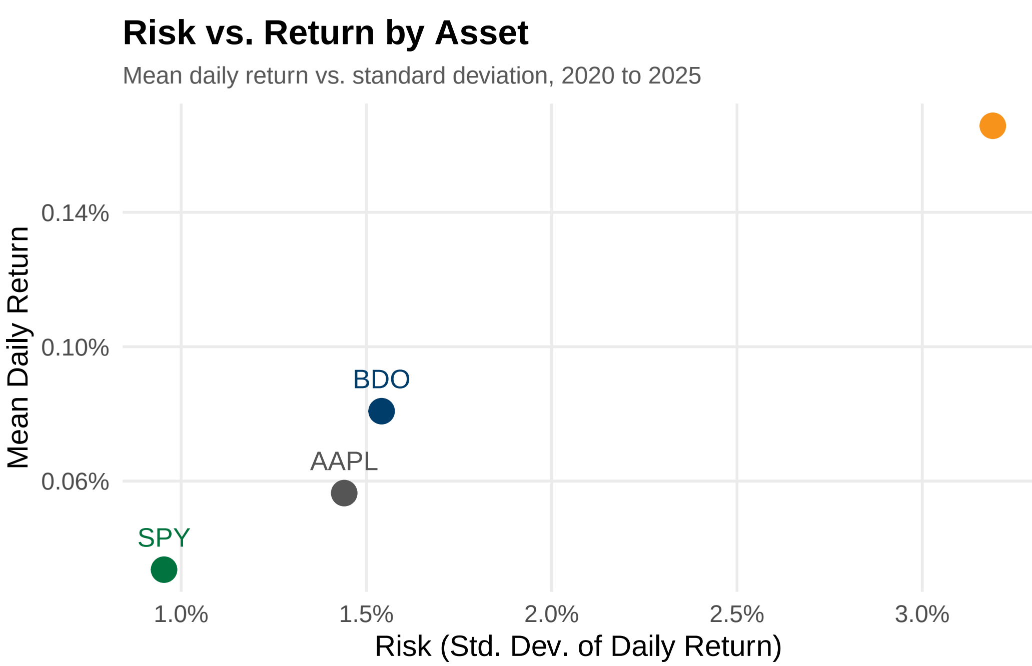

```r
ggsave("../reports/figures/fig_viz6.pdf", width = 7, height = 5, device =
↪  "pdf")
```

Viz Figure 7: Risk vs. Return by Asset

Scatter plot comparing mean daily return against standard deviation across all four assets.

```r
fig7_data <- sql_long |>
  filter(!is.na(Daily_Return)) |>
  group_by(Asset) |>
  summarise(Mean_Return = mean(Daily_Return), SD_Return = sd(Daily_Return),
   ↪  .groups = "drop")
```

```r
ggplot(fig7_data, aes(x = SD_Return, y = Mean_Return, color = Asset)) +
  geom_point(size = 4) +
  geom_text(aes(label = Asset), vjust = -1.2, size = 3.5, show.legend =
  ↪   FALSE) +
  scale_color_manual(values = c(AAPL = "#555555", BDO = "#003D6B",
                                BTC = "#F7931A", SPY = "#00733E")) +
  scale_x_continuous(labels = scales::percent_format()) +
  scale_y_continuous(labels = scales::percent_format()) +
  labs(title = "Risk vs. Return by Asset",
       subtitle = "Mean daily return vs. standard deviation, 2020 to 2025",
       x = "Risk (Std. Dev. of Daily Return)", y = "Mean Daily Return") +
  theme_minimal(base_size = 11) +
  theme(panel.grid.minor = element_blank(),
        axis.ticks = element_blank(),
        legend.position = "none",
        plot.title = element_text(face = "bold", size = 13),
        plot.subtitle = element_text(size = 9, color = "#595959"))
```




```r
ggsave("../reports/figures/fig_viz7.pdf", width = 7, height = 5, device =
↪   "pdf")
```

Viz Figure 8: Equal-Weighted Portfolio Risk Contribution

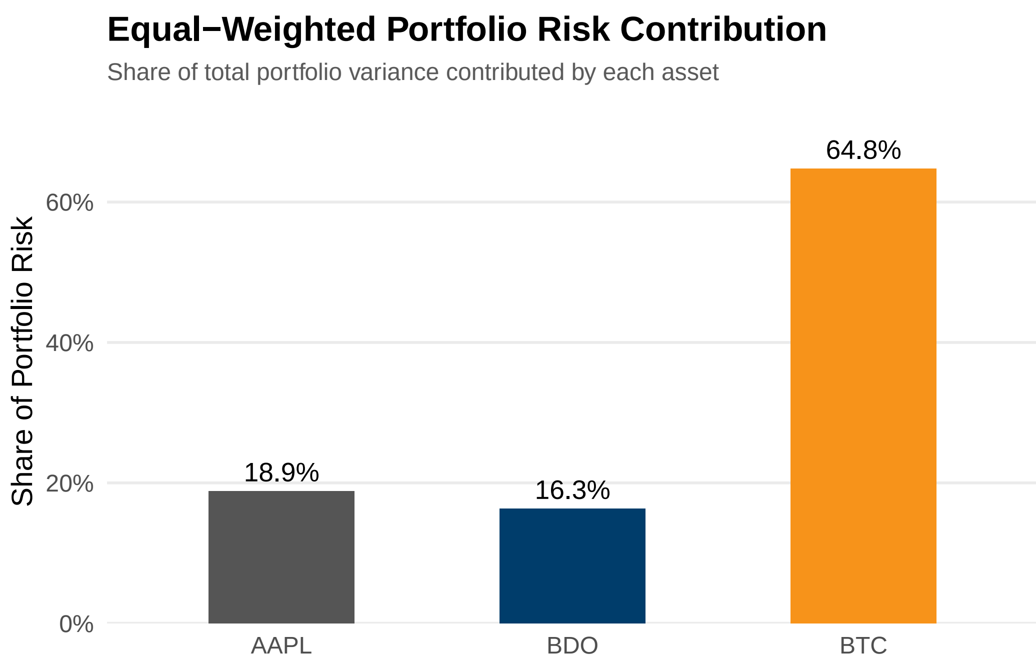
Decomposes total portfolio variance into each asset's contribution to show that equal dollar weights do not produce equal risk. SPY is excluded because Apple's 0.787 correlation with the S&P 500 means that including both AAPL and SPY would double-count the same equity factor exposure, inflating its risk contribution.

```r
returns_matrix <- sql_long |>
  filter(Asset %in% c("AAPL", "BDO", "BTC"), !is.na(Daily_Return)) |>
  select(Date, Asset, Daily_Return) |>
  pivot_wider(names_from = Asset, values_from = Daily_Return) |>
  select(AAPL, BDO, BTC) |>
  na.omit()

cov_matrix <- cov(returns_matrix)
weights <- c(AAPL = 1/3, BDO = 1/3, BTC = 1/3)
port_var <- as.numeric(t(weights) %*% cov_matrix %*% weights)
marginal_contrib <- cov_matrix %*% weights
risk_contrib <- as.numeric(weights * marginal_contrib / port_var)

fig8_data <- data.frame(Asset = names(weights), Risk_Contribution =
↪  risk_contrib)

ggplot(fig8_data, aes(x = Asset, y = Risk_Contribution, fill = Asset)) +
  geom_col(width = 0.5, show.legend = FALSE) +
  geom_text(aes(label = scales::percent(Risk_Contribution, accuracy = 0.1)),
            vjust = -0.5, size = 3.5) +
  scale_fill_manual(values = c(AAPL = "#555555", BDO = "#003D6B", BTC =
  ↪  "#F7931A")) +
  scale_y_continuous(labels = scales::percent_format(), expand =
  ↪  expansion(mult = c(0, 0.15))) +
  labs(title = "Equal-Weighted Portfolio Risk Contribution",
       subtitle = "Share of total portfolio variance contributed by each
       ↪  asset",
       x = NULL, y = "Share of Portfolio Risk") +
  theme_minimal(base_size = 11) +
  theme(panel.grid.major.x = element_blank(),
        panel.grid.minor = element_blank(),
        axis.ticks = element_blank(),
        plot.title = element_text(face = "bold", size = 13),
        plot.subtitle = element_text(size = 9, color = "#595959"))
```




```r
ggsave("../reports/figures/fig_viz8.pdf", width = 7, height = 4, device =
↪  "pdf")
```

Appendix A19 - Descriptive Statistics

Author: GALEDO, Enrique Lorenzo Hermoso Note: Corresponds to Section 6 (Descriptive and Visualizations). Code is being developed in a separate PR and will be merged here.

```r
# To be merged from GALEDO, Enrique Lorenzo Hermoso
```

Appendix A20 - Investment Recommendation

Author: SEECHUNG, Camille Castro Note: Corresponds to Section 7 (Investment Recommendation). Allocate USD 1,000,000.

```r
# Sec 7: Investment Recommendation -- USD 1,000,000 allocation among AAPL,
↪  BDO, BTC, cash -- SEECHUNG, Camille Castro
```

Appendix A21 - Executive Dashboard

Author: CRUZ, Ricardo Miguel Iñigo Note: Corresponds to Section 0 (Executive Summary).

```r
# Sec 0: Executive Dashboard -- key stats, charts, SQL findings,
↪  recommendation summary -- CRUZ, Ricardo Miguel Iñigo
```

B. Appendix B - Additional Tables and Figures

The following tables and figures present the full sequence of data extraction, cleaning, integration, and quantitative analysis (Q1-Q10) in chronological order.

Repository & Resources

The full source code, datasets, and rendered outputs for this project are available on GitHub:

https://github.com/sakudiff/FINSTAD-Case-1

The repository contains the complete R analysis scripts, the Quarto Markdown appendix, the SQLite database, and all supplementary files required to reproduce every table and figure in this report.